Lilongwe University of Agriculture and Natural Resources

Basic Sciences Department

Trigonometry and Introduction to Calculus - MAT 31201

Module

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Module Overview

This module has two parts namely trigonometry and Calculus. The main aim for this module is to extensively cover trigonometry and provide a comprehensive introduction to a branch of mathematics called **calculus**. For trigonometry you will learn functions, identities, graphs and equations. For calculus you will learn the main areas of calculus, namely, **differentiation** and **integration**. You will learn techniques and applications of both areas.

The first unit discusses **arcs, sectors and trigonometric functions**. The idea here is to introduce you to formulas for arc length and area of a sector. You will also learn the six trigonometric functions and how they relate to each other. The second unit covers **trigonometric identities and formulas**. Here we discuss the method of verifying identities. We also look at various formulas and how they are derived. In unit 3 we discuss **graphs, inverses and equations**. Here you learn how to sketch trigonometric graphs, how to calculate inverses and, finally, how to solve trigonometric equations.

The fourth unit discusses the **limit of a function** and **continuity of functions**. The focus is on calculating the limit of a function and identifying continuous function at a given number. The fifth unit gives the **definition** and **techniques of differentiation**. Unit 6 looks at **differentiation of transcendental functions**. In Unit 7, we look at **applications of differentiation** to other areas like rates of change and the shapes of graphs. In unit 8, we introduce **integration**. Both indefinite and definite integrals are discussed. Unit 9 covers **techniques of integration**. The substitution rule, integration by parts and integration of rational functions by partial fractions are discussed. Finally, unit 10, looks at one application of integration which is finding **areas** enclosed by curves.

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Part I

Trigonometry

Unit 1

Arcs, Sectors and Trigonometric Functions

1.1 Introduction

In this unit, you will learn that angles are measured in degrees and radians. You will also learn formulas for finding the length of the arc and area of a sector. The unit also covers trigonometric functions. You will learn the six functions and how they are related.

1.2 Objectives

By the end of this unit, you should be able to:

- convert angles between degrees and radians
- calculate arc length and area of a sector
- find angles which are co-terminal to a given angle
- find the reference angle of a given angle

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- arc
- sector
- trigonometric function
- radian measure
- positive and negative angles
- co-terminal angles
- reference angles
- the four quadrants
- standard angles

1.3 Angular Measure

Angles are measured in degrees and radians. The degree ($\circ$) is $\frac{1}{360}$ th of a complete revolution. The number 360 is only an arbitrary choice. A radian is defined as below.

Definition 1.1. *One radian is the measure of an angle made by an arc of length equal to the radius. (See Figure 1.1).*

Increasing the length of an arc to $2r$ results into an angle of 2 radians. In general, an angle of n radians is formed by an arc of length nr at the centre of the circle with radius r. Recall that the circumference of the circle of radius r has length $2\pi r$. Therefore, a complete revolution is 2π radians. So 2π radians $= 360^\circ$ giving π radians $= 180^\circ$, $\frac{\pi}{2}$ radians $= 90^\circ$, $\frac{\pi}{4}$ radians $= 45^\circ$, $\frac{\pi}{3}$ radians $= 60^\circ$ and $\frac{\pi}{6}$ radians $= 30^\circ$.

One degree is $\frac{\pi}{180}$ radians while one radian is $\frac{180}{\pi}$ degrees. We use this to convert between radians and degrees as in Table 1.1 below.

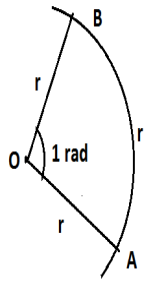

Figure 1.1: One radian

To change	Multiply by
Degrees to radians	$\pi/180°$
Radians to degrees	$180°/\pi$

Table 1.1: Converting between degrees and radians

The angle that measures α radians is written α rad or α^r, but normally just as α. However, if the angle measures β degrees, the degree symbol $\circ$ **must** be used as in $\beta°$.

Example 1.2. *Convert the following angles to degrees.*

1. $\dfrac{7\pi}{4}$
2. $\dfrac{5\pi}{6}$

Solution

We multiply by $\dfrac{180°}{\pi}$

1. $\dfrac{7\pi}{4} = \dfrac{7\pi}{4} \times \dfrac{180°}{\pi} = 315°.$
2. $\dfrac{5\pi}{6} = \dfrac{5\pi}{6} \times \dfrac{180°}{\pi} = 150°.$

Example 1.3. *Convert the following angles to radians.*

1. $150°$

2. $540°$

Solution

We multiply by $\dfrac{\pi}{180°}$.

1. $150° = 150° \times \dfrac{\pi}{180°} = \dfrac{5\pi}{6}$ rad.
2. $540° = 540° \times \dfrac{\pi}{180°} = 3\pi$ rad.

Table 1.2 gives radian measure for some standard angles.

Degrees	$30°$	$45°$	$60°$	$90°$	$270°$
Radians	$\dfrac{\pi}{6}$	$\dfrac{\pi}{4}$	$\dfrac{\pi}{3}$	$\dfrac{\pi}{2}$	$\dfrac{3\pi}{2}$

Table 1.2: Degrees and Radian Measures

1.4 Arc Length and Area of a Sector

In Figure 1.2, the arc AB of length s subtends an angle θ at the centre of the circle of radius r. We need to find the formula for calculating s given r and θ.

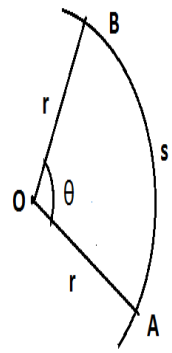

Figure 1.2: sector

By the definition of a radian, an arc of length nr subtends an angle of n radians at the centre of the circle. Hence the length of an arc s on a circle of radius r which

subtends an angle of θ radians is

$$s = r\theta. \tag{1.1}$$

This formula is derived from the definition of a radian. Therefore it is valid only when θ is given in radians.

Example 1.4. *In a circle of radius* 14 *cm, find;*

1. *the length of the arc which subtends an angle of* $\frac{3\pi}{7}$ *radians at the centre.*
2. *the angle subtended by an arc of length 6 cm.*

Solution

1. By Equation 1.1, $s = r\theta = 14 \times \frac{3\pi}{7} = 6\pi$ cm.
2. From $s = r\theta$, we have $\theta = \frac{s}{r} = \frac{6}{14} = 0.43$ rad.

Example 1.5. *Find the length of the arc which subtends an angle* $45°$ *at the centre in a circle of radius 8 cm.*

Solution

$r = 8$ and $\theta = 45° = \frac{\pi}{4}$.

Hence $s = r\theta = 8 \times \frac{\pi}{4} = 2\pi$ cm (≈ 6.3 cm).

Note that the area of the sector AOB in Figure 1.2 is proportional to the angle θ at the centre. Recall that the area of the circle with radius r is πr^2 and is obviously proportional to the central angle 2π. Now

$$\begin{aligned}
\frac{\text{area of sector} AOB}{\text{area of circle}} &= \frac{\theta}{2\pi} \\
\frac{A}{\pi r^2} &= \frac{\theta}{2\pi} \\
A &= \frac{\theta}{2\pi} \times \pi r^2 \\
&= \frac{1}{2} r^2 \theta.
\end{aligned}$$

Again this formula is valid only if θ is given in radians.

Example 1.6. 1. *Find the area of the sector with radius* 4 *cm and central angle* 2.5 *radians.*

2. *If the area of the sector is* 6.4 *cm*^{2} *and its angle is 0.8 rad, find its radius.*

3. *Find the area of the sector bounded by radii of 15 cm and arc of length* 20 *cm.*

Solution

1. $A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 4^2 \times 2.5 = 20 \text{ cm}^2$.

2. $A = \frac{1}{2}r^2\theta \Rightarrow 6.4 = \frac{1}{2}r^2 \times 0.8 \Rightarrow 6.4 = 0.4r^2 \Rightarrow r^2 = 16$ giving $r = 4$ cm.

3. $r = 15$, $s = 20$ and $\theta = \frac{s}{r} = \frac{20}{15} = \frac{4}{3}$.
   Hence $A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 15^2 \times \frac{4}{3} = \frac{1}{2} \times 225 \times \frac{4}{3} = 150 \text{ cm}^2$.

Recall that the perimeter of a closed geometric figure is a continuous line which forms the boundary of the figure. Clearly the perimeter of the sector in Figure 1.2 is

$$P = r + r + s = 2r + r\theta.$$

Example 1.7. *A sector of a circle has radius* r *and angle* θ*. Find the value of* θ *if the perimeter of the sector equals half the circumference of the circle.*

Solution

$2r + r\theta = \frac{1}{2}(2\pi r) \Rightarrow 2r + r\theta = \pi r \Rightarrow 2 + \theta = \pi$ giving $\theta = \pi - 2$ (≈ 1.142) radians.

Example 1.8. *A sector of angle* θ *in a circle of radius* r *cm has an area of* 5 *cm*^{2} *and its perimeter is* 9 *cm. Find the values of* r *and* θ*.*

$$A = \frac{1}{2}r^2\theta = 5 \tag{1.2}$$

$$P = 2r + r\theta = 9 \tag{1.3}$$

By Equation 1.3, $r\theta = 9 - 2r \Rightarrow \theta = \frac{9 - 2r}{r}$.

Substituting in Equation 1.2 we have $\frac{1}{2}r^2\left(\frac{9-2r}{r}\right) = 5 \Rightarrow 9r - 2r^2 = 10$

$\Rightarrow 2r^2 - 9r + 10 = 0$ giving $r = 2$ and $r = \frac{5}{2}$.

When $r = 2$, $\theta = \frac{9 - 2 \times 2}{2} = \frac{9-4}{2} = \frac{5}{2}$.

When $r = \frac{5}{2}$, $\theta = \frac{9 - 2 \times \frac{5}{2}}{\frac{5}{2}} = \frac{9-5}{\frac{5}{2}} = 4 \times \frac{2}{5} = \frac{8}{5}$.

Hence $r = 2$ when $\theta = \frac{5}{2}$ and $r = \frac{5}{2}$ when $\theta = \frac{8}{5}$.

Activity 1.9.

1. Convert the following radians to degrees: $\frac{10\pi}{6}$, $\frac{3\pi}{16}$, $\frac{7\pi}{4}$ and $\frac{7\pi}{9}$
2. Convert the following angles to radian measure as a multiple of π: 450°, 78°, 750°, 150° and 540°.
3. The area of a sector of a circle is 15 cm^2 and the length of its arc is 3 cm. Calculate the radius of the sector and its angle.
4. Find $\cos\frac{\pi}{3}, \tan 2, \sec\frac{\pi}{4}$ and $\csc 7$.
5. The area of a sector of a circle is $9cm^2$. If its radius is $6cm$ find the angle of the sector and the length of its arc.
6. If the area of a sector is $6.4cm^2$ and its angle is 0.8 rad, find its radius.
7. A chord in a circle of radius 8 cm is subtended by an angle of 40° at the centre. Find
   (a) the length of the chord.
   (b) the area of the minor segment.
8. Figure 1.3, shows part of a circle centre O of radius $6cm$.
   (a) If $\alpha = 0.8$ radians, find the length of the arc BC and the area of the sector BOC.
   (b) Find the value of the angle α (in radians) for which the arc length BC is equal to the sum of the arc length CA and the diameter.
9. A sector of a circle radius r has a total perimeter of $12cm$. If its area is Acm^2, show that $A = 6r - r^2$.

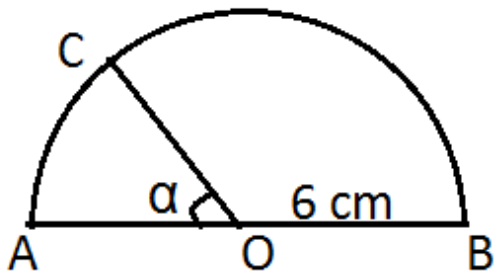

Figure 1.3: semicircle

1.5 Trigonometric Functions

The six trigonometric functions are sine, cosine, tangent, secant, cosecant and cotangent. Using the right angled triangle in Figure 1.4, we define these functions as follows;

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} \qquad \csc\theta = \frac{\text{hypotenuse}}{\text{opposite}}$$
$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} \qquad \sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}}$$
$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} \qquad \cot\theta = \frac{\text{adjacent}}{\text{opposite}}.$$

Note that here θ is an acute angle, i.e., $0 < \theta < \frac{\pi}{2}$ (or $0° < \theta < 90°$). Next we look at an approach which define trigonometric functions for any angles.

Figure 1.5 is the unit circle (circle of radius 1) drawn on an $x - y$ plane. The centre for this circle is actually the origin. The point P on the circumference of this circle is obtained by rotating OP from the positive x-axis in the counter clockwise direction. θ is the angle between OP and the x-axis. It is said to be in **standard position**. Note that θ is arbitrary (represents any angle).

Definition 1.10. *By the unit circle in Figure 1.5,* $\sin\theta = \frac{y}{1} = y$, $\cos\theta = \frac{x}{1} = x$, $\tan\theta = \frac{y}{x}$, $\csc\theta = \frac{1}{y}$, $\sec\theta = \frac{1}{x}$ *and* $\cot\theta = \frac{x}{y}$.

By Definition 1.10, coordinates of the point P in Figure 1.5 are $(x, y) = (\cos\theta, \sin\theta)$. Since trigonometric functions are defined by the unit circle, they are also called circular functions. Definition 1.10 shows that the six trigonometric functions are

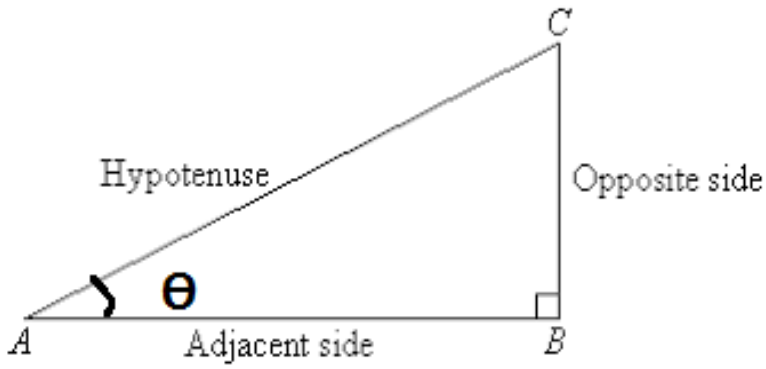

Figure 1.4: Right angled Triangle

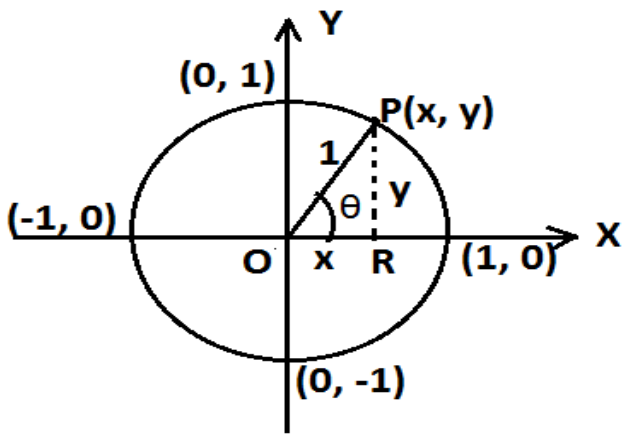

Figure 1.5: Unit Circle

related as follows; $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, $\csc\theta = \dfrac{1}{\sin\theta}$, $\sec\theta = \dfrac{1}{\cos\theta}$ and $\cot\theta = \dfrac{\cos\theta}{\sin\theta} = \dfrac{1}{\tan\theta}$.

Triangle PRO in Figure 1.5 is right angled at R. Therefore, by the Pythagoras theorem, $x^2 + y^2 = 1$. Since $x = \cos\theta$ and $y = \sin\theta$, we have,

$$\cos^2\theta + \sin^2\theta = 1.$$

This is true for any angle (remember θ is an arbitrary angle). Here $\cos^2\theta = (\cos\theta)^2$ (we find cosine of an angle then square the result) but $\cos^2\theta \neq \cos\theta^2$.

Again from the unit circle above, the maximum value of x is 1 and the minimum value is -1, i.e., $-1 \leq x \leq 1$. Similarly we have $-1 \leq y \leq 1$. Since $x = \cos\theta$ and $y = \sin\theta$, we have

$$-1 \leq \cos\theta \leq 1$$

and

$$-1 \leq \sin\theta \leq 1.$$

1.5.1 Positive and Negative Angles

The angle θ in Figure 1.5, is considered **positive** since it is formed by rotating the line OP from the positive x-axis in a counter clockwise direction. If OP is rotated in a clockwise direction, the resulting angle is negative (see Figure 1.6).

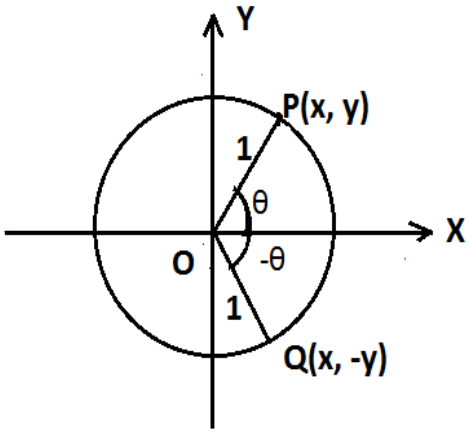

Figure 1.6: Positive and negative angles

Note that by Figure 1.6, $\cos(-\theta) = x = \cos\theta$, $\sin(-\theta) = -y = -\sin\theta$ and $\tan(-\theta) = -\tan\theta$. Therefore cosine function is an **even** function while sine and

tangent are **odd** functions. Consequently secant function is even but cosecant and cotangent are odd functions.

Example 1.11.

$$\begin{aligned}\sin(-24^\circ) &= -0.4067 = -\sin 24^\circ,\\ \cos(-24^\circ) &= 0.9135 = \cos 24^\circ,\\ \tan(-24^\circ) &= -0.4452 = -\tan 24^\circ.\end{aligned}$$

1.5.2 Co-terminal Angles

Recall that the angle θ in the unit circle in Figure 1.5 is formed by rotating the line OP from the positive x-axis in a counter clockwise direction. Suppose $\theta = 25^\circ$. Starting at the same positive x-axis, if we can make one revolution before coming to OP we have an angle of 385°. If we can make two revolutions we have an angle of 745°. If we take a different direction but still stop at the same OP, we have the angle -335°. The angles $25^\circ, 385^\circ$, -335° and 745° are **co-terminal**. Since the number of revolutions is infinite, there are infinitely many angles which are co-terminal to any given angle.

Definition 1.12. *Given the angle in standard position with measure θ, then the measures of the angles that are co-terminal with θ are given by*

$$\theta + 360^\circ k$$

where k is an integer.

By Definition 1.12, co-terminal angles are angles which differ by an integer multiple of 360° (or 2π if the angles are measured in radians).

Example 1.13. *If $\theta = 70^\circ$, find two positive and two negative angles which are co-terminal to θ.*

Solution

To find positive co-terminal angles we add to θ any positive integer multiple of 360°, for example 360° itself and 720°. This gives $360^\circ + 70^\circ = 430^\circ$ and $720^\circ + 70^\circ = 790^\circ$.

To find negative co-terminal angles, we add any negative integer multiple of 360° to 70°, for example -360° and -720°. This gives $-360^\circ + 70^\circ = -290^\circ$ and $-720^\circ + 70^\circ = -650^\circ$.

Example 1.14. *Find any two negative angles which are co-terminal to* $\frac{5\pi}{2}$.

Solution
$\frac{5\pi}{2} - 4\pi = -\frac{3\pi}{2}$ and $\frac{5\pi}{2} - 6\pi = -\frac{7\pi}{2}$.

Hence the angles which are co-terminal to $\frac{5\pi}{2}$ include $-\frac{3\pi}{2}$ and $-\frac{7\pi}{2}$.

1.5.3 The Four Quadrants

We partition the complete revolution into four quadrants. The first quadrant has angles between 0° and 90°. The second quadrant has angles between 90° and 180°. The third and fourth quadrants have angles between 180° and 270°, and 270° and 360° respectively (see Figure 1.7).

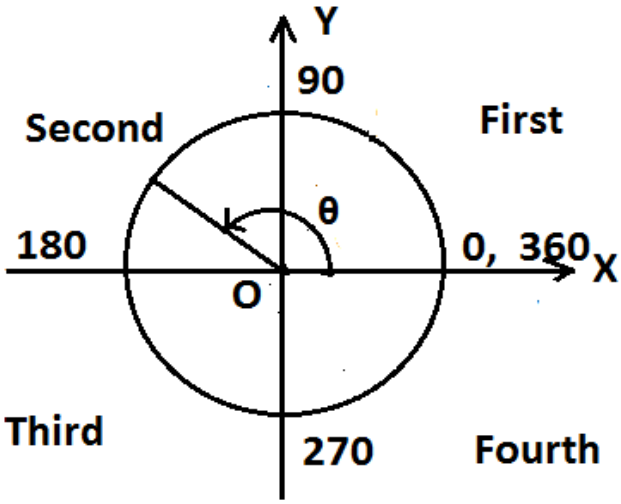

Figure 1.7: The Four Quadrants

A **quadrantal angle** is an angle whose final rotational point lies on the x-axis or y-axis. The angles $0°, 90°, 180°, 270°, 360°$ and their co-terminal angles are quadrantal angles.

The sign of a trigonometric function of an angle θ depends on the quadrant of θ. In the first quadrant all the three basic trig functions, $\sin\theta, \cos\theta$ and $\tan\theta$, are positive. In the second quadrant, $\sin\theta$ is positive but $\cos\theta$ and $\tan\theta$ are negative. In the third quadrant, $\tan\theta$ is positive while $\sin\theta$ and $\cos\theta$ are negative. In the fourth quadrant, $\cos\theta$ is positive but $\sin\theta$ and $\tan\theta$ are negative (see Figure 1.8).

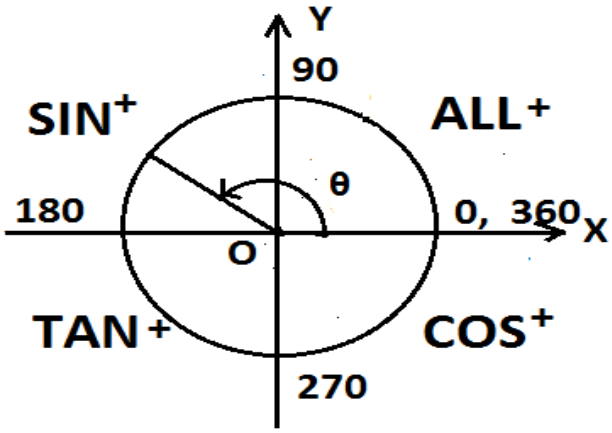

Figure 1.8: What function is positive in which quadrant

1.5.4 Reference Angles

Every angle in the first quadrant corresponds to exactly one angle in each of the other three quadrants. The first quadrant angle in this case is called the **reference angle** of the angle in the other quadrant.

Definition 1.15. *Let θ be a non-quadrantal angle in standard position. The reference angle for θ is the (first quadrant) acute angle β that the terminal side of θ makes with the x-axis.*

Figure 1.9 illustrates the reference angle in each quadrant. By Figure 1.9, if θ is in the first quadrant, then its reference angle is $\beta = \theta$. If θ is in the second quadrant, then $\beta = 180° - \theta$, if θ is in the third quadrant, then $\beta = \theta - 180°$ and $\beta = 360° - \theta$ if θ is in the fourth quadrant.

Table 1.3 gives a summary of the discussion on reference angles.

Quadrant of θ	Formula for β	
	Degrees	Radians
I	θ	θ
II	$180° - \theta$	$\pi - \theta$
III	$\theta - 180°$	$\theta - \pi$
IV	$360° - \theta$	$2\pi - \theta$

Table 1.3: Formulas for Reference Angles

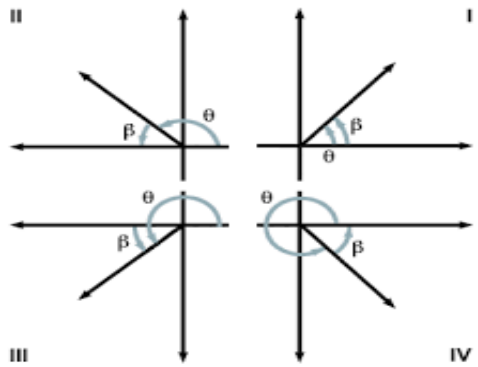

Figure 1.9: Reference Angles

Example 1.16. *Find the reference angle β for each angle θ below:*
(a) $\theta = 312°$ *(b)* $\theta = 245°$ *(c)* $\theta = -240°$ *(d)* $\theta = \dfrac{3\pi}{4}$ *(e)* $\theta = 3.5$

Solution

(a) The angle $\theta = 312°$ is in the fourth quadrant.
Therefore $\beta = 360° - \theta = 360° - 312° = 48°$.

(b) The angle $\theta = 245°$ is in the third quadrant.
Therefore $\beta = \theta - 180° = 245° - 180° = 65°$.

(c) The angle between $0°$ and $360°$ which is co-terminal to $-240°$ is

$$-240° + 360° = 120°$$

which is in the second quadrant. Hence $\beta = 180° - 120° = 60°$.

(d) The angle $\theta = \dfrac{3\pi}{4}$ is in the second quadrant.
Therefore $\beta = \pi - \theta = \pi - \dfrac{3\pi}{4} = \dfrac{\pi}{4}$.

(e) Since $\pi < 3.5 < \dfrac{3\pi}{2}$, the angle $\theta = 3.5$ is in the third quadrant.
Therefore $\beta = \theta - \pi = 3.5 - \pi$.

Working in reverse we can find angles in all the remaining three quadrants whose reference angle is the given first quadrant angle. For example, the angle $\beta = 30°$ is the reference angle to $180° - 30° = 150°$ in the second quadrant, $180° + 30° = 210°$ in the third quadrant and $360° - 30° = 330°$ in the fourth quadrant. The sine function values for these angles are related as follows;

$$\begin{aligned} \sin 150° &= 0.5 = \sin 30°, \\ \sin 210 &= -0.5 = -\sin 30°, \\ \sin 330° &= -0.5 = -\sin 30°. \end{aligned}$$

The cosine function values are related as follows;

$$\begin{aligned} \cos 150° &= -0.866 = -\cos 30°, \\ \cos 210° &= -0.866 = -\cos 30°, \\ \cos 330° &= 0.866 = \cos 30°. \end{aligned}$$

The tangent function values are related as follows;

$$\begin{aligned} \tan 150° &= -0.5774 = -\tan 30°, \\ \tan 210° &= 0.5774 = \tan 30°, \\ \tan 330° &= -0.5774 = -\tan 30°. \end{aligned}$$

The above discussion shows how reference angles can be used to find values of the trigonometric functions. If θ is a non-quadrantal angle in standard position, then to find the value of a trigonometric function at θ, find its value for the reference angle β and prefix the appropriate sign. The appropriate sign is determined by considering which function is positive or negative in the quadrant of the angle θ.

Example 1.17. *Find* $\cos 300°$ *given that* $\cos 60° = 0.5$.

Solution

$\theta = 300°$ is in the fourth quadrant and has reference angle $\beta = 60°$. Recall that cosine is positive in the fourth quadrant. Hence

$$\cos 300° = \cos 60° = 0.5.$$

1.5.5 Standard Angles

We consider trigonometric function values of angles which appear quite often. These are 45° 30°, 60° and quadrantal angles (0°, 90°, 180°, 270° and 360°). To begin with, we look at the 45° angle in Figure 1.10.

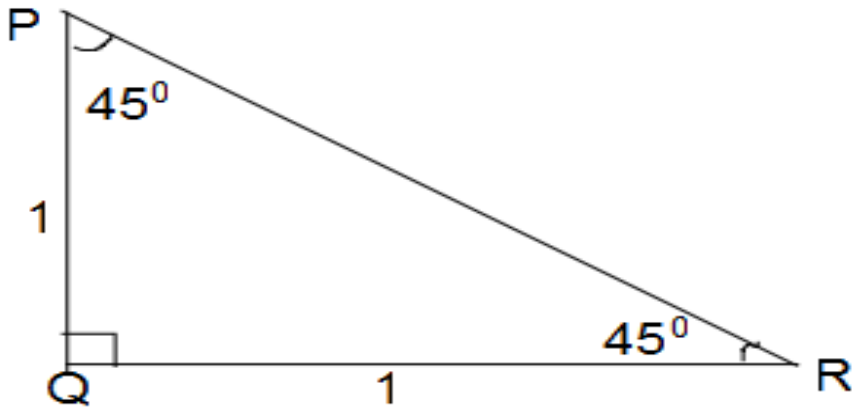

Figure 1.10: The 45° Angle

The triangle in Figure 1.10 is right angled at Q. Hence, by Pythagoras theorem, $PR = \sqrt{2}$. Now

$$\sin 45^\circ = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2},$$
$$\cos 45^\circ = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2},$$
$$\tan 45^\circ = \frac{\text{opposite}}{\text{adjacent}} = \frac{1}{1} = 1.$$

In Figure 1.11 we look at the angles 30° and 60°.

Triangle ABC in Figure 1.11 is equilateral with each side and angle measuring 2 and 60° respectively. We drop perpendicular bisector from A meeting BC at D. Then triangles ADB and ADC are both right angled at D. We use either triangle

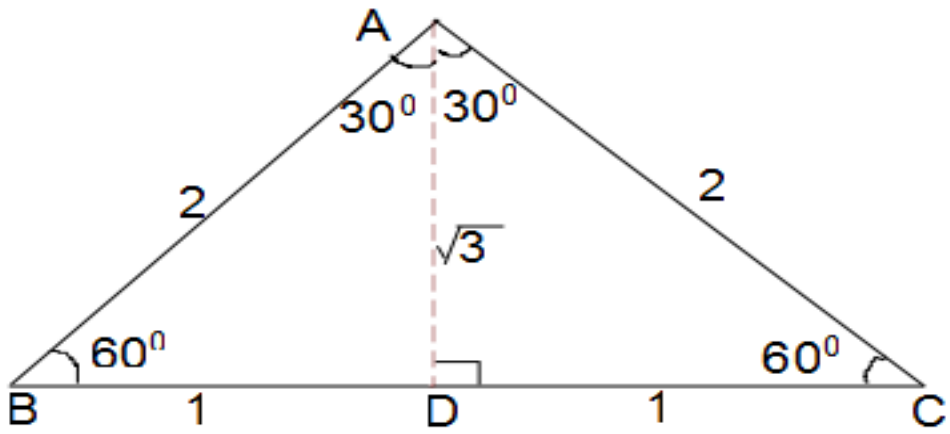

Figure 1.11: The 30° and 60° Angles

to get the following results for the 60° angle;

$$\sin 60^\circ = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{\sqrt{3}}{2},$$
$$\cos 60^\circ = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{1}{2},$$
$$\tan 60^\circ = \frac{\text{opposite}}{\text{adjacent}} = \sqrt{3}.$$

For the 30° angle we have;

$$\sin 30^\circ = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{2},$$
$$\cos 30^\circ = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\sqrt{3}}{2},$$
$$\tan 30^\circ = \frac{\text{opposite}}{\text{adjacent}} = \frac{1}{\sqrt{3}}.$$

Trigonometric function values for quadrantal angles are easily determined from the unit circle in Figure 1.5. For example,

$$\sin 180^\circ = y\text{-co-ordinate at } 180^\circ = 0,$$
$$\cos 180^\circ = x\text{-co-ordinate at } 180^\circ = 1,$$
$$\text{and } \tan 180^\circ = 0.$$

	$0°$	$30°$	$45°$	$60°$	$90°$	$180°$	$270°$	$360°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	U	0	U	0

Table 1.4: Trig Function Values for Standard Angles

These results are summarised in Table 1.4. U stands for undefined.

Note, from Table 1.4 that,

1. $\cos 30° = \sin 60° = \sin(90° - 30°) = \dfrac{\sqrt{3}}{2}$.
2. $\sin 30° = \cos 60° = \cos(90° - 30°) = 1/2$.

In general, we have,

1. $\cos\theta = \sin(90° - \theta)$.
2. $\sin\theta = \cos(90° - \theta)$.

For example, $\sin 25° = 0.4226 = \cos 65°$.

The above formulas are called **co-function** formulas for cosine and sine respectively. Proofs for these and other co-function formulas are provided in Section 2.

Example 1.18. *Find the exact values of* $\sin 315°$ *and* $\cos 240°$.

Solution

The angles $315°$ and $240°$ are in the fourth and third quadrants respectively. Their reference angles are $45°$ and $60°$ respectively. Recall that sine function is negative in the fourth quadrant and that cosine function is also negative in the third quadrant.

$$\sin 315° = -\sin(360 - 315)° = -\sin 45° = -1/\sqrt{2}$$

and

$$\cos 240° = -\cos(240 - 180)° = -\cos 60° = -1/2.$$

Example 1.19. *Find the exact values of* $\sin\dfrac{3\pi}{4}$ *and* $\tan\dfrac{11\pi}{6}$.

Solution

The angles $\dfrac{3\pi}{4}$ and $\dfrac{11\pi}{6}$ are in the second and fourth quadrants respectively. Their reference angles are $\pi - \dfrac{3\pi}{4} = \dfrac{\pi}{4}$ and $2\pi - \dfrac{11\pi}{6} = \dfrac{\pi}{6}$ respectively.

$$\sin\frac{3\pi}{4} = \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

and

$$\tan\frac{11\pi}{6} = \tan\left(-\frac{\pi}{6}\right) = -\tan\frac{\pi}{6} = -\frac{1}{\sqrt{3}}.$$

Activity 1.20.

1. State whether each statement is true or false.
   (a) The angles $-305°$ and $55°$ are co-terminal.
   (b) $\cot\theta \times \tan\theta \neq 1$.
   (c) The angle $764°$ lies in the fourth quadrant.
   (d) If $\sin A = \sin B$, then $A = B$.
   (e) The angle $\dfrac{13\pi}{3}$ lies in the second quadrant.
   (f) The angles $\dfrac{7\pi}{3}$ and $\dfrac{\pi}{3}$ are not co-terminal.
2. In which quadrant(s) is the angle θ if each of the following is satisfied?
   (a) $\cos\theta > 0$ but $\csc\theta < 0$
   (b) $\cot\theta < 0$, $\csc\theta > 0$, $\sin\theta > 0$
   (c) $\sin\theta > 0$ but $\sec\theta < 0$
   (d) $\cot\theta > 0$, $\csc\theta < 0$, $\cos\theta < 0$
3. Find the exact value of $\sin 120°, \sin(-300°)$, $\sin 240°$, $\cos 60°$, $\sin 420°$ and $\sec\dfrac{31\pi}{3}$ [*Use Table 1.4*].
4. Let $\theta = -\dfrac{7\pi}{3}$.

(a) Convert θ to degrees.

(b) Find two positive angles, in radians, which are co-terminal to θ.

5. Find two negative angles which are co-terminal to the angle $-55°$.

6. If the given angle is in standard position, find two positive coterminal angles and two negative coterminal angles; $121°$, $145°$, $243°$, $310°$ $620°$, $567°$, $\frac{5\pi}{4}$, $-\frac{\pi}{4}$, $-\frac{2\pi}{3}$ and $\frac{5\pi}{6}$.

7. Find the reference angle β if θ has the given measure; $240°$, $340°$, $-165°$, $275°$, $-400°$, $-\frac{3\pi}{4}$, $-\frac{\pi}{6}$, $\frac{4\pi}{3}$, 6, $-\frac{23\pi}{6}$, 5, $\frac{7\pi}{4}$ and 2.3.

Unit Summary

In this unit you have learnt the angular measure, arcs, sectors and trigonometric functions. You have learnt how to convert angles between radians and degrees. You have also learnt the six trigonometric functions, co-terminal angles, reference angles and standard angles.

Activity 1.21 (Unit Test).

1. If $\cos(x + 20°) = \sin x$, find x.

2. Convert $\theta = 0.2\pi$ to degrees.

3. Convert each of these radian angles to degrees; $-\frac{4\pi}{9}$, $\frac{\pi}{10}$, $-\frac{11\pi}{12}$, $\frac{7\pi}{15}$, $-\frac{3\pi}{11}$, 7.4, $-\frac{5\pi}{11}$ and 6.3.

4. Convert each angle to radians as a multiple of π; $225°$, $520°$, $315°$, $-405°$, $-75°$, $240°$, and $-210°$.

5. Find the area of the sector whose radius is 35 cm and perimeter 147 cm.

6. The area of a sector of a circle with a central angle of 4 rad is 8 m^2. Find the radius of the circle

Unit Answers

Suggested Solutions to Activity 1.9

1. $\dfrac{10\pi}{6} = 300°$, $\dfrac{3\pi}{16} = 33.75°$

2. $450° = \dfrac{5\pi}{2}$, $540° = 3\pi$

3. $r = 10$ and $\theta = \dfrac{3}{10}$

5. $\theta = 0.5$

6. $r = 4$

Suggested Solutions to Activity 1.20

1. (a) True (b) False (c) False (d) False (e) False (f) True

2. (a) Fourth (b) second (c) second (d) third

4. (a) $-420°$ (b) $\dfrac{5\pi}{3}$ and $\dfrac{11\pi}{3}$

5. $-415°$ and $-775°$

Suggested Solutions to Unit Test

1. $x = 35°$

2. $36°$

6. 2 m

Unit 2

Trigonometric Identities and Formulas

2.1 Introduction

In this unit, you will learn how to prove trigonometric identities. You will learn relevant proof techniques including things you need to avoid when proving identities. You will also learn various trigonometric formulas, how they are derived and how they are used to solve various problems.

2.2 Objectives

By the end of this unit, you should be able to:

- prove trigonometric identities
- derive trigonometric formulas
- use formulas to solve problems

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- trigonometric identities
- addition formula
- double angle formula
- half angle formula
- co-function formula
- product to sum formula
- sum to product formula

2.3 Proving Identities

An identity is a tautology, an equation or statement that is always true. For instance, the equation $\sin\theta = \dfrac{1}{\csc\theta}$ is an identity (of course it is a simple identity). Proving an identity means showing that both sides are actually having the same 'thing'. When proving an identity we do not take both sides at the same time. Instead, we use logical steps to show that one side of the equation can be transformed to the other side of the equation.

Identities are proved using other well known identities or those already proved. In Section 1.5 we derived a number of identities from the unit circle. To show their importance we list them below.

1. The **reciprocal identities** are

$$\sec\theta = \frac{1}{\cos\theta}, \quad \csc\theta = \frac{1}{\sin\theta} \quad \text{and} \quad \cot\theta = \frac{1}{\tan\theta}.$$

2. The **tangent and cotangent** identities are

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \text{and} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}.$$

3. For the identity $\cos^2\theta + \sin^2\theta = 1$ (or $\sin^2\theta = 1 - \cos^2\theta$ and $\cos^2\theta = 1 - \sin^2\theta$), if we divide both sides by $\sin^2\theta$ we obtain

$$\frac{\cos^2\theta}{\sin^2\theta} + \frac{\sin^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta} \Rightarrow \cot^2\theta + 1 = \csc^2\theta.$$

Diving both sides of the equation $\cos^2\theta + \sin^2\theta = 1$ by $\cos^2\theta$ gives

$$1 + \tan^2\theta = \sec^2\theta.$$

Hence the **Pythagorean identities** are

$$\cos^2\theta + \sin^2\theta = 1, \quad 1 + \cot^2\theta = \csc^2\theta \quad \text{and} \quad 1 + \tan^2\theta = \sec^2\theta.$$

Steps in proving identities;

1. Take one side and transform it to the expression on the other side. Start with the more complicated side of the identity, and transform it into the simpler side. If there is no direct way of connecting the two, transform both sides of an identity to a common expression.
2. Try algebraic operations such as multiplying, factoring, combining fractions, and splitting fractions.
3. If other steps fail, express each function in terms of sine and cosine functions, and then perform appropriate algebraic operations.
4. At each step, keep the other side of the identity in mind. This often reveals what you should do to get there.

Example 2.1. *Prove the identity* $\tan\theta + \cot\theta = \sec\theta\csc\theta$.

Solution

We take the expression on the Left Hand Side ($L.H.S$), simplify it and get the expression on the Right Hand Side ($R.H.S$).

$$\begin{aligned} L.H.S = \tan\theta + \cot\theta &= \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} \\ &= \frac{\sin^2\theta + \cos^2\theta}{\cos\theta\sin\theta} = \frac{1}{\cos\theta\sin\theta} \\ &= \left(\frac{1}{\cos\theta}\right)\left(\frac{1}{\sin\theta}\right) \\ &= \sec\theta\csc\theta = R.H.S. \end{aligned}$$

Example 2.2. *Prove the identity* $\dfrac{\cos^4\theta - \sin^4\theta}{\cos^2\theta - \sin^2\theta} = \dfrac{\csc\theta}{\sin\theta} - \dfrac{\cot\theta}{\tan\theta}$.

Solution

Notice that, by the difference of two squares,

$$\cos^4\theta - \sin^4\theta = (\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta).$$

Now

$$\begin{aligned} L.H.S &= \frac{(\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta)}{\cos^2\theta - \sin^2\theta} \\ &= \cos^2\theta + \sin^2\theta \\ &= 1. \end{aligned}$$

$$\begin{aligned} R.H.S &= \frac{1}{\sin\theta} \div \sin\theta - \frac{\cos\theta}{\sin\theta} \div \frac{\sin\theta}{\cos\theta} \\ &= \frac{1}{\sin\theta} \times \frac{1}{\sin\theta} - \frac{\cos\theta}{\sin\theta} \times \frac{\cos\theta}{\sin\theta} \\ &= \frac{1}{\sin^2\theta} - \frac{\cos^2\theta}{\sin^2\theta} = \frac{1-\cos^2\theta}{\sin^2\theta} \\ &= \frac{\sin^2\theta}{\sin^2\theta} = 1. \end{aligned}$$

Hence $L.H.S = R.H.S$

Example 2.3. *Show that* $(1-\cos^2\theta)(1+\cot^2\theta) = 1$.

Solution

$$\begin{aligned} L.H.S = (1-\cos^2\theta)(1+\cot^2\theta) &= (1-\cos^2\theta)\csc^2\theta \\ &= (1-\cos^2\theta)\left(\frac{1}{\sin^2\theta}\right) \\ &= \frac{\sin^2\theta}{\sin^2\theta} = 1 = R.H.S. \end{aligned}$$

Example 2.4. *Show that* $\dfrac{\cos\theta\cot\theta}{1-\sin\theta} - 1 = \csc\theta$.

Solution

$$
\begin{aligned}
L.H.S = \frac{\cos\theta\cot\theta}{1-\sin\theta} - 1 &= \frac{\cos\theta\left(\dfrac{\cos\theta}{\sin\theta}\right)}{1-\sin\theta} - 1 \\
&= \frac{\cos^2\theta}{\sin\theta} \times \frac{1}{1-\sin\theta} - 1 \\
&= \frac{1-\sin^2\theta}{\sin\theta(1-\sin\theta)} - 1 \\
&= \frac{(1+\sin\theta)(1-\sin\theta)}{\sin\theta(1-\sin\theta)} - 1 \\
&= \frac{1+\sin\theta}{\sin\theta} - 1 \\
&= \frac{1+\sin\theta-\sin\theta}{\sin\theta} \\
&= \frac{1}{\sin\theta} = \csc\theta = R.H.S.
\end{aligned}
$$

Example 2.5. *Prove the identity* $(\sec\theta + \tan\theta)(1-\sin\theta) = \cos\theta$.

Solution

$$
\begin{aligned}
L.H.S = (\sec\theta + \tan\theta)(1-\sin\theta) &= \left(\frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta}\right)(1-\sin\theta) \\
&= \frac{1+\sin\theta}{\cos\theta}(1-\sin\theta) \\
&= \frac{1-\sin^2\theta}{\cos\theta} \\
&= \frac{\cos^2\theta}{\cos\theta} \\
&= \cos\theta = R.H.S.
\end{aligned}
$$

Example 2.6. *Show that* $\dfrac{\cot\theta\sec^2\theta}{1+\cot^2\theta} = \tan\theta$.

Solution

$$
\begin{aligned}
L.H.S = \frac{\cot\theta\sec^2\theta}{1+\cot^2\theta} &= \frac{\cot\theta\sec^2\theta}{\csc^2\theta} \\
&= \left(\frac{\cos\theta}{\sin\theta} \times \frac{1}{\cos^2\theta}\right) \div \frac{1}{\sin^2\theta} \\
&= \left(\frac{1}{\sin\theta\cos\theta}\right) \times \sin^2\theta \\
&= \frac{\sin\theta}{\cos\theta} = \tan\theta = R.H.S.
\end{aligned}
$$

We can use different approaches to prove one identity and still get correct results provided logic steps are followed to the latter.

Example 2.7. *Prove the identity* $\dfrac{\tan\theta+\tan\beta}{\cot\theta+\cot\beta} = \tan\theta\tan\beta$.

Solution (Method One)

$$
\begin{aligned}
L.H.S &= \left(\frac{\sin\theta}{\cos\theta} + \frac{\sin\beta}{\cos\beta}\right) \div \left(\frac{\cos\theta}{\sin\theta} + \frac{\cos\beta}{\sin\beta}\right) \\
&= \left(\frac{\sin\theta\cos\beta+\sin\beta\cos\theta}{\cos\theta\cos\beta}\right) \div \left(\frac{\sin\beta\cos\theta+\sin\theta\cos\beta}{\sin\theta\sin\beta}\right) \\
&= \frac{\sin\theta\cos\beta+\sin\beta\cos\theta}{\cos\theta\cos\beta} \times \frac{\sin\theta\sin\beta}{\sin\beta\cos\theta+\sin\theta\cos\beta} \\
&= \frac{\sin\theta\sin\beta}{\cos\theta\cos\beta} \\
&= \left(\frac{\sin\theta}{\cos\theta}\right)\left(\frac{\sin\beta}{\cos\beta}\right) = \tan\theta\tan\beta = R.H.S.
\end{aligned}
$$

Solution (Method Two)

$$
\begin{aligned}
L.H.S = \frac{\tan\theta+\tan\beta}{\cot\theta+\cot\beta} &= (\tan\theta+\tan\beta) \div \left(\frac{1}{\tan\theta} + \frac{1}{\tan\beta}\right) \\
&= (\tan\theta+\tan\beta) \div \left(\frac{\tan\beta+\tan\theta}{\tan\theta\tan\beta}\right) \\
&= (\tan\theta+\tan\beta) \times \left(\frac{\tan\theta\tan\beta}{\tan\theta+\tan\beta}\right) \\
&= \tan\theta\tan\beta = R.H.S.
\end{aligned}
$$

Activity 2.8.

1. Simplify each of the following

   (a) $\cot x \sin x \sec x$.

   (b) $\dfrac{\sin x + \dfrac{\cos^2 x}{\sin x}}{\csc x}$.

   (c) $\dfrac{\sin x \sec x}{\cot x}$.

2. Prove each identity below.

   (a) $\dfrac{\cot y}{\tan y} + 1 = \csc^2 y$

   (b) $\dfrac{\cos\theta - \cos^3\theta}{\sin^3\theta} = \cot\theta$.

   (c) $\dfrac{\tan^2 x}{1+\tan^2 x} = \sin^2 x$

   (d) $\tan^2\theta - \sin^2\theta = \tan^2\theta\sin^2\theta$.

   (e) $\dfrac{1}{1+\cos\theta} + \dfrac{1}{1-\cos\theta}$
   $= 2\csc^2\theta$.

   (f) $\dfrac{\cos\theta}{\sin\theta+1} + \dfrac{\sin\theta+1}{\cos\theta} = 2\sec\theta$

   (g) $\dfrac{\cos x}{1-\sin x} - \tan x = \sec x$

   (h) $\dfrac{\csc\beta}{\sin\beta} - \dfrac{\cot\beta}{\tan\beta} = 1$

   (i) $\dfrac{1}{\cos^2 x} - \dfrac{1}{\cot^2 x} = 1$

   (j) $\sec^4 x - \sec^2 x = \tan^4 x + \tan^2 x$

   (k) $\sin x \cot x = \cos x$

   (l) $\dfrac{1-(\sin x - \cos x)^2}{\sin x} = 2\cos x$

   (m) $\cos^2\theta(\sec^2\theta - 1) = \sin^2\theta$

2.4 Addition and Co-function Formulas

We now look at identities that involve a trigonometric function of the sum or difference of two variables. Examples are $\sin(\theta+\beta)$, $\sin(\theta-\beta)$, $\cos(\theta+\beta)$ and $\cos(\theta-\beta)$. You may be tempted to think, for example, that $\cos(\theta+\beta) = \cos\theta + \cos\beta$. Unfortunately, this is totally wrong. Please keep in mind that the trigonometric functions cannot be 'distributed' through parentheses. We derive the correct formulas starting with the formulas for $\cos(\theta+\beta)$ and $\cos(\theta-\beta)$.

Addition and Subtraction Formulas for Cosine

Figure 2.1 is a unit circle. The lines OP and OQ are both rotated from the positive x-axis in a counter clockwise direction forming the angles β and $(\theta+\beta)$ with the x-axis respectively. OR is rotated from the same positive x-axis but in a clockwise direction forming angle $-\theta$ with the positive x-axis. Here we assume that $0 < \theta < \beta < 2\pi$ (but the result we are about to derive is true for any angles).

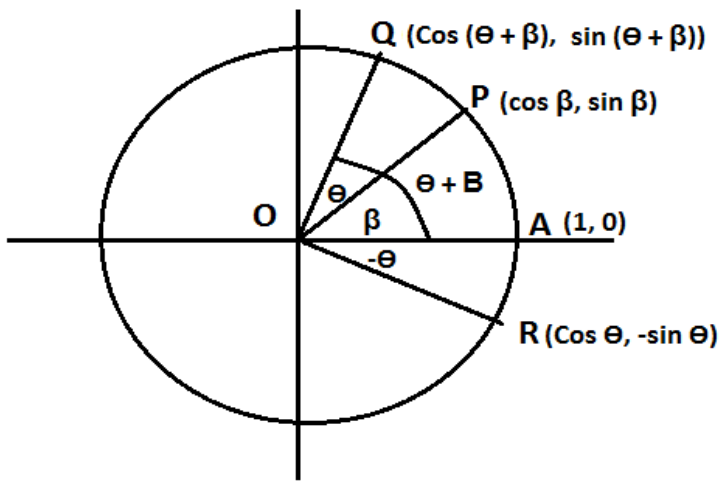

Figure 2.1: $\cos(\theta+\beta)$

Notice that angles AOQ and ROP are both equal to $\theta+\beta$. Consequently

$$\text{chord } AQ = \text{chord } RP.$$

Furthermore, $A = (1,0)$, $P = (\cos\beta, \sin\beta)$, $R = (\cos(-\theta), \sin(-\theta)) = (\cos\theta, -\sin\theta)$ and $Q = (\cos(\theta+\beta), \sin(\theta+\beta))$.

Using the distance formula we have

$$\begin{aligned} AQ &= (\cos(\theta+\beta)-1)^2 + (\sin(\theta+\beta))^2 \\ &= \cos^2(\theta+\beta) - 2\cos(\theta+\beta) + 1 + \sin^2(\theta+\beta) \\ &= \cos^2(\theta+\beta) + \sin^2(\theta+\beta) + 1 - 2\cos(\theta+\beta) \\ &= 2 - 2\cos(\theta+\beta) \end{aligned}$$

and

$$\begin{aligned} RP &= (\cos\theta - \cos\beta)^2 + (\sin\beta + \sin\theta)^2 \\ &= \cos^2\theta + \cos^2\beta - 2\cos\theta\cos\beta + \sin^2\beta + \sin^2\theta + 2\sin\beta\sin\theta \\ &= \cos^2\theta + \sin^2\theta + \cos^2\beta + \sin^2\beta - 2\cos\theta\cos\beta + 2\sin\beta\sin\theta \\ &= 2 - 2\cos\theta\cos\beta + 2\sin\beta\sin\theta. \end{aligned}$$

Since $AQ = RP$, we have

$$2 - 2\cos(\theta+\beta) = 2 - 2\cos\theta\cos\beta + 2\sin\beta\sin\theta$$

giving

$$\cos(\theta+\beta) = \cos\theta\cos\beta - \sin\theta\sin\beta. \tag{2.1}$$

Equation 2.1 is the **addition formula** for the cosine function. If we use $-\theta$ instead of θ we have

$$\cos(\theta-\beta) = \cos\theta\cos\beta + \sin\theta\sin\beta. \tag{2.2}$$

Equation 2.2 is called **subtraction formula** for the cosine function. Combining Equations 2.1 and 2.2 gives

$$\cos(\theta\pm\beta) = \cos\theta\cos\beta \mp \sin\theta\sin\beta. \tag{2.3}$$

Co-function Formulas

To obtain addition and subtraction formulas for the sine function, we first derive **co-function** formulas directly from the subtraction formula for cosine. The co-function formulas for sine and cosine are stated, without proof, in Section 1.5 on standard angles. We give the proofs below.

By Equation 2.2,

$$\begin{aligned} \cos\left(\frac{\pi}{2}-\beta\right) &= \cos\frac{\pi}{2}\cos\beta + \sin\frac{\pi}{2}\sin\beta \\ &= (0)\cos\beta + (1)\sin\beta \\ &= \sin\beta. \end{aligned}$$

The identity

$$\cos\left(\frac{\pi}{2} - \beta\right) = \sin\beta \tag{2.4}$$

is called the **co-function identity for cosine** where β is any real number or angle in radian measure. If β is in degree measure, replace $\frac{\pi}{2}$ with $90°$.

Now, if we put $\beta = \frac{\pi}{2} - \theta$ in Equation 2.4 we get

$$\cos\left(\frac{\pi}{2} - \left(\frac{\pi}{2} - \theta\right)\right) = \sin\left(\frac{\pi}{2} - \theta\right)$$

which gives

$$\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right).$$

The identity

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta \tag{2.5}$$

is called **co-function identity for sine**. θ is any real number or angle in radian measure. If θ is in degree measure, replace $\frac{\pi}{2}$ with $90°$.

The **co-function formula for the tangent** is

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta \tag{2.6}$$

and its proof is obtained by using the tangent identity, co-function formulas 2.4 and 2.5, and the cotangent identity;

$$\tan\left(\frac{\pi}{2} - \theta\right) = \frac{\sin\left(\frac{\pi}{2} - \theta\right)}{\cos\left(\frac{\pi}{2} - \theta\right)} = \frac{\cos\theta}{\sin\theta} = \cot\theta.$$

The following co-function identities can be proved in a similar manner:

$$\cot\left(\frac{\pi}{2} - \theta\right) = \tan\theta, \tag{2.7}$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = \csc\theta \tag{2.8}$$

and

$$\csc\left(\frac{\pi}{2} - \theta\right) = \sec\theta. \tag{2.9}$$

Addition and Subtraction Formulas for Sine

By Equation 2.4,

$$\begin{aligned}
\sin(\theta+\beta) &= \cos\left(\frac{\pi}{2}-(\theta+\beta)\right) \\
&= \cos\left(\left(\frac{\pi}{2}-\theta\right)-\beta\right) \\
&= \cos\left(\frac{\pi}{2}-\theta\right)\cos\beta+\sin\left(\frac{\pi}{2}-\theta\right)\sin\beta \\
&= \sin\theta\cos\beta+\cos\theta\sin\beta.
\end{aligned}$$

Hence the addition formula for the sine function is

$$\sin(\theta+\beta) = \sin\theta\cos\beta+\cos\theta\sin\beta. \tag{2.10}$$

If we put $\beta = -\beta$ we get the subtraction formula for sine;

$$\sin(\theta-\beta) = \sin\theta\cos\beta-\cos\theta\sin\beta. \tag{2.11}$$

Combining Equations 2.10 and 2.11 gives the formula

$$\sin(\theta\pm\beta) = \sin\theta\cos\beta\pm\cos\theta\sin\beta. \tag{2.12}$$

Addition and Subtraction Formulas for Tangent

We now derive the addition formula for the tangent function using the tangent identity and Equations 2.1 and 2.10.

$$\begin{aligned}
\tan(\theta+\beta) &= \frac{\sin(\theta+\beta)}{\cos(\theta+\beta)} \\
&= \frac{\sin\theta\cos\beta+\sin\beta\cos\theta}{\cos\theta\cos\beta-\sin\theta\sin\beta}.
\end{aligned}$$

Dividing both numerator and denominator by $\cos\theta\cos\beta$ gives;

$$\begin{aligned}
\tan(\theta+\beta) &= \frac{\dfrac{\sin\theta\cos\beta}{\cos\theta\cos\beta}+\dfrac{\sin\beta\cos\theta}{\cos\theta\cos\beta}}{\dfrac{\cos\theta\cos\beta}{\cos\theta\cos\beta}-\dfrac{\sin\theta\sin\beta}{\cos\theta\cos\beta}} \\
&= \frac{\dfrac{\sin\theta}{\cos\theta}+\dfrac{\sin\beta}{\cos\beta}}{1-\left(\dfrac{\sin\theta}{\cos\theta}\right)\left(\dfrac{\sin\beta}{\cos\beta}\right)} \\
&= \frac{\tan\theta+\tan\beta}{1-\tan\theta\tan\beta}.
\end{aligned}$$

Hence the equation

$$\tan(\theta + \beta) = \frac{\tan\theta + \tan\beta}{1 - \tan\theta\tan\beta} \tag{2.13}$$

is the addition formula for the tangent function. Similarly, the subtraction formula for the tangent function is

$$\tan(\theta - \beta) = \frac{\tan\theta - \tan\beta}{1 + \tan\theta\tan\beta}. \tag{2.14}$$

Combining Equations 2.13 and 2.14 gives

$$\tan(\theta \pm \beta) = \frac{\tan\theta \pm \tan\beta}{1 \mp \tan\theta\tan\beta}. \tag{2.15}$$

Example 2.9. *Find the exact values of:*

1. $\sin 75^\circ$
2. $\cos 120^\circ$.
3. $\tan 15^\circ$.

Solution

1.

$$\begin{aligned}
\sin 75^\circ &= \sin(45^\circ + 30^\circ) \\
&= \sin 45^\circ \cos 30^\circ + \sin 30^\circ \cos 45^\circ \\
&= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) \\
&= \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{6} + \sqrt{2}}{4}.
\end{aligned}$$

2.

$$\begin{aligned}
\cos 120^\circ &= \cos(90 + 30)^\circ \\
&= \cos 90^\circ \cos 30^\circ - \sin 90^\circ \sin 30^\circ \\
&= 0 \times \frac{\sqrt{3}}{2} - 1 \times \frac{1}{2} \\
&= -\frac{1}{2}.
\end{aligned}$$

3.

$$\begin{aligned}\tan 15^\circ &= \tan(60-45)^\circ \\ &= \frac{\tan 60^\circ - \tan 45^\circ}{1+\tan 60^\circ \tan 45^\circ} \\ &= \frac{\sqrt{3}-1}{1+\sqrt{3}}.\end{aligned}$$

Example 2.10. *Prove the following identities.*

1. $\sin(\theta + \frac{\pi}{2}) = \cos\theta$
2. $\cos(\theta + \pi) = -\cos\theta$
3. $\tan(\theta - \frac{\pi}{4}) = \dfrac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta}$

Solution

We use addition formulae for sine, cosine and tangent.

1. $\sin(\theta + \frac{\pi}{2}) = \sin\theta\cos\frac{\pi}{2} + \cos\theta\sin\frac{\pi}{2} = \sin\theta \times 0 + \cos\theta \times 1 = \cos\theta.$
2. $\cos(\theta + \pi) = \cos\theta\cos\pi - \sin\theta\sin\pi = \cos\theta \times (-1) - \sin\theta \times 0 = -\cos\theta.$
3.

$$\begin{aligned}\tan\left(\theta - \frac{\pi}{4}\right) &= \frac{\tan\theta - \tan\frac{\pi}{4}}{1+\tan\theta\tan\frac{\pi}{4}} = \frac{\tan\theta - 1}{1+\tan\theta} = \frac{\frac{\sin\theta}{\cos\theta} - 1}{1+\frac{\sin\theta}{\cos\theta}} \\ &= \frac{\sin\theta - \cos\theta}{\cos\theta} \div \frac{\cos\theta + \sin\theta}{\cos\theta} = \frac{\sin\theta - \cos\theta}{\cos\theta + \sin\theta}.\end{aligned}$$

2.5 Double Angle and Half Angle Formulas

We now look at formulas for trigonometric functions of twice an angle and half an angle, called **double angle and half angle formulas** respectively. First we consider double angle formulas.

Double Angle Formulas

We use addition formulae to derive double angle formulae.

In the addition formula for sine (in Equation 2.10) put

$\theta = \beta$;

$$\begin{aligned}\sin(\theta+\beta) &= \sin\theta\cos\beta + \sin\beta\cos\theta,\\ \sin(\theta+\theta) &= \sin\theta\cos\theta + \sin\theta\cos\theta,\\ \sin(2\theta) &= 2\sin\theta\cos\theta.\end{aligned}$$

The double angle formula for the sine function is

$$\sin(2\theta) = 2\sin\theta\cos\theta. \tag{2.16}$$

Similarly we set $\theta = \beta$ in Equation 2.1 to obtain the double angle formula for cosine;

$$\begin{aligned}\cos(\theta+\beta) &= \cos\theta\cos\beta - \sin\theta\sin\beta,\\ \cos(\beta+\beta) &= \cos(2\beta) = \cos^2\beta - \sin^2\beta.\end{aligned}$$

Now since $\cos^2\beta + \sin^2\beta = 1$, it follows that

$$\cos(2\beta) = \cos^2\beta - (1-\cos^2\beta) = 2\cos^2\beta - 1$$

and

$$\cos(2\beta) = 1 - \sin^2\beta - \sin^2\beta = 1 - 2\sin^2\beta.$$

The double angle formula for cosine is

$$\cos(2\beta) = \begin{cases}\cos^2\beta - \sin^2\beta,\\ 2\cos^2\beta - 1,\\ 1 - 2\sin^2\beta.\end{cases} \tag{2.17}$$

To derive the double angle formula for tangent, we put $\theta = \beta$ in Equation 2.13 to obtain

$$\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}. \tag{2.18}$$

Example 2.11.

Prove the identity $\tan\theta = \dfrac{\sin 2\theta}{\cos 2\theta + 1}$

Solution

$$\begin{aligned} R.H.S &= \frac{2\sin\theta\cos\theta}{(2\cos^2\theta - 1) + 1} \\ &= \frac{\sin\theta\cos\theta}{\cos\theta\cos\theta} \\ &= \frac{\sin\theta}{\cos\theta} \\ &= \tan\theta \\ &= L.H.S \end{aligned}$$

Example 2.12.

Write $\sin 3\theta$ in terms of $\sin\theta$.

Solution

$$\begin{aligned} \sin 3\theta &= \sin(2\theta + \theta) \\ &= \sin 2\theta\cos\theta + \cos 2\theta\sin\theta \\ &= (2\sin\theta\cos\theta)\cos\theta + (1 - 2\sin^2\theta)\sin\theta \\ &= 2\sin\theta\cos^2\theta + \sin\theta - 2\sin^3\theta \\ &= 2\sin\theta(1 - \sin^2\theta) + \sin\theta - 2\sin^3\theta \\ &= 2\sin\theta - 2\sin^3\theta + \sin\theta - 2\sin^3\theta \\ &= 3\sin\theta - 4\sin^3\theta. \end{aligned}$$

Half Angle Formulas

We use double angle formula for cosine to derive half angle formulas for both sine and cosine functions.

To obtain half angle formula for the cosine function, we set $\beta = \dfrac{\theta}{2}$ in the equation

$$\cos(2\beta) = 2\cos^2\beta - 1$$

and solve for $\cos\left(\dfrac{\theta}{2}\right)$ as follows;

$$\begin{aligned} \cos\theta &= 2\cos^2\left(\frac{\theta}{2}\right) - 1, \\ \cos\theta + 1 &= 2\cos^2\left(\frac{\theta}{2}\right), \\ \frac{\cos\theta + 1}{2} &= \cos^2\left(\frac{\theta}{2}\right) \end{aligned}$$

giving

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{\cos\theta + 1}{2}} \tag{2.19}$$

where the sign is determined by the quadrant of $\frac{\theta}{2}$.

In the equation

$$\cos(2\beta) = 1 - 2\sin^2\beta,$$

setting $\beta = \frac{\theta}{2}$ and solve for $\sin\left(\frac{\theta}{2}\right)$ gives the half angle formula for the sine function;

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 - \cos\theta}{2}} \tag{2.20}$$

where the sign is determined by the quadrant of $\frac{\theta}{2}$.

For the tangent we have,

$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin\left(\frac{\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right)} = \pm\sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}}$$

where, again, the sign is determined by the quadrant of $\frac{\theta}{2}$.

Multiplying $\sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}}$ by $\sqrt{\frac{1 + \cos\theta}{1 + \cos\theta}}$ gives

$$\sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}} \times \sqrt{\frac{1 + \cos\theta}{1 + \cos\theta}} = \sqrt{\frac{1 - \cos^2\theta}{(1 + \cos\theta)^2}} = \sqrt{\left(\frac{\sin\theta}{1 + \cos\theta}\right)^2} = \frac{\sin\theta}{1 + \cos\theta}.$$

The half angle formula for the tangent function is

$$\tan\left(\frac{\theta}{2}\right) = \begin{cases} \sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}}, \\ \frac{\sin\theta}{1 + \cos\theta}. \end{cases} \tag{2.21}$$

Example 2.13. *Show that* $2\sin^2 x = \tan x \sin 2x$.

Solution

$R.H.S = \frac{\sin x}{\cos x} \times 2\sin x \cos x = 2\sin^2 x = L.H.S.$

Example 2.14. *Prove the identity* $\cot x - \tan x = 2\cot 2x$.

Solution

$L.H.S = \cot x - \tan x = \dfrac{\cos x}{\sin x} - \dfrac{\sin x}{\cos x} = \dfrac{\cos^2 x - \sin^2 x}{\sin x \cos x}$.

$$R.H.S = 2\cot 2x = 2 \times \frac{1}{\tan 2x} = 2 \times \frac{1 - \tan^2 x}{2\tan x} = 2 \times \frac{1 - \frac{\sin^2 x}{\cos^2 x}}{2\frac{\sin x}{\cos x}}$$
$$= \frac{\frac{\cos^2 x - \sin^2 x}{\cos^2 x}}{\frac{\sin x}{\cos x}} = \frac{\cos^2 x - \sin^2 x}{\cos^2 x} \times \frac{\cos x}{\sin x} = \frac{\cos^2 x - \sin^2 x}{\sin x \cos x}.$$

Hence $L.H.S = R.H.S$.

Example 2.15. *Prove that* $\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$.

Solution

$\sin x = \sin\left(\frac{x}{2} + \frac{x}{2}\right) = \sin\left(2(\frac{x}{2})\right)$ and the result follows from the double angle formula for the sine function.

2.6 Products, Sums and the Expression $a\sin\theta + b\cos\theta$

In this sub section, we look at formulas which give the products of trigonometric functions as the sums and formulas which give sums as products. Finally we shall look at how to write the expression $a\sin\theta + b\cos\theta$ as a single sine function or cosine function. The formulas here involve only sine and cosine functions.

Product to Sum Formulae

We take a product of two trigonometric functions and rewrite it as a sum. Addition and subtraction formulae are used in deriving the formulae here.

Now by 2.1 and 2.2,

$$\cos(\theta + \beta) = \cos\theta\cos\beta - \sin\theta\sin\beta \text{ and}$$
$$\cos(\theta - \beta) = \cos\theta\cos\beta + \sin\theta\sin\beta.$$

Adding these equations gives

$$\cos(\theta + \beta) + \cos(\theta - \beta) = 2\cos\theta\cos\beta.$$

Rearranging we have

$$\cos\theta\cos\beta = \frac{1}{2}[\cos(\theta+\beta)+\cos(\theta-\beta)]. \tag{2.22}$$

The following product to sum identities can be derived in a similar manner;

$$\sin\theta\sin\beta = \frac{1}{2}[\cos(\theta-\beta)-\cos(\theta+\beta)], \tag{2.23}$$

$$\sin\theta\cos\beta = \frac{1}{2}[\sin(\theta+\beta)+\sin(\theta-\beta)] \tag{2.24}$$

and

$$\cos\theta\sin\beta = \frac{1}{2}[\sin(\theta+\beta)-\sin(\theta-\beta)]. \tag{2.25}$$

Sum to Product Formulae

When we introduced addition formulas, we observed that, for example,

$$\cos(\theta+\beta) \neq \cos\theta + \cos\beta.$$

We gave the formula for $\cos(\theta+\beta)$ but the formula for $\cos\theta+\cos\beta$ was not given. We now give the formulas which writes the sum, such as $\cos\theta+\cos\beta$, as the product of trigonometric functions.

Consider $\cos\theta+\cos\beta$. Let $\theta = A+B$ and $\beta = A-B$. Solving for A and B gives $A = \dfrac{\theta+\beta}{2}$ and $B = \dfrac{\theta-\beta}{2}$.

Now

$$\begin{aligned}
\cos\theta+\cos\beta &= \cos(A+B)+\cos(A-B)\\
&= \cos A\cos B - \sin A\sin B + \cos A\cos B + \sin A\sin B\\
&= 2\cos A\cos B\\
&= 2\cos\left(\frac{\theta+\beta}{2}\right)\cos\left(\frac{\theta-\beta}{2}\right).
\end{aligned}$$

Hence

$$\cos\theta+\cos\beta = 2\cos\left(\frac{\theta+\beta}{2}\right)\cos\left(\frac{\theta-\beta}{2}\right). \tag{2.26}$$

In the similar manner we can derive the following formulas;

$$\cos\theta-\cos\beta = -2\sin\left(\frac{\theta+\beta}{2}\right)\sin\left(\frac{\theta-\beta}{2}\right), \tag{2.27}$$

$$\sin\theta + \sin\beta = 2\sin\left(\frac{\theta+\beta}{2}\right)\cos\left(\frac{\theta-\beta}{2}\right) \tag{2.28}$$

and

$$\sin\theta - \sin\beta = 2\cos\left(\frac{\theta+\beta}{2}\right)\sin\left(\frac{\theta-\beta}{2}\right). \tag{2.29}$$

Example 2.16.

1. Find the exact value of $\sin(52.5°)\cos(7.5°)$.
2. Write the expression $4\cos 3\theta\cos 5\theta$ as a sum of two trigonometric functions.

Solution

1.

$$\begin{aligned}\sin(52.5°)\cos(7.5°) &= \frac{1}{2}[\sin(52.5+7.5)° + \sin(52.5-7.5)°] \\ &= \frac{1}{2}[\sin 60° + \sin 45°] \\ &= \frac{1}{2}\left(\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\right) = \frac{\sqrt{3}+\sqrt{2}}{4}.\end{aligned}$$

2.

$$\begin{aligned}4\cos 3\theta\cos 5\theta &= 4\left(\frac{1}{2}[\cos(3\theta+5\theta) + \cos(3\theta-5\theta)]\right) \\ &= 2\cos(8\theta) + 2\cos(-2\theta) \\ &= 2\cos(8\theta) + 2\cos(2\theta).\end{aligned}$$

Example 2.17.

Use sum to product formulae and other well known identities to prove the following

1. $\dfrac{\cos 4\theta - \cos 2\theta}{\sin 4\theta - \sin 2\theta} = -\tan 3\theta.$
2. $\dfrac{\sin 3\theta + \sin 7\theta}{2\sin^2\theta\sin 5\theta} = \csc^2\theta - 2.$

Solution

1.

$$\begin{aligned} L.H.S &= \frac{-2\sin[(4\theta+2\theta)/2]\sin[(4\theta-2\theta)/2]}{2\cos[(4\theta+2\theta)/2]\sin[(4\theta-2\theta)/2]} \\ &= \frac{-2\sin 3\theta \sin\theta}{2\cos 3\theta \sin\theta} \\ &= -\frac{\sin 3\theta}{\cos 3\theta} \\ &= -\tan 3\theta = R.H.S \end{aligned}$$

2.

$$\begin{aligned} L.H.S &= \frac{2\sin[(3\theta+7\theta)/2]\cos[(3\theta-7\theta)/2]}{2\sin^2\theta\sin 5\theta} \\ &= \frac{2\sin 5\theta\cos 2\theta}{2\sin^2\theta\sin 5\theta} = \frac{\cos 2\theta}{\sin^2\theta} \\ &= \frac{1-2\sin^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta} - \frac{2\sin^2\theta}{\sin^2\theta} \\ &= \csc^2\theta - 2 = R.H.S \end{aligned}$$

The Expression $a\sin\theta + b\cos\theta$

We simplify the expression $a\sin\theta + b\cos\theta$, where a and b are positive numbers, into a single trigonometric function. We hope to get

$$a\sin\theta + b\cos\theta = R\sin(\theta+\alpha) \tag{2.30}$$

since both $\sin\theta$ and $\cos\theta$ have period 2π just like $R\sin(\theta+\alpha)$. We need to find the values of R and α which satisfy equation 2.30.

Now

$$a\sin\theta + b\cos\theta = R\sin(\theta+\alpha) = R\cos\alpha\cos\theta + R\sin\alpha\cos\theta.$$

Equating coefficients of $\sin\theta$ we get $a = R\cos\alpha$....(i).
Equating coefficients of $\cos\theta$ we get $b = R\sin\alpha$....(ii).
Squaring and adding Equations (i) and (ii) we get

$$a^2 + b^2 = R^2\cos^2\alpha + R^2\sin^2\alpha = R^2(\cos^2\alpha + \sin^2\alpha) = R^2$$

giving $R = \sqrt{a^2+b^2}$.

Dividing (ii) by (i) gives

$$\frac{\sin\alpha}{\cos\alpha} = \frac{b}{a} \Rightarrow \tan\alpha = \frac{b}{a} \text{ giving } \alpha = \tan^{-1}\frac{b}{a}.$$

Hence

$$a\sin\theta + b\cos\theta = R\sin(\theta+\alpha) = \sqrt{a^2+b^2}\sin\left(\theta + \tan^{-1}\frac{b}{a}\right). \tag{2.31}$$

In the similar manner, we have

$$a\sin\theta - b\cos\theta = R\sin(\theta-\alpha) = \sqrt{a^2+b^2}\sin\left(\theta - \tan^{-1}\frac{b}{a}\right). \tag{2.32}$$

Expressing $a\sin\theta + b\cos\theta$ as a single cosine function gives the following (please verify using method similar to the one above);

$$a\sin\theta + b\cos\theta = R\cos(\theta-\alpha) \tag{2.33}$$

and

$$a\sin\theta - b\cos\theta = R\cos(\theta+\alpha) \tag{2.34}$$

where $R = \sqrt{a^2+b^2}$ (as before) but $\alpha = \tan^{-1}\dfrac{a}{b}$.

Example 2.18.

Express each of the following in the form $R\sin(\theta+\alpha)$.

1. $\sqrt{3}\cos\theta + \sin\theta$
2. $12\sin(3\theta) - 5\cos(3\theta)$

Solution

1. $a = 1$ and $b = \sqrt{3}$. $R = \sqrt{a^2+b^2} = \sqrt{1^2+\sqrt{3}} = \sqrt{4} = 2$.
   $\alpha = \tan^{-1}\dfrac{b}{a} = \tan^{-1}\dfrac{\sqrt{3}}{1} = \tan^{-1}\sqrt{3} = \dfrac{\pi}{3}$.
   Hence $\sqrt{3}\cos\theta + \sin\theta = 2\sin(\theta + \frac{\pi}{3})$.
2. $12\sin(3\theta) - 5\cos(3\theta) = \sqrt{12^2+5^2}\sin(3\theta+\beta) = 13\sin(3\theta+\beta)$.
   From $\tan\beta = -5/12$, $\beta = -22.62°$.
   Hence $12\sin(3\theta) - 5\cos(3\theta) = 13\sin(3\theta - 22.62°)$.

Example 2.19.

Express in the form $R\cos(\theta+\alpha)$ the expression $\cos\theta+\sin\theta$.

Solution

$a=b=1$ and $R=\sqrt{1^2+1^2}=\sqrt{2}$.

$\alpha=\tan^{-1}1=\dfrac{\pi}{4}$.

Hence $\cos\theta+\sin\theta=\sqrt{2}\cos\left(\theta-\dfrac{\pi}{4}\right)$.

Activity 2.20.

1. Find the exact value of $\sin(\theta/2)$ and $\cos(\theta/2)$ given that $\cos\theta=-1/8$ and that $\pi<\theta<3\pi/2$.
2. Find $\tan\theta/2$ if $\sin\theta=2/5$ and θ is in the second quadrant.
3. Write each product as a sum.
   (a) $\cos(3x)\sin x$
   (b) $\cos(5x)\cos(2x)$
   (c) $\sin 7x\sin 3x$
   (d) $\cos 6x\cos(-4x)$
   (e) $\sin(-4x)\cos 8x$
4. Find the exact value of
   (a) $\sin 22.5^\circ$ and $\tan 22.5^\circ$
   (b) $\sin 75^\circ+\sin 15^\circ$
   (c) $\cos 165^\circ-\cos 75^\circ$
   (d) $2\cos^2\left(\dfrac{\pi}{8}\right)-1$.
   (e) $\cos 105^\circ+\cos 15^\circ$
5. State a value of θ where $\dfrac{\sin 2\theta}{1+\cos 2\theta}$ is undefined.
6. If $\sin\theta=-\dfrac{4}{5}$, where θ is in quadrant III, find $\cos 2\theta, \tan\theta$ and $\sin 2\theta$.
7. Rewrite each sum as a product.
   (a) $\cos\theta+\cos 3\theta$

(b) $\sin 4x - \sin 8x$

(c) $\cos 5x - \cos 3x$

(d) $\sin 7\theta + \sin 3\theta$

8. If $\tan\alpha = -\dfrac{4}{3}$, where α is in quadrant IV, find the exact value of $\tan\dfrac{\alpha}{2}$.

9. Prove each identity below.

   (a) $\dfrac{\sin\theta + \sin 3\theta}{\cos\theta + \cos 3\theta} = \tan 2\theta$

   (b) $\dfrac{\sin 5\theta - \sin 3\theta}{\cos 5\theta + \cos 3\theta} = \tan\theta$

   (c) $\dfrac{\sin 5\theta + \sin 3\theta}{\sin 5\theta - \sin 3\theta} = \tan 4\theta \cot\theta$

   (d) $\dfrac{1 + \sin 2x + \cos 2x}{1 + \sin 2x - \cos 2x} = \cot x$

   (e) $\tan x + \cot x = 2\csc 2x$

10. Write each expression in the form $R\sin(\theta + \alpha)$.

    (a) $2\cos\theta + \sqrt{3}\sin\theta$

    (b) $\cos\theta - \sqrt{3}\sin\theta$

11. Write each expression in the form $R\cos(\theta + \alpha)$.

    (a) $2\cos\theta - 3\sin\theta$

    (b) $\cos\theta - \sin\theta$

Unit Summary

In this unit, you have learnt how to prove trigonometric identities. You have learnt that identities are proved by taking one side and show that it is equal to the other side. You have also learnt various trigonometric formulas including additional, double angle, half angle and sum to product formulas. You have learnt, not only how to derive these formulas, but also how to use them.

Activity 2.21 (Unit Test).

1. Given $\sin x = \frac{1}{5}$ and $\sin y = -\frac{2}{3}$, angle x is in quadrant II and angle y is in quadrant III, find the exact value of $\sin(x+y)$.

2. Simplify $2\cos(3x)\cos(2x) - \cos(x)$

3. Given that $\sin x = -\frac{1}{6}$ and $\cos y = -\frac{1}{3}$ and the co-terminal sides of x and y are in quadrant III. Find the exact value of $\cos(x+y)$.

4. Use the product-to-sum formula to write the product $\sin(x+y)\cos(x-y)$ as a sum.

5. Use the product-to-sum formula to evaluate $\cos\frac{11\pi}{12}\cos\frac{\pi}{12}$.

6. Use sum to product formula to express $\cos 8x + \cos 2x$ as the product.

7. Prove each identity below.

   (a) $\tan^2 x \cot^2 x - \cos^2 x = \sin x$

   (b) $\cos 2x = \cos^4 x - \sin^4 x$

   (c) $\cos 2x = \dfrac{1-\tan^2 x}{1+\tan^2 x}$

   (d) $\sin 2x = \dfrac{2\tan x}{1+\tan^2 x}$

   (e) $\dfrac{\sin 3\theta - \sin\theta}{\cos 3\theta + \cos\theta} = \tan\theta.$

   (f) $\dfrac{\sin 4\theta + \sin 6\theta}{\cos 4\theta - \cos 6\theta} = \cot\theta$

Unit Answers

Suggested Solutions to Activity 2.8

1. (a) 1 (b) 1 (c) $\tan^2 x$

Suggested Solutions to Activity 2.20

1. $\sin(\theta/2) = \frac{9}{16}$ and $\cos(\theta/2) = \frac{7}{16}$

3. (a) $\frac{1}{2}(\sin 4x - \sin 2x)$ (b) $\frac{1}{2}(\cos 7x + \cos 3x)$ (c) $\frac{1}{2}(\cos 4x - \cos 10x)$

4. (a) $\sin 22.5^\circ = \sqrt{\dfrac{2-\sqrt{2}}{2}}$ (b) $\dfrac{\sqrt{6}}{2}$ (d) $\dfrac{\sqrt{3}}{2}$

5. $\theta = \pi$

7. (a) $2\cos 2\theta \cos\theta$ (b) $-2\cos 6x \sin(-2x)$

10. (a) $\sqrt{7}\sin(\theta + 49^\circ)$ (b) $2\sin(\theta - \dfrac{\pi}{3})$

Suggested Solutions to Unit Test

1. $\dfrac{\sqrt{24}-\sqrt{5}}{15}$

2. $\cos(5x)$

3. $\dfrac{\sqrt{35}-\sqrt{8}}{18}$

4. $\dfrac{1}{2}(\sin 2x + \sin 2y)$

5. $\dfrac{-2-\sqrt{3}}{4}$

Unit 3

Graphs, Inverses and Equations

3.1 Introduction

In this unit, you will learn how to sketch trigonometric graphs. The general sine and cosine curves will be covered. You will also learn how to find the inverse of a given trigonometric function. Finally you will learn how to solve trigonometric equations.

3.2 Objectives

By the end of this unit, you should be able to:

- sketch trigonometric graphs
- find the inverse of a trigonometric function
- solve trigonometric equations

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- trigonometric graphs

- cosine curve
- sine curve
- tangent curve
- period
- phase shift
- amplitude
- vertical shift
- inverse trigonometric function
- trigonometric equation

3.3 Trigonometric Graphs

In this section we discuss the graphs of trigonometric functions such as $f(\theta) = \sin\theta$, $f(\theta) = \cos\theta$, $f(\theta) = \tan\theta$ and some related functions.

3.3.1 The Sine and Cosine Curves

To draw the graph of the sine function $f(\theta) = \sin\theta$, for $-2\pi \leq \theta \leq 2\pi$, first we note that $f(-2\pi) = 0$, $f\left(-\frac{3\pi}{2}\right) = 1$, $f(-\pi) = 0$, $f\left(-\frac{\pi}{2}\right) = -1$, $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 1$, $f(\pi) = 0$, $f\left(\frac{3\pi}{2}\right) = -1$ and $f(2\pi) = 0$. Table 3.1 is the table of values for the sine curve. The sine curve is given in Figure 3.1.

θ	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(\theta) = \sin\theta$	0	1	0	-1	0	1	0	-1	0

Table 3.1: Table of values for the sine curve

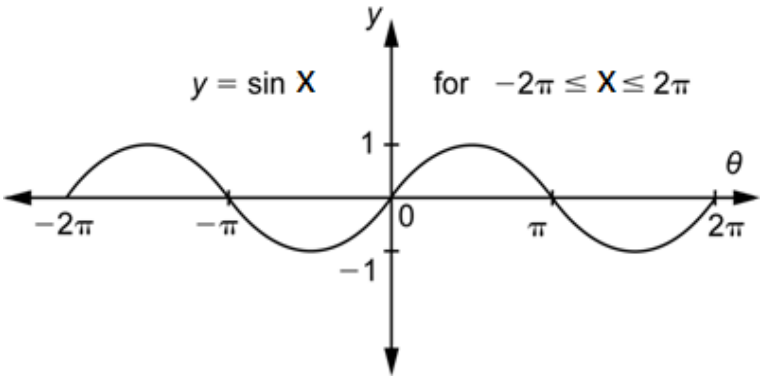

Figure 3.1: The sine curve

θ	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(\theta) = \cos\theta$	1	0	-1	0	1	0	-1	0	1

Table 3.2: Table of values for the cosine curve

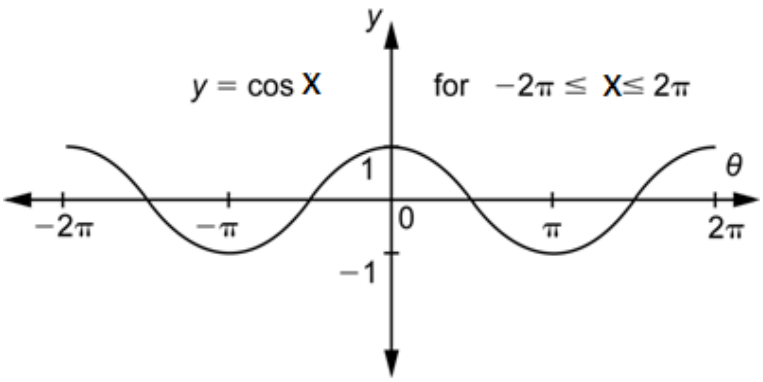

Figure 3.2: The cosine curve

The table of values for the cosine curve is given in Table 3.2. We have the cosine curve in Figure 3.2.

Notice that the cosine curve is identical to the sine curve. In fact the cosine curve is obtained by moving the sine curve by $\frac{\pi}{2}$ to the left. Similarly, the sine curve is obtained by moving the cosine curve by $\frac{\pi}{2}$ to the right. This difference is called the **phase difference** between the functions $\sin\theta$ and $\cos\theta$. As a consequence of this difference, we have the following results;

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta \tag{3.1}$$

and

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta. \tag{3.2}$$

In Figure 3.3 we have drawn the two curves on the same axes.

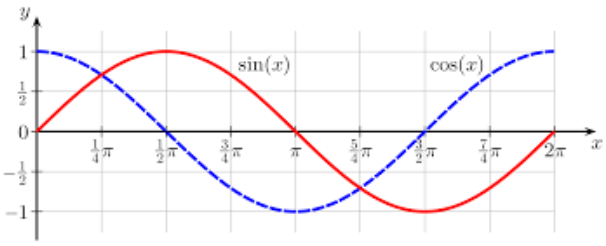

Figure 3.3: The cosine and sine curves

The sine and cosine functions have several distinct characteristics:

- They are periodic functions with a period of 2π. Hence for any angle θ we have

  $$\sin(\theta + 2n\pi) = \sin\theta \tag{3.3}$$

  and

  $$\cos(\theta + 2n\pi) = \cos\theta \tag{3.4}$$

  where n is an integer.

- The domain of each function is $(-\infty, \infty)$ and the range is $[-1, 1]$, i.e., each of $\sin\theta$ and $\cos\theta$ has a minimum value of -1 and a maximum value of 1.

- The graph of $y = \sin x$ is symmetric about the origin, because it is an odd function.
- The graph of $y = \cos x$ is symmetric about the y-axis, because it is an even function.

3.3.2 The Tangent Curve

Recall that $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$. The tangent function is undefined when $\cos\theta = 0$, i.e., when $\theta = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$ and their co-terminals. Therefore, the tangent function has a vertical asymptote (vertical lines which correspond to the zeros of the cosine function) whenever $\theta = \dfrac{\pi}{2}$ or $\dfrac{3\pi}{2}$ or their co-terminals.
Similarly, the tangent and sine functions each have zeros at integer multiples of π since $\tan\theta = 0$ when $\sin\theta = 0$. The tangent function has a period of π and each period in a tangent curve is separated by an asymptote. The tangent curve is given in Figure 3.4.

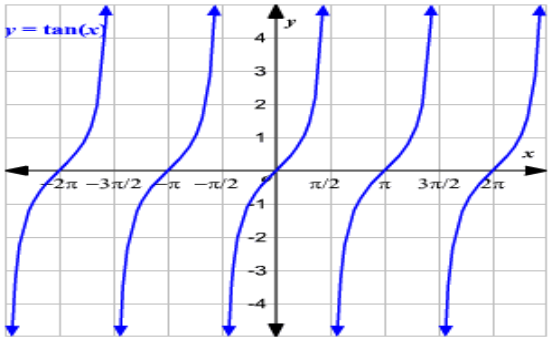

Figure 3.4: The tangent curve

3.3.3 Graphs of the Transformed Sine and Cosine Functions

Consider the functions $f(\theta) = d + a\sin(b\theta + c)$ and $f(\theta) = d + a\cos(b\theta + c)$. We look at the effects of each of the numbers a, b, c and d on the graphs of $f(\theta) = \sin\theta$ and $f(\theta) = \cos\theta$.

Example 3.1.

Sketch the graphs of $f(\theta) = 2\sin\theta$ and $f(\theta) = -2\sin\theta$ for $0 \leq x \leq 2\pi$.

Solution

The table of values for $f(\theta) = -2\sin\theta$ is given in Table 3.3.

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(\theta) = -2\sin\theta$	0	-2	0	2	0

Table 3.3: Table of values for the graph of the function $f(\theta) = -2\sin\theta$

The graph of $f(\theta) = -2\sin\theta$ is given in Figure 3.5. Notice that the maximum value for $f(\theta) = -2\sin\theta$ is 2 and the minimum value is -2. The graph of the function $f(\theta) = 2\sin\theta$ is left as an exercise.

Figure 3.5: The graph of $f(\theta) = -2\sin\theta$

The graph of $y = a\cos\theta$ (or $a\sin\theta$) ranges between a and $-a$ instead of between 1 and -1. If $|a| > 1$, the graph of $y = a\cos\theta$ ($y = a\sin\theta$) is a result of stretching the graph of $y = \cos\theta$ ($y = \sin\theta$) by a factor of $|a|$. If $|a| < 1$, the graph of $y = a\cos\theta$ is a result of compressing the graph of $y = \cos\theta$ ($y = \sin\theta$) by a factor of $1/|a|$.

The preceding discussion implies that the number a in $y = a\cos\theta$ (or $y = a\sin\theta$) changes the vertical length of the graph of $y = \cos\theta$ ($y = \sin\theta$). The number $|a|$, the absolute value of a, is called the **amplitude** of $y = a\cos\theta$ ($y = a\sin\theta$) and represents half the distance between the maximum and minimum values of the function.

The **period** of a function is the distance required for the function to complete one full cycle, after which the graph of the function begins repeating itself. Recall that each of the functions $f(\theta) = \sin\theta$ and $f(\theta) = \cos\theta$ has a period of 2π. The number b ($b \neq 1$) in $y = \cos b\theta$ ($y = \sin b\theta$) changes the period of the function $y = \cos\theta$ ($y = \sin\theta$). The period of $y = \cos b\theta$ ($y = \sin b\theta$) is given by

$$P = 2\pi/b.$$

If $b > 1$, the period of the function $y = \cos b\theta$ ($y = \sin b\theta$) is less than 2π and the graph of $y = \cos b\theta$ ($y = \sin b\theta$) represents a **horizontal shrinking** (compression) of the graph of $y = \cos\theta$ ($y = \sin\theta$). If $0 < b < 1$, the period of the function $y = \cos b\theta$ ($y = \sin b\theta$) is greater than 2π and the graph is a **horizontal stretching** of the graph of $y = \cos\theta$ ($y = \sin\theta$). If $b < 0$ we use identities $\sin(-\theta) = -\sin\theta$ and $\cos(-\theta) = \cos\theta$ to rewrite the function.

Example 3.2.

Find the amplitude and period for the graph of $y = -3\sin 2\theta$.

Solution

$a = -3$ and $b = 2$.

Amplitude $= |-3| = 3$.

Period $= P = \dfrac{2\pi}{2} = \pi$.

Example 3.3.

Find the amplitude and period for the graph of $y = 2\sin\dfrac{\theta}{2}$.

Solution

$a = 2$ and $b = \dfrac{1}{2}$.

Amplitude $= |2| = 2$.

Period $= P = \dfrac{2\pi}{(\frac{1}{2})} = 4\pi$.

In $y = \cos(b\theta + c)$, if we put $b\theta + c = 0$ and solve for θ, we get $\theta = \dfrac{-c}{b}$ called **phase shift** (also called **horizontal shift).** Both b and c affect the phase shift (or displacement) of $y = \cos(b\theta + c)$. The graph of $y = \cos(b\theta + c)$ is a result of shifting the graph of $y = \cos\theta$ to the left if the phase shift is negative or to the right if it is positive.

Example 3.4.

Find the amplitude, period and phase shift of $y = 3\sin\left(2\theta + \frac{\pi}{2}\right)$.

Solution

$a = 3, b = 2$ and $c = \frac{\pi}{2}$.

Amplitude $= 3$.

Period $= \pi$

Phase shift $= -\frac{(\pi/2)}{2} = -\frac{\pi}{4}$.

Example 3.5.

A water wave is created in a wave tank. It has an amplitude of 3 and a period of $\frac{4\pi}{5}$. Write the equation of this wave as a sine function.

Solution

The required equation takes the form $y = a\sin bx$.

Now $a = \pm 3$ and $\frac{2\pi}{b} = \frac{4\pi}{5}$ giving $b = \frac{5}{2}$.

Hence the equation is $y = \pm 3\sin\frac{5}{2}x$.

The number d in $y = d + a\cos(b\theta + c)$ is called the **vertical shift**. It shifts d steps the graph of $y = a\cos(b\theta + c)$. If $d > 0$, the shift is d steps upwards and if $d < 0$, it is d steps downwards. The graph of $y = d + a\cos(b\theta + c)$ oscillates about the horizontal line $y = d$ instead of about the x-axis.

Example 3.6.

Find the amplitude, period, phase shift and vertical shift for $y = -2 + 5\cos(2x + 2\pi)$.

Solution

$a = 5, b = 2, c = 2\pi$ and $d = -2$.

Amplitude $= 5$ and Period $= \frac{2\pi}{b} = \frac{2\pi}{2} = \pi$.

Phase shift $= -\frac{c}{b} = -\frac{2\pi}{2} = -\pi$ and Vertical shift $= -2$.

3.3.4 Sketching Trigonometric Graphs

It is sufficient to sketch one period of a particular graph. The reason is that after one period, the graph just repeats itself.

To find coordinates first we set $x = -\frac{c}{b}$, the phase shift as the x coordinate for the starting point. To determine x-coordinates for the other key points, we divide the period by 4 and add the result to successive x values beginning with the phase shift. Therefore,an interval containing one cycle is

$$x \in \left[\frac{-c}{b}, \frac{-c}{b} + \frac{2\pi}{b}\right].$$

Example 3.7.

Sketch the graph of $y = 2 + 3\sin\left(2x + \frac{\pi}{2}\right)$.

Solution

$a = 3, b = 2, c = \frac{\pi}{2}$ and $d = 2$.

Amplitude $= 3$, Phase shift $= \frac{-\pi/2}{2} = -\frac{\pi}{4}$, Period $= P = \frac{2\pi}{2} = \pi$ and $\frac{P}{4} = \frac{\pi}{4}$.

x-coordinate for the starting point is $-\frac{\pi}{4}$. x-coordinates on key points are $-\frac{\pi}{4}, 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}$

Please verify that corresponding y coordinates are $2, 5, 2, -1$ and 2 respectively. The table of values is given in Table 3.4.

x	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$
y	2	5	2	-1	2

Table 3.4: Table of values for the function $y = 2 + 3\sin\left(2x + \frac{\pi}{2}\right)$

Figure 3.6 is the graph of $y = 2 + 3\sin\left(2x + \frac{\pi}{2}\right)$.

Example 3.8.

Sketch the graph of $y = \frac{1}{2}\sin\left(x - \frac{\pi}{3}\right)$.

Solution

$a = \frac{1}{2}$, $b = 1$ and $c = -\frac{\pi}{3}$.

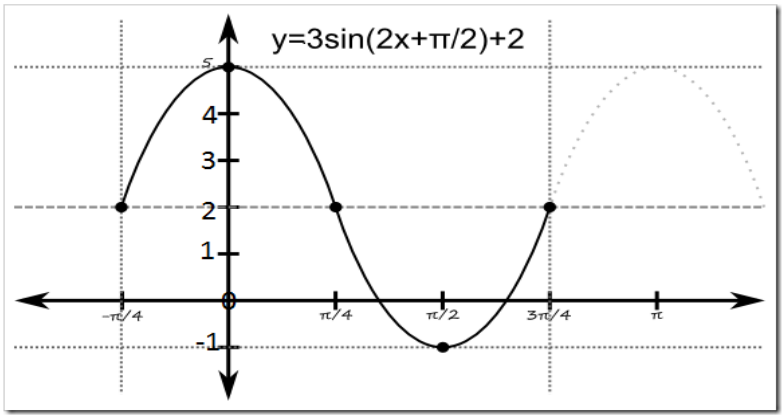

Figure 3.6: The graph of $y = 2 + 3\sin\left(2x + \frac{\pi}{2}\right)$

Amplitude $= \frac{1}{2}$, phase shift $= \frac{\pi}{3}$, Period $= P = 2\pi$ and $\frac{P}{4} = \frac{\pi}{2}$.

Table 3.5 is the table of values for the function $y = \frac{1}{2}\sin\left(x - \frac{\pi}{3}\right)$. The graph is given in Figure 3.7.

x	$\frac{\pi}{3}$	$\frac{5\pi}{6}$	$\frac{8\pi}{6}$	$\frac{11\pi}{6}$	$\frac{14\pi}{6}$
$y = \frac{1}{2}\sin\left(x - \frac{\pi}{3}\right)$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0

Table 3.5: Table of values for the function $y = 2 + 3\sin\left(2x + \frac{\pi}{2}\right)$

Activity 3.9.

1. Find the amplitude, period, phase shift and vertical shift and sketch the graph of each function [*at least one period from phase shift*];

   (a) $y = -3\sin(2x)$

   (b) $y = 4 + 3\sin(2x - \pi)$

   (c) $y = -2\cos(3x)$

   (d) $y = 2 + 3\sin(2x)$

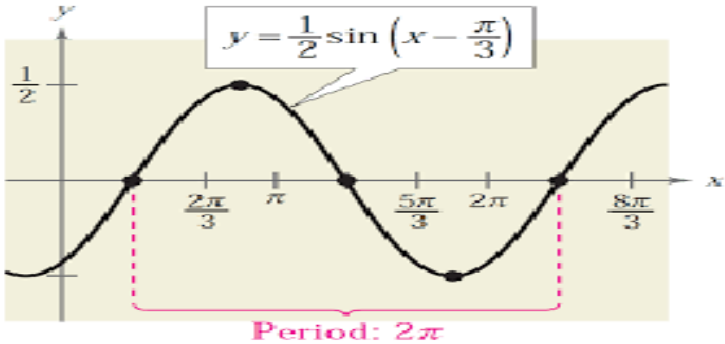

Figure 3.7: The graph of $y = \dfrac{1}{2}\sin\left(x - \dfrac{\pi}{3}\right)$

(e) $y = 3 + 4\cos(3x - 2\pi)$

(f) $y = \sin(3x - \pi/2)$

(g) $y = 3\cos(4x - \pi/3)$

(h) $y = -2 + \cos\left(\dfrac{1}{2}x + \dfrac{\pi}{4}\right)$

(i) $y = -5 + 2\sin\left(2x - \dfrac{\pi}{4}\right)$

(j) $3 - 2\sin\left(\dfrac{1}{2}x + \dfrac{\pi}{2}\right)$

(k) $4 + 4\sin\left(\dfrac{1}{3}x - \dfrac{\pi}{3}\right)$

(l) $4 - 5\cos\left(\dfrac{1}{3}x + \dfrac{\pi}{6}\right)$

(m) $5\cos\left(3x - \dfrac{\pi}{2}\right)$

3.4 Inverse Trigonometric Functions

Recall that for a function to have an inverse, we require that it should be one to one. Also recall that a function is one to one if, whenever $a \neq b$ in the domain, $f(a) \neq f(b)$ in the co-domain. This is equivalent to saying $f(a) = f(b)$ in the co-domain implies $a = b$ in the domain. In other words, a function is one to one if and only if no two elements in the domain are mapped to the same element in the co-domain.

Consider the angles whose sine is $1/2$. They include $30°, 150°, 390°$ and $-210°$. Hence, in general, the sine function is not one-to-one. However, restricting the domain of the sine function to $[-90°, 90°]$ gives a one-to-one and hence invertible function. We use this new function with domain $[-90°, 90°]$ and range $[-1, 1]$ to define the inverse sine function.

Definition 3.10. *The inverse sine function, denoted* $\sin^{-1}$*, is defined by*

$$y = \sin^{-1} x \text{ if and only if } x = \sin y$$

for $-1 \leq x \leq 1$ *and* $-90° \leq y \leq 90°$.

Note that here $\sin^{-1} x \neq \dfrac{1}{\sin x}$. Recall that $\dfrac{1}{\sin x} = \csc x$, a different function all together.

By Definition 3.10, it is essential to look for an angle of $\sin^{-1}$ in the range $[-90°, 90°]$. For example, even though $\sin 150° = 1/2$, $\sin^{-1} 1/2 \neq 150°$ since $150°$ is outside the stated range. In fact $\sin^{-1} 1/2 = 30°$.

The inverse sine function is well defined on scientific calculators. If the calculator is set in degree mode, entering *shift* $\to$ sin $\to$ 0.5 $\to$ =, should result into the calculator displaying $30°$.

Example 3.11.

Evaluate $\sin^{-1} 0.75$.

Solution

Using calculator in degree mode (*shift* $\to$ sin $\to$ 0.75 $\to$ =), $\sin^{-1} 0.75 = 48.59°$.

Now that we have defined inverse sine function, we know that properties which hold for inverse functions are also true for this function. For example, if $-1 \leq x \leq 1$, we have

$$\sin\left(\sin^{-1} x\right) = x.$$

Similarly

$$\sin^{-1}(\sin y) = y$$

if $-90° \leq y \leq 90°$.

Example 3.12.

1. $\sin\left(\sin^{-1} 0.5\right) = 0.5$ since $-1 \leq 0.5 \leq 1$.
2. $\sin^{-1}(\sin 45°) = 45°$ since $-90° \leq 45° \leq 90°$.

Example 3.13.

Evaluate $\sin^{-1}\dfrac{\sqrt{3}}{2}$.

Solution

Let $\sin^{-1}\dfrac{\sqrt{3}}{2} = y$. Then $\sin y = \dfrac{\sqrt{3}}{2}$ and $-90^\circ \leq y \leq 90^\circ$. Since $\sin 60^\circ = \dfrac{\sqrt{3}}{2}$, $y = 60^\circ$. Hence $\sin^{-1}\dfrac{\sqrt{3}}{2} = 60^\circ$.

The inverse sine function is also called the **arcsine function**, and $\arcsin x$ may be used in place of $\sin^{-1} x$. If $y = \arcsin x$, then $\sin y = x$.

Now we define the inverse cosine function. Here we restrict the domain of the cosine function to $[0^\circ, 180^\circ]$ but the range remains $[-1, 1]$.

Definition 3.14. *The inverse cosine function, denoted* $\cos^{-1}$*, is defined by*

$$y = \cos^{-1} x \text{ if and only if } x = \cos y$$

for $-1 \leq x \leq 1$ *and* $0 \leq y \leq 180^\circ$.

The inverse cosine function is also called the **arccosine function** and the notation $\arccos x$ is used in place of $\cos^{-1} x$. If $-1 \leq x \leq 1$, we have

$$\cos\left(\cos^{-1} x\right) = x.$$

Similarly, if $0 \leq y \leq 180^\circ$ we have

$$\cos^{-1}(\cos y) = y.$$

Example 3.15.

1. $\cos\left(cos^{-1} - 0.5\right) = -0.5$ since $-1 \leq -0.5 \leq 1$.
2. $\cos^{-1}[\sin(-30^\circ)] = \cos^{-1}(-0.5) = 120^\circ$ since $-90^\circ \leq -30^\circ \leq 90^\circ$.

We now define the inverse tangent function. To get the inverse of the tangent function, we restrict the domain of the same to an open interval $(-90^\circ, 90^\circ)$. The range remains the entire set of real numbers (recall that the domain of a function is the range of its inverse).

Definition 3.16. *The* **inverse tangent function***, or* **arctan function***, denoted by* $\tan^{-1}$ *or arctan, is defined by*

$$y = \tan^{-1} x = \arctan x \textit{ if and only if } x = tany$$

for any real number x and for $-90° < y < 90°$.

From the properties of an inverse function we have

$$\tan\left(\tan^{-1}\right) = x$$

for every x. Also

$$\tan^{-1}(\tan y) = y$$

if $-90° \leq y \leq 90°$.

Example 3.17.

1. $\tan\left(\tan^{-1} 1\right) = 1$
2. $\tan^{-1}(\tan 45°) = 45°$ since $-90° < 45° < 90°$.

Example 3.18.

Show that $\sin^{-1} x + \cos^{-1} x = 90°$ for $x \in [-1, 1]$.

Solution

Let $y = \cos^{-1} x$. Then $\cos y = x$ and $y \in [0, 180°]$.

By Equation 2.5, $\sin(90° - y) = \cos y$. Since $\cos y = x$, we have

$$\sin(90° - y) = \cos y = x \text{ giving } 90° - y = \sin^{-1} x.$$

Now $\sin^{-1} + \cos^{-1} x = (90° - y) + y = 90°$.

Activity 3.19.

1. Evaluate each expression without using a calculator.
   (a) $\tan^{-1}(-1)$
   (b) $\arctan \sqrt{3}$
   (c) $\cos(\sin^{-1} 0.5)$
   (d) $\sin\left(\dfrac{3\pi}{2} + \arccos \dfrac{\sqrt{2}}{2}\right)$

(e) $\sin\left(2\sin^{-1}\dfrac{\sqrt{2}}{2}\right)$

(f) $\tan\left[\dfrac{\pi}{4}+\tan^{-1}(-1)\right]$

(g) $\cos^{-1}[\sin(\tan^{-1}\sqrt{3})]$

2. Without using a calculator, show that $\tan^{-1}\dfrac{1}{3}+\tan^{-1}\dfrac{1}{2}=\dfrac{\pi}{4}$.

3.5 Trigonometric Equations

A basic trigonometric equation takes one of the forms $\sin x = k, \cos x = k$ or $\tan x = k$ where k is a constant. We will look at the method of solving a basic trigonometric equation. All other seemingly complicated equations can be reduced to one or more forms of the basic trigonometric equation. A basic trigonometric equation has at most two solutions for $0° \leq x \leq 360°$.

To solve an equation such as $\sin x = k, \cos x = k$ or $\tan x = k$,

1. Identify the quadrants in which x lies.
2. Find the first quadrant angle y for which $\sin y = |k|, \cos y = |k|$ or $\tan y = |k|$ (remember in the first quadrant sine, cosine and tangent are all positive).
3. Use the value of angle y to determine the corresponding values of x in the quadrants identified in 1. (Each x is an angle whose reference angle is y).

Example 3.20.

Solve each equation below for $0° \leq x \leq 360°$.

1. $\cos x = \dfrac{1}{2}$.
2. $\sin x = -0.45$.

Solution

1. Cosine is positive in the first and fourth quadrants. These are the quadrants in which the solutions lie.

   The first quadrant angle is $\cos^{-1}\dfrac{1}{2} = 60°$.

   The fourth quadrant angle is $360° - 60° = 300°$.

   Hence $x = 60°$ or $300°$.

2. Sine is negative in the third and fourth quadrants.

   Now $\sin y = 0.45$ gives $y = \sin^{-1} 0.45 = 26.7°$.

   Hence $x = 180° + 26.7° = 206.7°$ or $x = 360° - 26.7° = 333.3°$.

Example 3.21.

Solve each equation below for $0° \leq x \leq 360°$ (or $0 \leq x \leq 2\pi$ in radian measure).

1. $2\cos^2 x + \cos x = 0$.
2. $\cos x + 2\sec x = -3$.
3. $4\sin^2 x \tan x - \tan x = 0$
4. $\tan^2 x + \tan x - 6 = 0$

Solution

1. $2\cos^2 x + \cos x = 0 \Rightarrow \cos x(2\cos x + 1) = 0$ giving $\cos x = 0$ or $2\cos x + 1 = 0$.

   For $\cos x = 0$ we have $x = 90°$ and $270°$.

   $2\cos x + 1 = 0 \Rightarrow \cos x = -1/2$ and x is in the second and third quadrants. The first quadrant angle is $y = \cos^{-1} 1/2 = 60°$. Therefore $x = 180° - 60° = 120°$ and $x = 180° + 60° = 240°$.

   Hence $x = 90°, 120°, 240°$ and $270°$.

2. $\cos x + 2\sec x = -3 \Rightarrow \cos x + \dfrac{2}{\cos x} + 3 = 0 \Rightarrow \cos^2 x + 3\cos x + 2 = 0 \Rightarrow (\cos x + 2)(\cos x + 1) = 0$ giving $\cos x = -2$ and $\cos x = -1$.

   $\cos x = -2$ has no solution since $-1 \leq \cos x \leq 1$ for any angle x.

   For $\cos x = -1$ we have $x = 180°$ (and this is the only solution to the given equation).

3. $4\sin^2 x \tan x - \tan x = 0 \Rightarrow \tan x\left(4\sin^2 x - 1\right) = 0$ giving $\tan x = 0$ and $\sin^2 x = 1/4$. For $\tan x = 0$, $x = 0$ and $x = \pi$.

   $\sin^2 x = 1/4 \Rightarrow \sin x = \pm\sqrt{1/4} = \pm 1/2$ giving $\sin x = 1/2$ and $\sin x = -1/2$.

   For $\sin x = 1/2$, x is in the first and second quadrants. The first quadrant angle is $\dfrac{\pi}{6}$ and the second quadrant angle is $\pi - \dfrac{\pi}{6} = \dfrac{5\pi}{6}$.

   For $\sin x = -1/2$, x is in the third and fourth quadrants. The third quadrant angle is $\dfrac{\pi}{6} + \pi = \dfrac{7\pi}{6}$ and the fourth quadrant angle is $2\pi - \dfrac{\pi}{6} = \dfrac{11\pi}{6}$.

Hence the solutions are $x = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}$ and $\frac{11\pi}{6}$.

In degree measure the solutions are $0°, 30°, 150°, 180°, 210°$ and $330°$.

4. $\tan^2 x + \tan x - 6 = 0 \Rightarrow (\tan x - 2)(\tan x + 3) = 0$ giving $\tan x = 2$ and $\tan x = -3$.

   For $\tan x = 2$, x is in the first and third quadrants. The first quadrant angle is $\tan^{-1} 2 = 63.4°$ and the third quadrant angle is $63.4° + 180° = 243.4°$.

   For $\tan x = -3$, x is in the second and fourth quadrant. The first quadrant (reference) angle is $\tan^{-1} 3 = 71.6°$. The second quadrant angle is $180° - 71.6° = 108.4°$ and the fourth quadrant angle is $360° - 71.6° = 288.4°$.

   Hence the solutions are $x = 63.4°, 108.4°, 243.4°$ and $288.4°$.

Since each trigonometric function is periodic, each basic equation has infinitely many solutions. We may not list all solutions, instead we construct the **general solution** by adding integer multiples of the period to a particular solution.

Both sine and cosine functions have period 2π. Therefore the general solution for $\sin x = k$ or $\cos x = k$ is

$$x = \theta + 2n\pi \text{ (or } x = \theta + 360°n \text{ if } \theta \text{ is given in degrees)}$$

where n is an integer. Tangent function has period π, so the general solution for the equation $\tan x = k$ is

$$x + n\pi \text{ (or } \theta + 180°n \text{ if } \theta \text{ is given in degrees)}$$

where n is an integer.

Example 3.22.

Find the general solutions to each trigonometric equation below.

1. $\tan x = -1.5$
2. $2\sin^2 x - \sin x - 1 = 0$
3. $2\tan^2 x - \sec^2 x - 5 = 0$

Solution

1. Tangent is negative in the second and fourth quadrants.

   The corresponding first quadrant angle is $\tan^{-1} 1.5 = 56.3°$.

   So $x = 180° - 56.3° = 123.7°$ or $x = 360° - 56.3° = 303.7°$.

   Hence the general solutions are $123.7° + 180°n$ and $303.7° + 180°n$.

2. $2\sin^2 x - \sin x - 1 = 0 \Rightarrow 2\sin^2 x - 2\sin x + \sin x - 1 = 0$
   $\Rightarrow (2\sin x + 1)(\sin x - 1) = 0$ giving $\sin x = -1/2$ and $\sin x = 1$.

   For $\sin x = 1$, $x = \dfrac{\pi}{2}$.

   For $\sin x = -1/2$, x is in the third and fourth quadrants. The first quadrant (reference) angle is $\sin^{-1} 1/2 = \dfrac{\pi}{6}$. Therefore $x = \pi + \dfrac{\pi}{6} = \dfrac{7\pi}{6}$ and $x = 2\pi - \dfrac{\pi}{6} = \dfrac{11\pi}{6}$.

   Hence solutions to the given equation are $\dfrac{\pi}{2}$, $\dfrac{7\pi}{6}$ and $\dfrac{11\pi}{6}$. The general solutions are

   $$\frac{\pi}{2} + 2n\pi,\ \frac{7\pi}{6} + 2n\pi \text{ and } \frac{11\pi}{6} + 2n\pi$$

   where n is any integer. In degree measure we have

   $$90° + 360°n,\ 210° + 360°n \text{ and } 330° + 360°n.$$

3. Using the Pythagorean identity $\tan^2 x + 1 = \sec^2 x$, we substitute for $\sec^2 x$ in the given equation.

   $2\tan^2 x - \sec^2 x - 5 = 0 \Rightarrow 2\tan^2 x - (\tan^2 x + 1) - 5 = 0 \Rightarrow 2\tan^2 - 6 = 0$
   $\Rightarrow 2\tan^2 x = 6 \Rightarrow \tan^2 x = 3 \Rightarrow \tan x = \pm\sqrt{3}$ giving $\tan x = \sqrt{3}$ and $\tan x = -\sqrt{3}$.

   For $\tan x = \sqrt{3}$, x is in the first and third quadrants. The first quadrant angle is $\dfrac{\pi}{3}$ and the third quadrant angle is $\dfrac{\pi}{3} + \pi = \dfrac{4\pi}{3}$.

   For $\tan x = -\sqrt{3}$, x is in the second and fourth quadrants. The second quadrant angle is $\pi - \dfrac{\pi}{3} = \dfrac{2\pi}{3}$ and the fourth quadrant angle is $2\pi - \dfrac{\pi}{3} = \dfrac{5\pi}{3}$.

   Hence $x = \dfrac{\pi}{3}, \dfrac{2\pi}{3}, \dfrac{4\pi}{3}$ and $\dfrac{5\pi}{3}$.

   For the general solutions recall that tangent function has period π. Since $\dfrac{5\pi}{3} = \dfrac{2\pi}{3} + \pi$ and $\dfrac{4\pi}{3} = \dfrac{\pi}{3} + \pi$, we need to use only $\dfrac{\pi}{3}$ and $\dfrac{2\pi}{3}$ to construct the general solutions.

   Hence the general solutions are

$$x = \frac{\pi}{3} + \pi n \text{ and } \frac{2\pi}{3} + \pi n$$

where n is any integer.

Activity 3.23.

1. Solve each equation for $0° \leq x \leq 360°$.
   (a) $\tan x = 6$
   (b) $\sin^2 x + \sin x = 6$
   (c) $\sin x + \cos x \cot x = \csc x$
   (d) $\sin x = 5 \cos x$
   (e) $\tan^2 x + \tan x - 12 = 0$
   (f) $15 \sin^2 x - 26 \sin x + 8 = 0$
   (g) $2 \sin^2 x - \cos 2x = 0$
   (h) $5 \sin^2 x + 13 \sin x - 6 = 0$
   (i) $\sin 2x + 2 \cos x = 0$
   (j) $\sin(x/2) = 1/2$
2. Solve each equation over the given interval;
   (a) $\sec x = -2$; $270° \leq x \leq 360°$
   (b) $2 \cos^2 x = \cos x$; $\pi \leq x < 2\pi$
   (c) $(2 \cos x + 1)(\tan x - 1) = 0$; $2\pi < x < 3\pi$
   (d) $\sin^2 x = \sin x$; $0 \leq x \leq \frac{\pi}{2}$
3. Find all the general solutions of the following equations;
   (a) $2 \tan x - \sec^2 x = 0$
   (b) $2 \sin^2 x - \cos x - 1 = 0$
   (c) $\csc^2 x - \csc x = 2$
   (d) $2 \cos^2 x + \cos x = 1$
   (e) $4 \csc^2 \theta - 7 = 4 \cot \theta$
   (f) $\tan \theta = 3 \sin \theta$

(g) $2\cos(2x) + 1 = 0$

(h) $5\sin\dfrac{3\theta}{4} + 4 = 0$

(i) $\sec^2\left(\dfrac{\theta}{3} - 50^\circ\right) = 1.2$

(j) $3\sin\theta + 5\cot\theta = \csc\theta$

Unit Summary

In this unit you have learnt graphs, inverses and equations. You have learnt how to sketch graphs of trigonometric functions. You have also learnt that in general, trigonometric functions are not invertible but we get inverses after restricting the ranges of functions. Finally you have learnt how to solve trigonometric equations. Both particular solutions and general solutions have been discussed.

Activity 3.24 (Unit Test).

1. Sketch, at least one period from phase shift, the graph of each equation below.

   (a) $y = 2\sin\dfrac{1}{2}x$

   (b) $y = 3\sin\left(2x + \dfrac{\pi}{2}\right)$

   (c) $y = 6 - \cos(3x - \pi)$

   (d) $y = -3 + \sin\left(\dfrac{1}{2}x + \dfrac{\pi}{4}\right)$

   (e) $y = -2 - \cos(3x - \pi)$

   (f) $y = 1 + \sin(2x - \pi)$

   (g) $y = 3 + 4\cos\left(x - \dfrac{\pi}{4}\right)$

2. Find the exact value of the expression whenever it is defined.

   (a) $\sin^{-1}\left(\sin\dfrac{\pi}{4}\right)$

   (b) $\sin^{-1}\left(\sin\dfrac{2\pi}{3}\right)$

(c) $\sin\left(\sin^{-1}\frac{1}{2}\right)$

(d) $\cos^{-1}(\cos 3.14)$

(e) $\sin^{-1}\left(\tan\frac{3\pi}{4}\right)$

(f) $\cos^{-1}\left(\cos\frac{5\pi}{6}\right)$

(g) $\tan\left(\sin^{-1}\frac{4}{5}+\tan^{-1}\frac{3}{4}\right)$

(h) $\cos\left(2\sin^{-1}\frac{15}{17}\right)$

(i) $\tan\left(\tan^{-1}\frac{4}{3}+\cos^{-1}\frac{8}{17}\right)$

3. Find the general solution for each equation below.

(a) $2\cos^2 x-\cos x=0$

(b) $3\cos^2 x+8\sin x=7$

(c) $5\sin(2x-5)=-3.045$

(d) $\sin^2=\frac{1}{2}\sin 2x$

(e) $\sin 2x=\sin x$

(f) $\cos 2x=4\cos x-2$

Unit Answers

Suggested Solutions to Activity 3.9

	(a)	(b)	(c)	(d)	(e)	(f)
Amplitude	3	3	2	3	4	1
Phase Shift	0	$\frac{\pi}{2}$	0	0	$\frac{2\pi}{3}$	$\frac{\pi}{6}$
Period	π	π	$\frac{2\pi}{3}$	π	$\frac{2\pi}{3}$	$\frac{2\pi}{3}$
Vertical Shift	0	4	0	2	3	0

Suggested Solutions to Activity 3.19

1. (a) $-\frac{\pi}{4}$ (b) 90° (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{2}}{2}$

Suggested Solutions to Activity 3.23

1. (a) $x = 80.5^\circ$ and $x = 260.5^\circ$ (b) No Solution (c) No solution (h) $x = 23.6^\circ$ and $x = 156.4^\circ$ (i) 90° and 270°

2. (a) 240° (b) $\frac{11\pi}{6}$ and $\frac{3\pi}{2}$

Suggested Solutions to Unit Test

2. (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{1}{2}$ (d) 3.14 (e) $\frac{\pi}{2}$

3. (a) $\frac{\pi}{3} + 2n\pi$, $\frac{\pi}{2} + 2n\pi$, $\frac{3\pi}{2} + 2n\pi$, $\frac{5\pi}{2} + 2n\pi$
   (b) $0.7297 + 2n\pi$, $2.4119 + 2n\pi$
   (c) $111.259 + 2n\pi$ and $163.742 + 2n\pi$
   (d) $2n\pi$, $\frac{\pi}{4} + 2n\pi$, $\pi + 2n\pi$, $\frac{5\pi}{4} + 2n\pi$

Part II

Calculus

Unit 4

Limits and Continuity of Functions

4.1 Introduction

In this unit, we will discuss the limit of a function. We will look at the definition of a limit before discussing limit laws. We will also consider one sided limits, limits at infinity and infinite limits. The difference between continuous and discontinuous functions will be discussed. Finally, we will look at tangent lines to the curve.

4.2 Objectives

By the end of this unit, you should be able to:

- define the limit of a function.
- find the limit of a given function, if it exists.
- identify continuous and discontinuous functions.
- find the slope and equation of the tangent line to the given curve at the given point.

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- limit
- direct substitution property
- one-sided limits
- limits at infinity
- infinite limits
- asymptote
- continuous function
- tangent line to the curve

4.3 Definition and Notation

Definition 4.1. *Suppose $f(x)$ is defined when x is near the number a, i.e., f is defined on some open interval that contains a except possibly at a itself. Then the limit of $f(x)$, as x approaches a, equals L, written*

$$\lim_{x \to a} f(x) = L$$

if the values of $f(x)$ are arbitrarily close to L as x is sufficiently close to a (on either side of a) but not equal to a.

Suppose $f(x) = \frac{1}{2}(3x - 1)$ and consider $a = 4$. We are interested in values of $f(x)$ when x is close to 4 from both sides but not necessarily equal to 4.

Some values are given in Figure 4.1. As x becomes closer to 4 from either side, $f(x)$ becomes closer to 5.5.

Hence $\lim_{x \to 4} \left[\frac{1}{2}(3x - 1)\right] = 5.5$.

Alternative notation for

$$\lim_{x \to a} f(x) = L$$

x ⟶ 4 4 ⟵ x

X	3.9	3.99	3.999	3.9999	4	4.0001	4.001	4.01	4.1
f(x)	5.35	5.485	5.4985	5.49985	5.5	5.50015	5.5015	5.515	5.65

Figure 4.1: Table of Values for $f(x) = \frac{1}{2}(3x - 1)$

is $f(x) \to L$ as

$$x \to a.$$

4.3.1 Limit Laws

Suppose that c, L and M are constants and that $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$. Then

1. $\lim_{x \to a} c = c$

2. $\lim_{x \to a} [cf(x)] = c \lim_{x \to a} f(x) = cL.$
   The limit of a constant times a function is the constant times the limit of the function.

3. $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x) = L + M.$
   The limit of the sum of two functions is the sum of their limits.

4. $\lim_{x \to a} [f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x) = L - M.$
   The limit of the difference of two functions is the difference of their limits.

5. $\lim_{x \to a} [f(x)g(x)] = \left[\lim_{x \to a} f(x)\right] \left[\lim_{x \to a} g(x)\right] = LM.$
   The limit of a product of two functions is the product of their limits.

6. $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} = \frac{L}{M}$, provided $\lim_{x \to a} g(x) = M \neq 0.$

The limit of a quotient of two functions is the quotient of their limits, provided the limit of the denominator is not zero.

7. $\lim\limits_{x\to a}[f(x)]^n = [\lim_{x\to a} f(x)]^n = L^n$

8. $\lim\limits_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim\limits_{x\to a} f(x)} = \sqrt[n]{L}$ where n is a positive integer.

Example 4.2.

Suppose that $\lim\limits_{x\to 4} f(x) = 5$ and $\lim\limits_{x\to 4} g(x) = -2$, find

1. $\lim\limits_{x\to 4} f(x)g(x)$

2. $\lim\limits_{x\to 4}(f(x) + 3g(x))$

3. $\lim\limits_{x\to 4} \dfrac{g(x)}{f(x) - 1}$

Solution

We use limit laws.

1. $\lim\limits_{x\to 4} f(x)g(x) = \left[\lim\limits_{x\to 4} f(x)\right] \times \left[\lim\limits_{x\to 4} g(x)\right] = 5 \times -2 = -10.$

2. $\lim\limits_{x\to 4}(f(x) + 3g(x)) = \lim\limits_{x\to 4} f(x) + 3\lim\limits_{x\to 4} g(x) = 5 + 3(-2) = 5 - 6 = -1.$

3. $\lim\limits_{x\to 4} \dfrac{g(x)}{f(x) - 1} = \dfrac{\lim\limits_{x\to 4} g(x)}{\lim\limits_{x\to 4} f(x) - 1} = \dfrac{-2}{5 - 1} = -\dfrac{2}{4} = -\dfrac{1}{2}.$

Apart from limit laws, we can use Theorem 4.3 in evaluating limit of some functions. Sometimes we can use both limit laws and Theorem 4.3.

Theorem 4.3 (Direct substitution property)**.** *If f is a polynomial or a rational function and a is in the domain of f, then*

$$\lim_{x\to a} f(x) = f(a).$$

Suppose a and n are real numbers. By Theorem 4.3, the following properties of limits are immediate:

1. $\lim\limits_{x\to a} x = a$

2. $\lim_{x \to a} x^n = a^n$

3. $\lim_{x \to a} \sqrt[n]{x} = \sqrt[n]{a}$

Example 4.4.

Evaluate the following limits:

1. $\lim_{x \to 5} (12x^2 - 7x + 14)$

2. $\lim_{x \to -2} \dfrac{2x+7}{x^3-3}$

3. $\lim_{x \to 3} \dfrac{x-1}{x-3}$

Solution

1. We first use limit laws and then substitute 5 for x.

$$\begin{aligned}
\lim_{x \to 5} (12x^2 - 7x + 14) &= \lim_{x \to 5} 12x^2 + \lim_{x \to 5} (-7x) + \lim_{x \to 5} 14 \\
&= 12 \lim_{x \to 5} x^2 - 7 \lim_{x \to 5} x + 14 \\
&= 12(5)^2 - 7(5) + 14 \\
&= 12 \times 25 - 35 + 14 \\
&= 279.
\end{aligned}$$

2. Using direct substitution we have

$$\lim_{x \to -2} \frac{2x+7}{x^3-3} = \frac{2(-2)+7}{(-2)^3-3} = \frac{-4+7}{-8-3} = -\frac{3}{11}.$$

3. Substituting directly we have,

$$\lim_{x \to 3} \frac{x-1}{x-3} = \frac{3-1}{3-3} = \frac{2}{0}.$$

Hence $\lim_{x \to 3} \dfrac{x-1}{x-3}$ does not exist.

Not all limits can be evaluated by direct substitution. For example, directly substituting 2 for x in evaluating $\lim_{x\to 2} \frac{x^2-4}{x-2}$ gives $\frac{0}{0}$ which is undefined. It looks like the limit does not exist.

However, factorising the numerator and then simplifying, we have

$$\begin{aligned}\lim_{x\to 2} \frac{x^2-4}{x-2} &= \lim_{x\to 2} \frac{(x+2)(x-2)}{x-2} \\ &= \lim_{x\to 2}(x+2) \\ &= 2+2 = 4.\end{aligned}$$

In general, for some functions, preliminary algebra is used to write the function in a different but equivalent form before substitution is done. That is, we do some algebraic manipulations to the function until it is in the form in which substitution will make sense (See Proposition 4.5). Depending on the function given, we can do any of the following to either the denominator or the numerator;

1. Factorisation,
2. Expansion,
3. Rationalisation.

Proposition 4.5. *If $f(x) = g(x)$ when $x \neq a$, then*

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x),$$

provided the limits exist.

Example 4.6.

Evaluate each limit if it exists.

1. $\lim_{x\to 0} \left(\frac{2x^2-3x+4}{x} + \frac{5x-4}{x} \right)$
2. $\lim_{h\to 0} \frac{(h+4)^2-16}{h}$
3. $\lim_{y\to 0} \frac{\sqrt{4+y}-2}{y}$
4. $\lim_{r\to 1} \left(\frac{r^2}{r-1} - \frac{1}{r-1} \right)$

Solution

1.

$$\begin{aligned}
\lim_{x\to 0}\left(\frac{2x^2-3x+4}{x}+\frac{5x-4}{x}\right) &= \lim_{x\to 0}\frac{(2x^2-3x+4)+(5x-4)}{x}\\
&= \lim_{x\to 0}\frac{2x^2+2x}{x}\\
&= \lim_{x\to 0} 2x+2\\
&= 2(0)+2\\
&= 2.
\end{aligned}$$

2.

$$\begin{aligned}
\lim_{h\to 0}\frac{(h+4)^2-16}{h} &= \lim_{h\to 0}\frac{(h^2+8h+16)-16}{h}\\
&= \lim_{h\to 0}\frac{h^2+8h}{h}\\
&= \lim_{h\to 0}\frac{h(h+8)}{h}\\
&= \lim_{h\to 0} h+8\\
&= 8.
\end{aligned}$$

3.

$$\begin{aligned}
\lim_{y\to 0}\frac{\sqrt{4+y}-2}{y} &= \lim_{y\to 0}\frac{\sqrt{4+y}-2}{y}\times\frac{\sqrt{4+y}+2}{\sqrt{4+y}+2}\\
&= \lim_{y\to 0}\frac{4+y+2\sqrt{4+y}-2\sqrt{4+y}-4}{y(\sqrt{4+y}+2)}\\
&= \lim_{y\to 0}\frac{y}{y(\sqrt{4+y}+2)}\\
&= \lim_{y\to 0}\frac{1}{\sqrt{4+y}+2}\\
&= \frac{1}{\sqrt{4+0}+2}\\
&= \frac{1}{2+2}=\frac{1}{4}.
\end{aligned}$$

4.

$$\begin{aligned}\lim_{r\to 1}\left(\frac{r^2}{r-1}-\frac{1}{r-1}\right) &= \lim_{r\to 1}\frac{r^2-1}{r-1}\\ &= \lim_{r\to 1}\frac{(r-1(r+1)}{r-1}\\ &= \lim_{r\to 1}(r+1)\\ &= 1+1=2.\end{aligned}$$

4.3.2 One Sided Limits

We write

$$\lim_{x\to a^-} f(x) = L$$

and say **the limit of $f(x)$ as x approaches a from the left is equal to L** if we can make the values of $f(x)$ arbitrarily close to L by taking x sufficiently close to a and $x < a$. If we require that x be greater than a, we get **the right hand limit of $f(x)$ as x approaches a equal to L** and we write

$$\lim_{x\to a^+} f(x) = L.$$

Theorem 4.7 describes existence of the limit of a function based on the left and right hand limits.

Theorem 4.7. $\lim_{x\to a} f(x) = L$ *if and only if* $\lim_{x\to a^+} f(x) = L$ *and* $\lim_{x\to a^-} f(x) = L$.

By Theorem 4.7, a two-sided limit exists if and only if both of the one-sided limits exist and are equal.

Example 4.8.

1. Show that $\lim_{x\to 0}\frac{|x|}{2x}$ does not exist.

2. If $f(x) = \begin{cases}\sqrt{x-2} & \text{if } x > 2\\ 4-2x & \text{if } x < 2\end{cases}$ determine whether $\lim_{x\to 2} f(x)$ exists or not.

Solution

1. We need to show that the left hand limit and the right hand limit are not equal. Recall that $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$.

   Now
$$\lim_{x\to 0^+} \frac{|x|}{2x} = \lim_{x\to 0^+} \frac{x}{2x} = \lim_{x\to 0^+} \frac{1}{2} = \frac{1}{2}$$
   and
$$\lim_{x\to 0^-} \frac{|x|}{2x} = \lim_{x\to 0^-} \frac{-x}{2x} = \lim_{x\to 0^-} \left(-\frac{1}{2}\right) = -\frac{1}{2}.$$
   Since $\lim\limits_{x\to 0^+} \frac{|x|}{2x} \neq \lim\limits_{x\to 0^-} \frac{|x|}{2x}$, $\lim\limits_{x\to 0} \frac{|x|}{2x}$ does not exist.

2. For $x > 2$, we have $f(x) = \sqrt{x-2}$.

   Therefore, $\lim\limits_{x\to 2^+} f(x) = \lim\limits_{x\to 2^+} \sqrt{x-2} = \sqrt{2-2} = 0$.

   For $x < 2$, we have $f(x) = 4 - 2x$.

   Therefore, $\lim\limits_{x\to 2^-} f(x) = \lim\limits_{x\to 2^-} (4-2x) = 4 - 2(2) = 0$.

   Since $\lim\limits_{x\to 2^-+} f(x) = \lim\limits_{x\to 2^-} f(x) = 0$, $\lim\limits_{x\to 2} f(x)$ exists and $\lim\limits_{x\to 2} f(x) = 0$.

Activity 4.9.

1. Evaluate each limit if it exists.

   (a) $\lim\limits_{x\to -3} 200$

   (b) $\lim\limits_{x\to 2} (x^2 + 5x + 1)$

   (c) $\lim\limits_{t\to 0} \dfrac{4 - \sqrt{16+t}}{t}$

   (d) $\lim\limits_{x\to 5} \dfrac{x-5}{x^2-25}$

   (e) $\lim\limits_{t\to 0^+} \dfrac{\sqrt{t^3}}{\sqrt{t}}$

   (f) $\lim\limits_{x\to 0} \dfrac{x}{\sqrt{x+3} - \sqrt{3}}$

   (g) $\lim\limits_{x\to 3} \dfrac{x^2 - 7x + 12}{x-3}$

(h) $\lim_{r \to 1} \dfrac{r^2 - 3r + 2}{r - 1}$

(i) $\lim_{h \to 0} \dfrac{(9+h)^{-1} - 9^{-1}}{h}$

2. If $f(x) = \begin{cases} 1 + x^2 & \text{if } x < 2 \\ x^3 & \text{if } x \geq 2 \end{cases}$, find $\lim_{x \to 2^-} f(x)$.

3. If $f(x) = \begin{cases} 4x^2 - 1 & \text{if } x < 1 \\ 3x + 2 & \text{if } x \geq 1 \end{cases}$, find $\lim_{x \to 1^+} f(x)$.

4. Suppose that $f(t) = \begin{cases} -t & \text{if } t < 1 \\ t^2 & \text{if } t \geq 1 \end{cases}$, find $\lim_{x \to 1} f(t)$.

4.3.3 Limits at Infinity and Infinite Limits

Let f be a function defined on some interval (a, ∞). Then $\lim_{x \to -\infty} f(x) = L_1$ means the values of $f(x)$ can be made arbitrarily close to L_1 by taking x sufficiently large negative. $\lim_{x \to \infty} f(x) = L_2$ means that the values of $f(x)$ can be made arbitrarily close to L_2 by taking x sufficiently large.

Definition 4.10. *If f is a function and L_1 and L_2 are real numbers, then the statements*

$$\lim_{x \to -\infty} f(x) = L_1$$

and

$$\lim_{x \to \infty} f(x) = L_2$$

denote the **limits at infinity.** *The first statement is read 'the limit of $f(x)$ as x approaches $-\infty$ is L_1' and the second is read 'the limit of $f(x)$ as x approaches ∞ is L_2'.*

Recall that ∞ is not a number but represents the idea that something never ends.

Definition 4.11. *Let $k \in \mathbb{R}$. Then the line $y = k$ is called a* **horizontal asymptote** *of the curve $y = f(x)$ if either $\lim_{x \to \infty} f(x) = k$ or $\lim_{x \to -\infty} f(x) = k$.*

Theorem 4.12 is used when calculating limits at infinity.

Theorem 4.12. *Let r be a rational number. If $r > 0$, then*

$$\lim_{x \to \infty} \frac{1}{x^r} = \lim_{x \to -\infty} \frac{1}{x^r} = 0.$$

By Theorem 4.12, the following statements are true.

1. $\lim_{x\to\infty} \frac{1}{x^2} = 0.$

2. $\lim_{x\to-\infty} \frac{1}{x} = 0.$

3. $\lim_{x\to\infty} \frac{2}{x^3} = 2 \lim_{x\to\infty} \frac{1}{x^3} = 2 \times 0 = 0.$

Limits at infinity involving rational functions are evaluated by dividing by the highest power of x that occurs in the denominator in both numerator and denominator. After that, where possible, we can use limit laws and Theorem 4.12.

Example 4.13.

Evaluate the following limits:

1. $\lim_{x\to\infty} \frac{3x^3 + 4x^2 + 6x - 7}{2x^2 - x + 2}$

2. $\lim_{x\to-\infty} \left(4 - \frac{3}{x^2}\right)$

3. $\lim_{x\to\infty} \frac{-2x^3 + 3}{3x^2 + 1}$

4. $\lim_{x\to\infty} \frac{x + 1}{2x^2 + 2x + 1}$

5. $\lim_{x\to-\infty} \frac{\sqrt{2x^2 + 3}}{2x + 3}$

Solution

1. We divide by x^2, the highest power of x occurring in the denominator.

$$\begin{aligned}
\lim_{x\to\infty} \frac{3x^3+4x^2+6x-7}{2x^2-x+2} &= \lim_{x\to\infty} \frac{\dfrac{3x^3}{x^2}+\dfrac{4x^2}{x^2}+\dfrac{6x}{x^2}-\dfrac{7}{x^2}}{\dfrac{2x^2}{x^2}-\dfrac{x}{x^2}+\dfrac{2}{x^2}} \\
&= \lim_{x\to\infty} \frac{3x+4+\dfrac{6}{x}-\dfrac{7}{x^2}}{2-\dfrac{1}{x}+\dfrac{2}{x^2}} \\
&= \frac{\lim\limits_{x\to\infty} 3x + \lim\limits_{x\to\infty} 4 + 6\lim\limits_{x\to\infty}\dfrac{1}{x} - 7\lim\limits_{x\to\infty}\dfrac{1}{x^2}}{2\lim\limits_{x\to\infty}\dfrac{1}{x^2} - \lim\limits_{x\to\infty}\dfrac{1}{x} + \lim\limits_{x\to\infty} 2} \\
&= \frac{\lim\limits_{x\to\infty} 3x + \lim\limits_{x\to\infty} 4 + 6(0) - 7(0)}{2(0)-0+2} \\
&= \frac{\lim\limits_{x\to\infty}(3x+4)}{2} \\
&= \lim_{x\to\infty} \frac{3x+4}{2}.
\end{aligned}$$

As $x \to \infty$, the numerator $(3x+4) \to \infty$. The denominator remains 2.

Hence $\displaystyle\lim_{x\to\infty} \frac{3x^3+4x^2+6x-7}{2x^2-x+2} = \lim_{x\to\infty} \frac{3x+4}{2} = \infty.$

2. $\displaystyle\lim_{x\to-\infty}\left(4-\frac{3}{x^2}\right) = \lim_{x\to-\infty} 4 - 3\lim_{x\to-\infty}\frac{1}{x^2} = 4-3(0) = 4.$

Hence the line $y=4$ is the horizontal asymptote to the curve $y = 4 - \dfrac{3}{x^2}$.

3.

$$\begin{aligned}
\lim_{x\to\infty} \frac{-2x^3+3}{3x^2+1} &= \lim_{x\to\infty} \frac{\dfrac{-2x^3}{x^2}+\dfrac{3}{x^2}}{\dfrac{3x^2}{x^2}+\dfrac{1}{x^2}} \\
&= \lim_{x\to\infty} \frac{-2x+\dfrac{3}{x^2}}{3+\dfrac{1}{x^2}} \\
&= \frac{\lim\limits_{x\to\infty}(-2x)+\lim\limits_{x\to\infty}\dfrac{3}{x^2}}{\lim\limits_{x\to\infty}3+\lim\limits_{x\to\infty}\dfrac{1}{x^2}} \\
&= \frac{\lim\limits_{x\to\infty}(-2x)}{\lim\limits_{x\to\infty}3} \\
&= \lim_{x\to\infty}\frac{(-2x)}{3}.
\end{aligned}$$

As $x \to \infty$, the numerator, $-2x$ is large negative and the denominator remains 3.

Hence $\lim\limits_{x\to\infty} \dfrac{-2x^3+3}{3x^2+1} = -\infty$.

4.

$$\begin{aligned}
\lim_{x\to\infty} \frac{x+1}{2x^2+2x+1} &= \lim_{x\to\infty} \frac{\dfrac{x}{x^2}+\dfrac{1}{x^2}}{\dfrac{2x^2}{x^2}+\dfrac{2x}{x^2}+\dfrac{1}{x^2}} \\
&= \lim_{x\to\infty} \frac{\dfrac{1}{x}+\dfrac{1}{x^2}}{2+\dfrac{2}{x}+\dfrac{1}{x^2}} \\
&= \frac{\lim\limits_{x\to\infty}\dfrac{1}{x}+\lim\limits_{x\to\infty}\dfrac{1}{x^2}}{\lim\limits_{x\to\infty}2+\lim\limits_{x\to\infty}\dfrac{2}{x}+\lim\limits_{x\to\infty}\dfrac{1}{x^2}} \\
&= \frac{0+0}{2+0+0} \\
&= 0.
\end{aligned}$$

5.

$$\begin{aligned}
\lim_{x\to-\infty}\frac{\sqrt{2x^2+3}}{2x+3} &= \lim_{x\to-\infty}\frac{\sqrt{\dfrac{2x^2}{x^2}+\dfrac{3}{x^2}}}{\dfrac{2x}{x}+\dfrac{3}{x}} \\
&= \lim_{x\to-\infty}\frac{\sqrt{2+\dfrac{3}{x^2}}}{2+\dfrac{3}{x}} \\
&= \frac{\sqrt{\lim\limits_{x\to-\infty}2+\lim\limits_{x\to-\infty}\dfrac{3}{x^2}}}{\lim\limits_{x\to-\infty}2+\lim\limits_{x\to-\infty}\dfrac{3}{x}} \\
&= \frac{\sqrt{2+0}}{2+3(0)} \\
&= \frac{\sqrt{2}}{2}.
\end{aligned}$$

When calculating a limit of a rational function, three cases occur. Its either the denominator has a higher degree than the numerator, or the degree of the numerator is greater than that of the denominator. In the third case, the numerator and the denominator have equal degrees. The results obtained in each case are given in Proposition 4.14.

Proposition 4.14. *Let $f(x)$ and $g(x)$ be polynomials.*

1. *If the degree of $f(x)$ is less than the degree of $g(x)$, then $\lim\limits_{x\to\pm\infty}\dfrac{f(x)}{g(x)}=0$.*

2. *If the degree of $f(x)$ is greater than the degree of $g(x)$ then $\lim\limits_{x\to\pm\infty}\dfrac{f(x)}{g(x)}=\infty$ or $\lim\limits_{x\to\pm\infty}\dfrac{f(x)}{g(x)}=-\infty$ depending on the coefficients of leading terms.*

3. *If the degree of $f(x)$ is equal to the degree of $g(x)$ then $\lim\limits_{x\to\pm\infty}\dfrac{f(x)}{g(x)}$ is the quotient of the leading coefficients.*

By Proposition 4.14, $\lim\limits_{x\to\infty}\dfrac{x^6+x^3}{2x^2+3}=\infty$, $\lim\limits_{x\to\infty}\dfrac{2x^2+3}{x^6+4x^3}=\lim\limits_{x\to-\infty}\dfrac{2x^2+3}{x^6+4x^3}=0$ and $\lim\limits_{x\to-\infty}\dfrac{x^6+4x^3}{3x^6+9x^4}=\dfrac{1}{3}$.

If the values of a function $f(x)$ outgrow all positive bounds as x approaches a finite number a, we say that $\lim_{x\to a} f(x) = \infty$. If the values of f become large and negative, exceeding all negative bounds as $x \to a$, we say that $\lim_{x\to a} f(x) = -\infty$. A limit which is ∞ or $-\infty$ is called an **infinite limit**.

Definition 4.15.

The line $x = a$ is a **vertical asymptote** *of the graph of a function $y = f(x)$ if either*

$$\lim_{x\to a^+} f(x) = \pm\infty \text{ or } \lim_{x\to a^-} f(x) = \pm\infty.$$

Example 4.16.

Evaluate the limit and identify the vertical asymptote of the graph of each function.

1. $\lim_{x\to 4^+} \dfrac{5x}{x-4}$.
2. $\lim_{x\to 2^+} \dfrac{7}{2-x}$
3. $\lim_{x\to 2^-} \dfrac{3x}{2x-4}$.

Solution

1. If x is close to 4 but larger than 4, the denominator $x - 4$ is small positive and $5x$ is close to 10. So $\dfrac{5x}{x-4}$ is a large positive number.

   Thus $\lim_{x\to 4^+} \dfrac{5x}{x-4} = \infty$ and the line $x = 4$ is a vertical asymptote.

2. If x is close to 2 but greater than 2, the denominator $2-x$ is a small negative number but the numerator remains 7. So $\dfrac{7}{2-x}$ is a large negative number.

   Hence $\lim_{x\to 2^+} \dfrac{7}{2-x} = -\infty$ and the line $x = 2$ is a vertical asymptote of the curve $y = \dfrac{7}{2-x}$.

3. Similarly, $\lim_{x\to 2^-} \dfrac{3x}{2x-4} = -\infty$ and the equation of the vertical asymptote of the curve $y = \dfrac{3x}{2x-4}$ is $x = 2$.

Activity 4.17.

Evaluate each limit blow if it exists.

1. $\lim_{x\to-\infty}\left(\frac{1}{x}+2\right)$
2. $\lim_{x\to\infty}\frac{x^4+x^2+13}{x^3+8x+9}$
3. $\lim_{x\to6^+}\frac{8}{6-x}$
4. $\lim_{x\to\infty}\frac{2x^2}{(x+2)^3}$
5. $\lim_{x\to6^+}\frac{8}{x-6}$
6. $\lim_{x\to6^-}\frac{8}{6-x}$
7. $\lim_{x\to-\infty}\frac{3x^2+2x+1}{5-2x^2+3x}$
8. $\lim_{x\to\infty}\frac{x^2+x+1}{(3x+2)^2}$
9. $\lim_{x\to3^-}\frac{5x}{6-2x}$
10. $\lim_{x\to-\infty}\frac{3x^3-1}{4x+3}$

4.4 Continuous Functions

Definition 4.18. *A function f is* **continuous** *at a number c if the following conditions are satisfied:*

(i) *f is defined on an open interval containing c, i.e., $f(c)$ exists.*

(ii) $\lim_{x\to c} f(x)$ *exists.*

(iii) $\lim_{x \to c} f(x) = f(c)$.

To show that a function f is continuous at a number c, we need to show that f satisfies all the three conditions in Definition 4.18. To show that a function is not continuous at a number c, we need to show that it does not satisfy any one condition out of the three in Definition 4.18.

If a function f is not continuous at c, we say that it is **discontinuous** at c or that it has a **discontinuity** at c

Example 4.19.

1. Is the function $f(x) = \begin{cases} x^2 + 3x + 5 & \text{if } x < 1 \\ 6x + 3 & \text{if } x \geq 1 \end{cases}$ continuous at $x = 1$?

2. Is the function $f(x) = \begin{cases} 2x - 3 & \text{if } x < 5 \\ 4x - 1 & \text{if } x > 5 \end{cases}$ continuous at $x = 5$?

3. Is the function $f(x) = \begin{cases} x + 1 & \text{if } x < 3 \\ 2x + 1 & \text{if } x \geq 3 \end{cases}$ continuous at $x = 3$?

4. Is the function $f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ x^2 & \text{if } x = 2 \\ 2x - 1 & \text{if } x > 2 \end{cases}$ continuous at $x = 2$?

Solution

1. (i) $f(1) = 6(1) + 3 = 9$.
   Therefore the first condition is satisfied.

   (ii)
   $$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (6x + 3) = 6(1) + 3 = 9$$
   and
   $$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (x^2 + 3x + 5) = 1^2 + 3(1) + 5 = 9.$$
   Since
   $$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^-} f(x) = 9,$$
   $$\lim_{x \to 1} f(x)$$
   exists and is equal to 9.
   Therefore the second condition is satisfied.

(iii) From the above steps, $\lim_{x\to 1} f(x) = f(1) = 9$.

Therefore the third condition is satisfied.

Hence $f(x)$ is continuous at $x = 1$.

2. (i) The function f is defined if x is greater than or less than 5, but not when x is equal to 5. Therefore $f(5)$ does not exist.

Hence the function is not continuous at $x = 5$ since the first condition fails.

3. (i) $f(3) = 2(3) + 1 = 7$.

Hence the first condition is satisfied.

(ii)

$$\lim_{x\to 3^+} f(x) = \lim_{x\to 3^+} (2x + 1) = 2(3) + 1 = 7$$

and

$$\lim_{x\to 3^-} f(x) = \lim_{x\to 3^-} (x + 1) = 3 + 1 = 4.$$

Since

$$\lim_{x\to 3^+} f(x) \neq \lim_{x\to 3^-} f(x),$$

$$\lim_{x\to 3} f(x)$$

dos not exist.

Hence f is discontinuous at $x = 3$ since f fails to satisfy the second condition.

4. (i) $f(2) = 2^2 = 4$.

(ii)

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} (x + 1) = 2 + 1 = 3$$

and

$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} (2x - 1) = 2(2) - 1 = 3.$$

Since

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = 3,$$

$$\lim_{x\to 2} f(x)$$

exists and is equal to 3.

(iii) $\lim_{x\to 2} f(x) = 3$ but $f(2) = 4$. Since these are not equal, the third condition fails.

Hence the function is discontinuous at $x = 3$.

Example 4.20.

For what value of a is the function $f(x) = \begin{cases} ax+5 & \text{if } x < 4 \\ x^2 - x & \text{if } x \geq 4 \end{cases}$ continuous at $x = 4$?

Solution

Since $f(4) = 4^2 - 4 = 12$, the function satisfies the first condition at $x = 4$.

For the second condition to be satisfied, $\lim_{x \to 4^-} f(x) = 4a + 5$ must equal $\lim_{x \to 4^+} f(x) = 12$.

Setting $4a + 5 = 12$ gives $a = \frac{7}{4}$.

Hence the value $a = \frac{7}{4}$ makes the function continuous at $x = 4$.

4.5 Tangent Lines to the Curve

A tangent line to a curve is a straight line that touches the curve at only one point (See Figure 4.2). We will look at how to find the equation of a tangent line to the curve at a given point. First we define the slope of the tangent line.

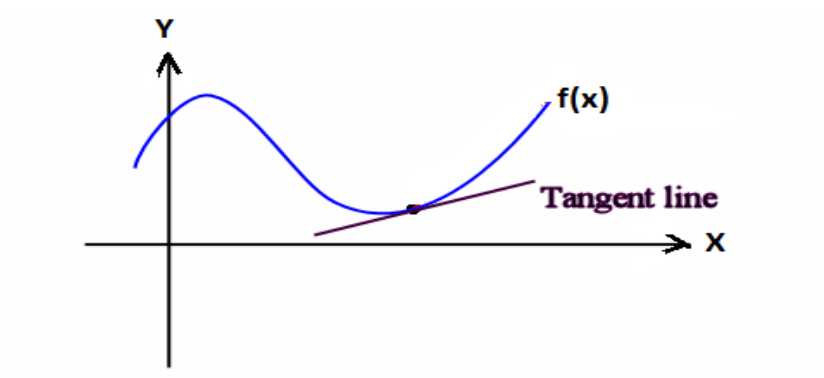

Figure 4.2: The tangent line

Definition 4.21. *The slope of the tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is given by*

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

provided that the limit exists.

Once the slope of the tangent line is found, we can find the equation using **point-slope** form of the equation of straight line discussed in Module 1 of Mat 111.

Example 4.22.

1. Find the slope of the tangent line to the curve $y = x^3 + 1$ at the point $P(1, 2)$.
2. Find equation of the tangent line to the curve $y = x^2 + 4$ at the point $P(-1, 5)$.
3. Find the slope of the tangent line to the curve $y = 2x^2 - 4$ at $x = 2$.

Solution

1. $a = 1$, $f(x) = x^3 + 1$ and $f(a) = f(1) = 2$. Notice that

$$x^3 - 1 = (x - 1)(x^2 + x + 1).$$

   Hence the slope is

$$m = \lim_{x \to 1} \frac{(x^3 + 1) - 2}{x - 1} = \lim_{x \to 1} \frac{(x - 1)(x^2 + x + 1)}{(x - 1)} = \lim_{x \to 1}(x^2 + x + 1) = 1 + 1 + 1 = 3..$$

2. $a = -1$, $f(x) = x^2 + 4$ and $f(-1) = 5$.

   The slope is

$$m = \lim_{x \to -1} \frac{(x^2 + 4) - 5}{x - (-1)} = \lim_{x \to -1} \frac{x^2 - 1}{x + 1} = \lim_{x \to 1-} \frac{(x + 1)(x - 1)}{(x + 1)} = \lim_{x \to -1}(x - 1) = -2.$$

   Using the point $(-1, 5)$ and slope $m = -2$, the equation becomes

$$y - 5 = -2(x - (-1)) \Rightarrow y = -2x - 2 + 5 \Rightarrow y = -2x + 3.$$

3. $a = 2$, $f(x) = 2x^2 - 4$ and $f(a) = f(2) = 2(2)^2 - 4 = 8 - 4 = 4$.
   The slope is

$$\begin{aligned} m &= \lim_{x\to 2} \frac{(2x^2-4)-4}{x-2} = \lim_{x\to 2} \frac{2x^2-8}{x-2} \\ &= \lim_{x\to 2} \frac{2(x^2-4)}{x-2} = \lim_{x\to 2} \frac{2(x-2)(x+2)}{x-2} \\ &= \lim_{x\to 2} 2(x+2) = 2(2+2) = 8. \end{aligned}$$

Activity 4.23.

1. Is the function $f(x) = \begin{cases} \dfrac{x^2-x-2}{x-2} & \text{if } x \neq 2 \\ 7 & \text{if } x = 2 \end{cases}$ continuous at $x = 2$?

2. Is the function $f(x) = \begin{cases} x+1 & \text{if } x < 2 \\ 2x-1 & \text{if } x \geq 2 \end{cases}$ continuous at $x = 2$?

3. Suppose $f(x) = \begin{cases} bx & \text{if } x \leq 3 \\ 5 & \text{if } x > 3 \end{cases}$. Find the value of b such that the function $f(x)$ is continuous at $x = 3$.

4. Find the value of a such that the function $f(x) = \begin{cases} a+x & \text{if } x < 2 \\ 1+x^2 & \text{if } x \geq 2 \end{cases}$ is continuous at $x = 2$.

5. Find all values of a such that the function $f(x) = \begin{cases} x^2+2x & \text{if } x < a \\ -1 & \text{if } x \geq a \end{cases}$ is continuous everywhere.

6. Find the slope of the tangent line to the curve $y = \sqrt{x}$ at $x = 9$.

7. Find the equation of the tangent line to the given curve at the given point.
   (a) $y = x^3 - x^2 + 2x + 1$ at $(2, 9)$ [**Hint:** *Factorise* $x^3 - x^2 + 2x + 1$].
   (b) $y = \dfrac{x+2}{3-x}$ at $(2, 4)$.
   (c) $y = x^2$ at $x = 3$.
   (d) $y = x^2 + 2x + 3$ at $x = 1$.

Unit Summary

In this unit, you have learnt the limit of a function. Both definition and existence of a limit were discussed. You also learnt continuous functions and tangent lines to a given curve at a given point.

Activity 4.24 (Unit Test).

1. If $f(x) = \begin{cases} x^2 + 8x + 15 & \text{if } x \leq 2 \\ 4x + 7 & \text{if } x > 2 \end{cases}$, find $f(2)$.

2. If $f(x) = \begin{cases} x^3 - 2 & \text{if } x \geq 2 \\ 1 + x^2 + 7 & \text{if } x < 2 \end{cases}$, find $\lim\limits_{x \to 2^+} f(x)$.

3. Suppose that $f(t) = \begin{cases} -t & \text{if } t < 1 \\ t^2 & \text{if } t \geq 1 \end{cases}$. Find $\lim\limits_{t \to 1} f(t)$.

4. Evaluate each limit.

   (a) $\lim\limits_{x \to 4} \dfrac{x^2 + x - 20}{x - 4}$

   (b) $\lim\limits_{x \to 3^-} \dfrac{|4x - 12|}{x - 3}$

   (c) $\lim\limits_{t \to \infty} \dfrac{3}{1 + t^2}$

   (d) $\lim\limits_{x \to 3^+} \dfrac{5x}{2x - 6}$

   (e) $\lim\limits_{x \to 1/4} \dfrac{4x - 1}{1 - 16x^2}$

   (f) $\lim\limits_{x \to 9} \dfrac{x^2 - 81}{3 - \sqrt{x}}$

   (g) $\lim\limits_{x \to \infty} \dfrac{-6x^4 + 2x^3 - 7}{12x^4 + x^5 + 3x^6}$

   (h) $\lim\limits_{x \to \infty} \dfrac{\sqrt{2 + x} - \sqrt{2}}{x}$

5. Find the value of a such that the function $f(x) = \begin{cases} x & \text{if } x \leq 3 \\ a + \dfrac{x}{2} & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$.

6. Suppose the function $f(x) = \begin{cases} 2x^2+3 & \text{if } x \le 3 \\ 3x+b & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$. Find the value of b.

7. Find the equation of the tangent line to the curve $y = x^3 - 1$ at $x = -1$.

8. Find the slope of the tangent line to the curve $y = \dfrac{x+1}{x-3}$ at $x = 2$.

Unit Answers

Suggested Solutions to Activity 4.9

(1) (a) 200 (b) 15 (c) $-\dfrac{1}{8}$ (d) $\dfrac{1}{10}$ (e) 0 (g) -1 (h) -1.

(2) 5 (3) 5 (4) The limit does not exist.

Suggested Solutions to Activity 4.17

(1) 2 (2) ∞ (3) $-\infty$ (4) 0 (5) ∞

(7) $-\dfrac{3}{2}$ (8) $\dfrac{1}{9}$ (9) ∞ (10) $-\infty$

Suggested Solutions to Activity 4.23

1. The function is not continuous at $x = 2$.
2. The function is continuous at $x = 2$.
3. $b = \dfrac{5}{3}$ 4. $a = 3$ 5. $a = -1$ 6. $m = \dfrac{1}{6}$
7. (a) $y = 10x - 11$
   (b) $y = 5x - 6$
   (c) $y = 6x - 9$
   (d) $y = 4x + 2$

Suggested Solutions to Unit Test

1. $f(2) = 35$ 2. $\lim\limits_{x \to 2^+} f(x) = 6$ 3. $\lim\limits_{x \to 1} f(t)$ does not exist.

4. (a) 9 (b) 4 (c) 0 (d) ∞ (e) $-\frac{1}{2}$ (h) 0

5. $a = \frac{3}{2}$ 6. $b = 12$ 7. $y = 3x + 1$ 8. $y = 5x - 4$

Unit 5

Definition and Techniques of Differentiation

5.1 Introduction

In this unit, we will introduce the derivative of an algebraic function. We will define the differentiation and use the definition to find the derivative of a function. We will also look at other techniques of finding the derivative of a function besides the definition. These include the product rule, the power rule, and the chain rule. Implicit differentiation and higher order derivatives will also be discussed.

5.2 Objectives

By the end of this unit, you should be able to:

- define the derivative of a differentiable function.
- find the derivative of a function from first principles.
- find the derivative using other techniques such as power rule, product rule, quotient rule and chain rule.
- perform implicit differentiation.
- find second order and third order derivatives.

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- derivatives
- differentiation from first principles
- power rule
- product rule
- quotient rule
- chain rule
- constant rule
- constant multiple rule
- sum and difference rule
- implicit differentiation
- higher order derivatives

5.3 Differentiation from First Principles

Differentiation is the process of finding the **derivative** of a function. We define the derivative of a function in terms of limits.

Definition 5.1. *The derivative of a function f at a number a is given by*

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

Finding the derivative of a function using Definition 5.1 is called **differentiation from first principles**.

In Definition 4.21, we defined the slope of the tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ as

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

Now if we put $x = a + h$, then we have $h = x - a$ and h approaches 0 if and only if x approaches a. So the definitions 4.21 and 5.1 are equivalent.

Therefore the slope of the tangent line to $y = f(x)$ at $(a, f(a))$ is the derivative of f at a.

Example 5.2.

Find the derivative of each of the following functions;

1. $f(x) = 2x^2$ at the number $a = 2$.
2. $f(x) = x^2 - 8x + 9$ at the number $a = 5$

Solution

1. $f(x) = 2x^2$ and $a = 2$.

   $f(a + h) = f(2 + h) = 2(2 + h)^2 = 2(4 + 4h + h^2) = 8 + 8h + 2h^2$.

   $f(a) = f(2) = 2(2)^2 = 2 \times 4 = 8$.

   Hence

$$\begin{aligned} f'(a) &= \lim_{h \to 0} \frac{f(a + h) - f(a)}{h} \\ f'(2) &= \lim_{h \to 0} \frac{f(2 + h) - f(2)}{h} \\ &= \lim_{h \to 0} \frac{(8 + 8h + 2h^2) - 8}{h} \\ &= \lim_{h \to 0} \frac{8h + 2h^2}{h} \\ &= \lim_{h \to 0} 8 + 2h = 8. \end{aligned}$$

2.

$$\begin{aligned}
f'(5) &= \lim_{h\to 0} \frac{f(5+h)-f(5)}{h} \\
&= \lim_{h\to 0} \frac{[(5+h)^2 - 8(5+h)+9] - [5^2 - 8(5)+9]}{h} \\
&= \lim_{h\to 0} \frac{25+10h+h^2-40-8h+9-25+40-9}{h} \\
&= \lim_{h\to 0} \frac{10h+h^2-8h}{h} \\
&= \lim_{h\to 0} 10+h-8 = 2.
\end{aligned}$$

5.3.1 The Derivative Function

In Definition 5.1, if the number a is not specified, then the derivative of a function $f(x)$ becomes another function, say, $g(x)$.

Definition 5.3. *The derivative function of $f(x)$, denoted $f'(x)$, is given by*

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

provided the limit exits.

Example 5.4.

Find the derivative of each of the following functions;

1. $f(x) = 2x^2$.
2. $f(x) = x^2 - 8x + 9$ at the number $a = 5$

Solution

1. $f(x+h) = 2(x+h)^2 = 2(x^2+2xh+h^2) = 2x^2+4xh+2h^2$.
   Hence

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0} \frac{(2x^2+4xh+2h^2)-2x^2}{h} \\
&= \lim_{h\to 0} \frac{4xh+2h^2}{h} \\
&= \lim_{h\to 0} (4x+2h) \\
&= 4x.
\end{aligned}$$

2. We will first find $f'(x)$ and then substitute 5 for x.

   Now $f(x+h) = (x+h)^2 - 8(x+h) + 9 = x^2 + 2xh + h^2 - 8x - 8h + 9$.

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{(x^2 + 2xh + h^2 - 8x - 8h + 9) - (x^2 - 8x + 9)}{h} \\
&= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - 8x - 8h + 9 - x^2 + 8x - 9}{h} \\
&= \lim_{h\to 0} \frac{2xh + h^2 - 8h}{h} \\
&= \lim_{h\to 0} (2x + h - 8) \\
&= 2x - 8.
\end{aligned}$$

   Hence $f'(5) = 2(5) - 8 = 10 - 8 = 2$.

Apart from $f'(x)$, other alternative notations for the derivative of the function $y = f(x)$ with respect to x are; y', $\frac{dy}{dx}$, $\frac{df}{dx}$, $\frac{d}{dx}[f(x)]$, $Df(x)$ and $D_x f(x)$.

Activity 5.5.

1. Differentiate from first principles each function:
   (a) $f(x) = x^2 + x - 2$
   (b) $f(x) = 4x^3 - 4x + 4$
   (c) $g(x) = 12x^2 - 4$
   (d) $h(x) = 9x^2 + 12x + 16$
   (e) $f(x) = x^2 - 19x - 7$
   (f) $g(x) = 13x^3 + 7x^2 - 5x + 10$
2. Find the derivative (from first principles) of $7x^3 - 3$ at the number $x = 2$.
3. Find the derivative of $8x^3 - x^2 + 4x - 6$ at the number $x = 1$.
4. If $f(x) = 3x^3 + 17x$, find $f'(4)$ from first principles.

5.4 Formulae and Techniques of Differentiation

5.4.1 The Derivative of a Constant Function

Recall that a constant function takes the form $f(x) = a$ where a is a real number.

Theorem 5.6. *Let c be any constant. Then the derivative of c with respect to x is zero, i.e.,* $\frac{d}{dx}(c) = 0$.

Example 5.7. By Theorem 5.6, $\frac{d}{dx}(20) = 0$ *and* $\frac{d}{dx}(-0.35) = 0$.

5.4.2 The Power Rule

We look at the derivative of a polynomial function x^n where n is a real number.

Theorem 5.8. *If n is any real number, then* $\frac{d}{dx}(x^n) = nx^{n-1}$.

Example 5.9.

By Theorem 5.8, we have

1. $\frac{d}{dx}(x^7) = 7x^{7-1} = 7x^6$.
2. $\frac{d}{dx}(x^{2/3}) = \frac{2}{3}x^{2/3-1} = \frac{2}{3}x^{-1/3}$.
3. $\frac{d}{dx}(x) = 1$.

5.4.3 The Constant Multiple Rule

This rule given the derivative of a function which is the product of two functions of which one of them is a constant.

Theorem 5.10. *If c is a constant and f is a differentiable function, then* $\frac{d}{dx}[cf(x)] = c\frac{d}{dx}[f(x)]$.

In Theorem 5.10, by differentiable function we mean a function whose derivative exists.

Example 5.11.

By Theorem 5.10, we have the following:

1. If $f(x) = 2x^3$, then $\dfrac{d}{dx}(f(x)) = 2\dfrac{d}{dx}(x^3) = 2(3x^{3-1}) = 6x^2$.
2. If $f(x) = 17x^{-2}$, then $f'(x) = -34x^{-3}$.

5.4.4 The Sum and Difference Rule

Theorem 5.12. *Let f and g be differentiable functions. Then* $\dfrac{d}{dx}[f(x) \pm g(x)] = \dfrac{d}{dx}[f(x)] \pm \dfrac{d}{dx}[g(x)]$.

In words, Theorem 5.12, means, *the derivative of the sum of two functions, is the sum of the derivatives of the two functions* and *the derivative of the difference between two functions is the difference between the derivatives of the two functions.*

Example 5.13.

Find the derivative of each function:

1. $f(x) = 8x^3 + 4x^2 + 6x - 9$.
2. $f(x) = 17x^4 + 9x^3 + x^2 - 7x$.
3. $f(x) = 7x^{-7} + 6x^6 - 5\sqrt[3]{x} - 2$

Solution

1. $f'(x) = 24x^2 + 8x + 6$.
2. $f'(x) = 68x^3 + 27x^2 + 2x - 7$.
3. Note that $f(x) = 7x^{-7} + 6x^6 - 5x^{1/3} - 2$.
   Therefore

$$\begin{aligned} f'(x) &= -49x^{-8} + 36x^5 - 5\left(\frac{1}{3}x^{1/3-1}\right) \\ &= -49x^{-8} + 36x^5 - \frac{5}{3}x^{-2/3} \\ &= -49x^{-8} + 36x^5 - \frac{5}{3}\left(\frac{1}{\sqrt[3]{x^2}}\right). \end{aligned}$$

5.4.5 The Product Rule

Recall that we can multiply two functions to find their product. The product rule gives the derivative of a function which is the product of two functions.

Theorem 5.14. *Let f and g be differentiable functions. Then*

$$\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)].$$

In words, the product rule is, *The derivative of a product of two functions is the first function times the derivative of the second function plus the second function times the derivative of the first function.*

Example 5.15.

1. Differentiate $f(x) = x^2(2x^3 - 1)$.
2. If $f(x) = 8x(3x^2 + x)$, find $f'(1)$.
3. Find y' given that $y = (2x + 7)(x^2 - 5x + 2)$.

Solution

1. By the product rule,

$$\begin{aligned} f'(x) &= x^2\frac{d}{dx}[2x^3 - 1] + (2x^3 - 1)\frac{d}{dx}[x^2] \\ &= x^2(6x^2) + (2x^3 - 1)(2x) \\ &= 6x^4 + 4x^4 - 2x \\ &= 10x^4 - 2x. \end{aligned}$$

2. By the product rule,

$$\begin{aligned} f'(x) &= 8x\frac{d}{dx}[3x^2 + x] + (3x^2 + x)\frac{d}{dx}[8x] \\ &= 8x(6x + 1) + (3x^2 + x)(8) \\ &= 48x^2 + 8x + 24x^2 + 8x \\ &= 72x^2 + 16x. \end{aligned}$$

Hence $f'(1) = 72(1)^2 + 16(1) = 88$.

3.

$$\begin{aligned} y' &= (2x+7)\frac{d}{dx}[x^2-5x+2] + (x^2-5x+2)\frac{d}{dx}[2x+7] \\ &= (2x+7)(2x-5) + (x^2-5x+2)2 \\ &= (2x+7)(2x-5) + 2(x^2-5x+2). \end{aligned}$$

5.4.6 The Quotient Rule

The quotient rule gives the derivative of the function which is the result of dividing two functions.

Theorem 5.16. *If f and g are differentiable functions, then*

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}[f(x)] - f(x)\frac{d}{dx}[g(x)]}{[g(x)]^2}.$$

In words the quotient rule is, *the derivative of the quotient is the denominator times the derivative of the numerator minus the numerator times the derivative of the denominator, all divided by the square of the denominator.*

Example 5.17.

1. Find $f'(x)$ for $f(x) = \dfrac{x}{x^2-1}$.
2. Differentiate $y = \dfrac{x^2}{2x+1}$ with respect to x.
3. Find y' given that $y = \dfrac{x^4(2x+7)}{(x^2+2)}$.

Solution

1.

$$\begin{aligned} f'(x) &= \frac{(x^2-1)\frac{d}{dx}[x] - x\frac{d}{dx}[x^2-1]}{(x^2-1)^2} \\ &= \frac{(x^2-1) - x(2x)}{(x^2-1)^2} \\ &= \frac{x^2-1-2x^2}{(x^2-1)^2} \\ &= \frac{-(x^2+1)}{(x^2-1)^2}. \end{aligned}$$

2.

$$\begin{aligned} y' &= \frac{(2x+1)\frac{d}{dx}[x^2] - x^2\frac{d}{dx}[2x+1]}{(2x+1)^2} \\ &= \frac{(2x+1)2x - x^2(2)}{(2x+1)^2} \\ &= \frac{4x^2+2x-2x^2}{(2x+1)^2} \\ &= \frac{2x^2+2x}{(2x+1)^2}. \end{aligned}$$

3. $y' = \dfrac{(x^2+2)\frac{d}{dx}[x^4(2x+7)] - x^4(2x+7)\frac{d}{dx}[x^2+2]}{(x^2+2)^2}.$

   By the product rule,

$$\begin{aligned} \frac{d}{dx}[x^4(2x+7)] &= x^4\frac{d}{dx}[2x+7] + (2x+7)\frac{d}{dx}[x^4] \\ &= x^4(2) + (2x+7)(4x^3) \\ &= 2x^4 + 4x^3(2x+7). \end{aligned}$$

   Hence

$$\begin{aligned} y' &= \frac{(x^2+2)(2x^4+4x^3(2x+7)) - x^4(2x+7)(2x)}{(x^2+2)^2} \\ &= \frac{(x^2+2)[2x^4+4x^3(2x+7)] - 2x^5(2x+7)}{(x^2+2)^2}. \end{aligned}$$

5.4.7 The Chain Rule

Recall that the **composition** of two functions $f(x)$ and $g(x)$ is

$$f \circ g = f(g(x)).$$

The composition is also called the **composite function**. The Chain rule is used to find the derivative of a composite function.

Theorem 5.18. *If $y = f(u)$, $u = g(x)$, and the derivatives $\frac{dy}{du}$ and $\frac{du}{dx}$ both exist, then the derivative of the composite function defined by $y = f(g(x))$ is given by*

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} = f'(u)g'(x) \\ &= f'(g(x))g'(x). \end{aligned}$$

Example 5.19.

If $y = u^4$ and $u = x^3 + x$, find $\dfrac{dy}{dx}$.

Solution

By the chain rule,

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= 4u^3 \times (3x^2 + 1) \\ &= 4(x^3 + x)^3(3x^2 + 1).\end{aligned}$$

Let $g(x)$ be any function. The function $[g(x)]^n$ where n is a real number is an example of a composite function. The derivative of this function is found by combining both chain rule and power rule.

Theorem 5.20. *If g is a differentiable function and n is any real number, then*

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \times \frac{d}{dx}[g(x)].$$

Example 5.21.

1. Find $f'(x)$ if $f(x) = (x^5 - 4x + 8)^7$.
2. Find $\dfrac{dy}{dx}$ for $y = (x^2 - 3x + 1)^4$.
3. Given that $y = (2x + 8)^4(x^2 - 6x)^6$, find y'.

Solution

1.

$$\begin{aligned}f'(x) &= 7(x^5 - 4x + 8)^6 \frac{d}{dx}[x^5 - 4x + 8] \\ &= 7(x^5 - 4x + 8)^6(5x^4 - 4).\end{aligned}$$

2.

$$\begin{aligned}\frac{dy}{dx} &= 4(x^2 - 3x + 1)^3 \frac{d}{dx}[x^2 - 3x + 1] \\ &= 4(x^2 - 3x + 1)^3(2x - 3).\end{aligned}$$

3. By the product rule,

$$y' = (2x+8)^4 \frac{d}{dx}\left[(x^2-6x)^6\right] + (x^2-6x)^6 \frac{d}{dx}\left[(2x+8)^4\right].$$

We will use the chain rule to find the derivatives of $(x^2-6x)^6$ and $(2x+8)^4$. Now

$$\begin{aligned} y' &= (2x+8)^4 \left[6(x^2-6x)^5 \frac{d}{dx}(x^2-6x)\right] + (x^2-6x)^6 \left[4(2x+8)^3 \frac{d}{dx}[2x+8]\right] \\ &= (2x+8)^4 \left[6(x^2-6x)^5(2x-6)\right] + (x^2-6x)^6 \left[4(2x+8)^3(2)\right] \\ &= 6(2x+8)^4(x^2-6x)^5(2x-6) + 8(x^2-6x)^6(2x+8)^3. \end{aligned}$$

Activity 5.22.

1. Find $\dfrac{dy}{dx}$ for each function below.

   (a) $y = 24343978$

   (b) $y = x^6 + 4x^4 - x^3 + 14x^2 - 8$

   (c) $y = 16x^5 - 9x^4 + 3x^3 - 2x^2 + 18x - 72$

   (d) $y = x^{200} - x^{3/4}$

   (e) $y = (x-4)^3(3x-1)^3$

   (f) $y = x^3 - x^2 + \dfrac{1}{x}$

   (g) $y = \dfrac{x^4-16}{x-2}$

   (h) $y = (3x-3)(2x+5)$

   (i) $y = \dfrac{3x^4 - 2x^3 + 6x}{12x}$

   (j) $y = (3x^2+4x)^8 - (2x-11)^3$

   (k) $y = (x^{1/3}-2)(x^{2/3}+2x^{1/3}+4)$

   (l) $y = (x+1)\sqrt{x}$

   (m) $y = \left(\dfrac{9x^3+3x^2+2x+1}{4x^2+5x}\right)^6$

   (n) $y = \dfrac{9x^3+3x^2+2x+1}{(4x^2+5x)^6}$

(o) $y = \sqrt[3]{5x^2 + 3x + 4}$

(p) $y = \dfrac{1}{\sqrt{2x+3}}$

(q) $y = (4x^2 + 15x)^{19}(9x^4 + 4x^{3/2})^{21}$

2. Find $g'(0)$ given that $g(x) = (4x^5 - 3x^3 - 2x^2 + x + 10)^4$.

5.5 Implicit Differentiation

We have so far dealt with functions of the general form $y = f(x)$ in which one variable y is explicitly expressed in terms of the other variable x. For example $y = 2x + 1$ and $y = \sqrt[3]{4x^2 - 3}$. Some functions are defined implicitly by a relation between x and y. For example $x^3 + y^3 - x^2y = 6xy$ and $y^4 + 3y - x^3 = 5x$. Implicit differentiation gives the derivatives of functions with two variables.

To find the derivative of a function such as $2x^2+4y^3-6x+y^2 = 3xy$, we differentiate each variable x and y as before, but the result of differentiating y is multiplied by y' $\left(\text{or } \dfrac{dy}{dx}\right)$. Finally we solve for y'.

Example 5.23.

1. If $2x^2 + 4y^3 - 6x + y^2 = 3x^2y$, find y'.
2. Differentiate with respect to x, $y^4 + 3y - 4x^3 = 5x + 1$.
3. Find the slope of the tangent line to the graph of $y^4 + 3y - 4x^3 = 5x + 1$ at the point $(1, -2)$.

Solution

1. We differentiate both sides:

$$\begin{aligned}
4x + 12y^2y' - 6 + 2yy' &= 3x^2\frac{d}{dx}[y] + y\frac{d}{dx}[3x^2] \\
4x + 12y^2y' - 6 + 2yy' &= 3x^2y' + 6xy \\
12y^2y' + 2yy' - 3x^2y' &= 6xy - 4x + 6 \\
(12y^2 + 2y - 3x^2)y' &= 6xy - 4x + 6 \\
y' &= \frac{6xy - 4x + 6}{12y^2 + 2y - 3x^2}.
\end{aligned}$$

Notice that we have used the product rule in finding the derivative of $3x^2y$.

2.

$$\begin{aligned}\frac{d}{dx}(y^4+3y-4x^3) &= \frac{d}{dx}(5x+1)\\ 4y^3y'+3y'-12x^2 &= 5\\ (4y^3+3)y' &= 5+12x^2\\ y' &= \frac{12x^2+5}{4y^3+3}.\end{aligned}$$

3. Recall that the slope of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$ is equal to $f'(a)$, the derivative of f at a.

   From 2 above, $y' = \dfrac{12x^2+5}{4y^3+3}$.

   Hence slope $= m = \dfrac{12(1)^2+5}{4(-2)^3+3} = -\dfrac{17}{29}$.

Activity 5.24.

1. Find $\dfrac{dy}{dx}$ for each function below.
   (a) $x^3 + y^3 = 3xy$
   (b) $x^2 + 2xy^2 = 3y + 4$
   (c) $1 - xy = x - y$
   (d) $9x^2 + 25y^2 = 225$
   (e) $y^2 = x^3$
   (f) $xy^2 - x^3y = 6$
   (g) $\sqrt{x+y} = 1 + x^3 + y^2$
   (h) $2y^3 + y^2 - 6y^5 = x^4 - 2x^3 + x^5$
2. Find the slope of the tangent line to the curve $y^3x + x^2y^2 = 6$ at the point $(2, 1)$.

5.6 Higher Derivatives

Differentiating a differentiable function f, gives its derivative f'. Let us call f' the **first derivative** of f. Suppose that f' is also differentiable. Then the derivative

of f' is the **second derivative** of f. The third derivative of f is the derivative of the second derivative of f.

Continuing in this manner, we can find the **fourth** and **fifth** derivatives of a given function. The derivatives of a function from the second going up are called **higher derivatives**.

5.6.1 The Second Derivative

The second derivative of a function f, denoted $(f')' = f''$, is obtained by differentiating f', the derivative of f. Other notations for f'' include: $f^{(2)}$, $\frac{d}{dx}\left(\frac{dy}{dx}\right)$ and $\frac{d^2y}{dx^2}$.

Example 5.25.

1. If $y = 3x^4 + 5x^2 + 6x - 7$, find y''.
2. Find $f''(2)$ given that $f(x) = 19x^3 + 2x^2 - 7x + 8$.

Solution

1.

$$\begin{aligned} y' &= 12x^3 + 10x + 6 \\ y'' &= 36x^2 + 10. \end{aligned}$$

2.

$$\begin{aligned} f'(x) &= 57x^2 + 4x - 7. \\ f''(x) &= 114x + 4, \\ f''(2) &= 114(2) + 4 = 232. \end{aligned}$$

5.6.2 The Third Derivative

The third derivative of a function f, denoted $(f'')' = f'''$, is obtained after differentiating the second derivative and is denoted by; $f^{(3)}$, $\frac{d}{dx}\left(\frac{d^2y}{dx^2}\right)$ and $\frac{d^3y}{dx^3}$.

Example 5.26.

1. If $y = 4x^5 + 6x^4 - 7x^3 + 8x^2 + 10x - 17$, find y'''.

2. Find $f'''(1)$ given that $f(x) = 7x^6 + 9x^4 - 3x$.

Solution

1.

$$\begin{aligned} y' &= 20x^4 + 24x^3 - 21x^2 + 16x + 10 \\ y'' &= 80x^3 + 72x^2 - 42x + 16 \\ y''' &= 240x^2 + 144x - 42. \end{aligned}$$

2.

$$\begin{aligned} f'(x) &= 42x^5 + 36x^3 - 3 \\ f''(x) &= 210x^4 + 108x^2 \\ f'''(x) &= 840x^3 + 216x \\ f'''(1) &= 840(1)^3 + 216(1) = 1056. \end{aligned}$$

Activity 5.27.

1. Let $f(x) = ax^2+bx+c$ where $a, b, c \in \mathbb{R}$. We know that $f(2) = 26$, $f'(2) = 23$ and $f''(2) = 14$. Find $f(1)$.

2. Find $\dfrac{dy}{dx}$, $\dfrac{d^2y}{dx^2}$ and $\dfrac{d}{dx}\left(\dfrac{d^2y}{dx^2}\right)$ for each function below.

   (a) $y = 18x^{20} - 9x^2 + 7x$

   (b) $y = (4x + 12)(6x - 9)$

   (c) $y = 21x^5 - 4x^2 + 9x - 2$

   (d) $y = \dfrac{5}{\sqrt{x}} + 6\sqrt[3]{x}$

3. Let $f(x) = 5x^3 - 2x^2 + 6x + 1$. Find $f'(2)$ and $f'''(1)$.

4. Let $g(x) = 15x^6 - 7x^4 + 13x^2 - 98$. Find $g''(1) - g'''(2)$.

Unit Summary

In this unit, you have learnt techniques of differentiation. These include differentiation from first principles, product rule, quotient rule and chain rule. You have also learnt implicit differentiation and higher order derivatives.

Activity 5.28 (Unit Test).

1. Differentiate $12x^3 - 6x^2 + 3x - 1$ from first principles.

2. Find $\dfrac{dy}{dx}$ for each function below.

   (a) $y = 2x^5(x^2 - 5x + 7)^7$

   (b) $y^4 + 5y^2 = x^4 - 5x^2$

   (c) $3(x^2 + y^2)^2 = 25(x^2 - y^3)$

   (d) $y = \dfrac{2}{\sqrt[3]{x^2 + 2x + 6}}$

   (e) $y = \dfrac{2x^2 - 7x + 3}{\sqrt[3]{x^2 + 2x + 6}}$

   (f) $y = (x + 2)(x^2 - x - 1)$

   (g) $y = (x + 2)^{16}(x^2 - x - 1)^5$

   (h) $y = (1 + \sqrt{x})^9$

3. For each of the following f, find $f''(a)$ and $f'''(b)$ for the given a and b.

   (a) $f(x) = 7x^6 - 8x^5 + 15x$, $a = 1$ and $b = 3$

   (b) $f(x) = \dfrac{x^2}{1 - 2x}$, $a = -2$ and $b = 2$

   (c) $f(x) = (2 + 3x)^2$, $a = 0$ and $b = 1$

   (d) $f(x) = x^3 - 3x^2 + 4x - 1$, $a = 8$ and $b = -10$.

Unit Answers

Suggested Solutions to Activity 5.5

1. (a) $2x + 1$ (b) $12x^2 - 4$ (c) $24x$ (d) $18x + 12$ (e) $2x - 19$

(f) $39x^2 + 14x - 5$

2. 21 3. 24

Suggested Solutions to Activity 5.22

1. (a) 0 (b) $6x^5+16x^3-3x^2+28x$ (e) $9(x-4)^3(3x-1)^2+3(3x-1)^3(x-4)^2$
   (g) $\dfrac{4x^3(x-2)-(x^4-16)}{(x-2)^2}$ (h) $2(3x-3)+3(2x+5)$ (i) $\dfrac{3}{4}x^2-\dfrac{1}{3}x$
   (j) $8(3x^2+4x)^7(6x+4)-6(2x-11)^2$

2. 4000

Suggested Solutions to Activity 5.24

1. (a) $\dfrac{3y-3x^2}{3y^2-3x}$ (b) $\dfrac{2x+2y^2}{3-4xy}$ (c) $\dfrac{y+1}{1-x}$
   (e) $\dfrac{3x^2}{2y}$ (h) $\dfrac{4x^3-6x^2+5x^4}{6y^2+2y-30y^4}$

2. $-\dfrac{5}{14}$

Suggested Solutions to Activity 5.27

1. $a = 7$, $b = -5$ and $c = 8$. So $f(x) = ax^2 + bx + c = 7x^2 - 5x + 8$ giving $f(1) = 10$.

2. (a) $\dfrac{dy}{dx} = 360x^{19}-18x+7$, $\dfrac{d^2y}{dx^2} = 6840x^{18}-18$ and $\dfrac{d}{dx}\left(\dfrac{d^2y}{dx^2}\right) = 123120x^{17}$.
   (b) $\dfrac{dy}{dx} = 6(4x+12)+4(6x-9) = 48x+36$, $\dfrac{d^2y}{dx^2} = 48$ and $\dfrac{d}{dx}\left(\dfrac{d^2y}{dx^2}\right) = 0$.
   (c) $\dfrac{dy}{dx} = 105x^4 - 8x + 9$, $\dfrac{d^2y}{dx^2} = 420x^3 - 8$ and $\dfrac{d}{dx}\left(\dfrac{d^2y}{dx^2}\right) = 1260x^2$.

3. $f'(2) = 58$ and $f'''(1) = 30$.

Suggested Solutions to Unit Test

1. $36x^2 - 12x + 3$

2. (a) $10x^4(x^2-5x+7)^7+14x^5(x^2-5x+7)^6(2x-5)$

   (c) $\dfrac{50x-12x^3-12xy^2}{12x^2y+12y^3+75y^2}$

   (d) $-\dfrac{4x+4}{\sqrt[3]{(x^2+2x+6)^4}}$

   (f) $(x^2-x-1)+(2x-1)(x+2)$

3. (a) $f''(1)=50$ and $f'''(3)=18360$ (c) $f''(0)=18$ and $f'''(1)=0$.

Unit 6

Derivatives of Transcendental Functions

6.1 Introduction

In Unit 2, we discussed derivatives of algebraic functions, i.e., functions which can be expressed in terms of polynomials. In this unit, you will learn how to find derivatives of transcendental functions. Recall that transcendental functions are non-algebraic functions, i.e., functions which can not be expressed in terms of polynomials. Transcendental functions whose derivatives are discussed in this unit are trigonometric, exponential and logarithmic functions

6.2 Objectives

By the end of this unit, you should be able to:

- differentiate trigonometric functions.
- differentiate natural exponential functions.
- differentiate logarithmic functions.

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- differentiation
- trigonometric functions
- natural exponential functions
- logarithmic functions

6.3 Trigonometric Functions

Derivatives of basic trigonometric functions are given in Table 6.1. The formulae are valid only when x is given in radians.

$f(x)$	$\sin x$	$\cos x$	$\tan x$	$\cot x$	$\csc x$	$\sec x$
$\frac{d}{dx}[f(x)]$	$\cos x$	$-\sin x$	$\sec^2 x$	$-\csc^2 x$	$-\csc x \cot x$	$\sec x \tan x$

Table 6.1: Derivatives of trigonometric functions

Example 6.1.

Find the derivative of each function with respect to x:

1. $y = \sqrt{x} \cot x$
2. $y = \tan x^3$
3. $y = \sin 3x$
4. $y = \cos(5x + 4)$
5. $y = \cot(x^2 + x)$
6. $y = \sin(\sin x)$

Solution

1. By the Product Rule,

$$\begin{aligned} y' &= \cot x \frac{d}{dx}[\sqrt{x}] + \sqrt{x}\frac{d}{dx}[\cot x] \\ &= (\cot x)\left(\frac{1}{2}x^{-1/2}\right) + \sqrt{x}(-\csc^2 x) \\ &= \frac{\cot x}{2\sqrt{x}} - \sqrt{x}\csc^2 x. \end{aligned}$$

2. By the Chain Rule,

$$\begin{aligned} y' &= (\sec^2 x^3)\frac{d}{dx}[x^3] \\ &= 3x^2 \sec^2 x^3. \end{aligned}$$

3. By the Chain Rule,

$$\begin{aligned} y' &= (\cos 3x)\frac{d}{dx}[3x] \\ &= 3\cos 3x. \end{aligned}$$

4. By the Chain Rule,

$$\begin{aligned} y' &= (-\sin(5x+4)\frac{d}{dx}[5x+4] \\ &= -5\sin(5x+4). \end{aligned}$$

5.

$$\begin{aligned} y' &= -\csc^2(x^2+x)\frac{d}{dx}[x^2+x] \\ &= -(2x+1)\csc^2(x^2+x). \end{aligned}$$

6.

$$\begin{aligned} y' &= \cos(\sin x)\frac{d}{dx}[\sin x] \\ &= \cos x \cos(\sin x). \end{aligned}$$

Example 6.2.

Use the Quotient Rule of differentiation to show that

1. $\dfrac{d}{dx}[\tan x] = \sec^2 x$

2. $\dfrac{d}{dx}[\sec x] = \sec x \tan x$

Solution

1.

$$\begin{aligned}
\frac{d}{dx}[\tan x] &= \frac{d}{dx}\left[\frac{\sin x}{\cos x}\right] \\
&= \frac{\cos x \dfrac{d}{dx}[\sin x] - \sin x \dfrac{d}{dx}[\cos x]}{\cos^2 x} \\
&= \frac{\cos x(\cos x) - \sin x(-\sin x)}{\cos^2 x} \\
&= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} \quad (\cos^2 x + \sin^2 x = 1) \\
&= \sec^2 x.
\end{aligned}$$

2.

$$\begin{aligned}
\frac{d}{dx}[\sec x] &= \frac{d}{dx}\left[\frac{1}{\cos x}\right] \\
&= \frac{\cos x \dfrac{d}{dx}[1] - 1 \dfrac{d}{dx}[\cos x]}{\cos^2 x} \\
&= \frac{0 - (-\sin x)}{\cos^2 x} \\
&= \frac{\sin x}{\cos^2 x} \\
&= \left(\frac{1}{\cos x}\right)\left(\frac{\sin x}{\cos x}\right) \\
&= \sec x \tan x.
\end{aligned}$$

Activity 6.3.

1. If $y = \cos x + \sin x$, show that $y'' = -y$.

2. Use the Quotient Rule of differentiation to show that

(a) $\dfrac{d}{dx}[\cot x] = -\csc^2 x$

(b) $\dfrac{d}{dx}[\csc x] = -\csc x \cot x$

3. Differentiate each function with respect to x.

   (a) $f(x) = 5\sin x + 3\cos x - \tan \pi x$

   (b) $f(x) = 4\sec x - 7\sin x$

   (c) $f(x) = \dfrac{2x + \cos x}{\sin x}$

   (d) $f(x) = \sqrt{x}\csc x + x^{10}\tan x$

   (e) $f(x) = 5x\cos x - x^2 \sin x$

   (f) $f(x) = \sin(\sqrt{x^2+1})$

   (g) $f(x) = \cos x - \sqrt{2}\csc x$

   (h) $f(x) = \cos^2(3x) - \cos^4(2x^2)$

   (i) $f(x) = \tan^4 x - \tan x^4$

6.4 Exponential Functions

Recall that an exponential function takes the general form

$$y = a^{f(x)}$$

where a is a constant, $a \neq 1$ and $a > 0$. We will look at the derivative of natural exponential functions, i.e., exponential functions with base e. Also recall that e is an irrational number whose value is approximately 2.7182818285.

Theorem 6.4. *Let $y = e^{f(x)}$ where $f(x)$ is a differentiable function. Then*

$$y' = f'(x)e^{f(x)}.$$

If $f(x) = x$, then $e^{f(x)} = e^x$ and $f'(x) = 1$. By Theorem 6.4,

$$\frac{d}{dx}[e^x] = e^x,$$

i.e., the derivative of e^x is e^x itself.

Example 6.5.

Find the derivative of each function with respect to x.

1. $y = e^{-2x}$
2. $y = e^{\sqrt{x}}$
3. $y = e^{3x} - e^{-3x}$
4. $y = e^{\sqrt{1+x^2}}$
5. $y = 4x^3 e^{(4x+1)}$

Solution

1. $y' = (e^{-2x})\dfrac{d}{dx}(-2x) = -2e^{-2x}$.
2. $y' = e^{\sqrt{x}}\dfrac{d}{dx}[\sqrt{x}] = \dfrac{1}{2\sqrt{x}}e^{\sqrt{x}}$.
3. $$\begin{aligned} y' &= e^{3x}\frac{d}{dx}[3x] - e^{-3x}\frac{d}{dx}[-3x] \\ &= e^{3x}(3) - e^{-3x}(-3) \\ &= 3e^{3x} + 3e^{-3x} = 3(e^{3x} + e^{-3x}). \end{aligned}$$
4. $$\begin{aligned} y' &= e^{\sqrt{1+x^2}}\frac{d}{dx}[\sqrt{1+x^2}] \\ &= e^{\sqrt{1+x^2}}\frac{1}{2}(1+x^2)^{-1/2}\frac{d}{dx}[1+x^2] \\ &= e^{\sqrt{1+x^2}}\frac{2x}{2\sqrt{1+x^2}} \\ &= \frac{x}{\sqrt{1+x^2}}e^{\sqrt{1+x^2}}. \end{aligned}$$
5. We use the product rule as follows:
$$\begin{aligned} y' &= 4x^3\frac{d}{dx}\left[e^{4x+1}\right] + e^{4x+1}\frac{d}{dx}[4x^3] \\ &= 4x^3e^{4x+1}\frac{d}{dx}[4x+1] + 12x^2e^{4x+1} \\ &= 16x^3e^{4x+1} + 12x^2e^{4x+1} \\ &= 4x^2e^{4x+1}(4x+3). \end{aligned}$$

Activity 6.6.

Differentiate each function with respect to x.

1. $y = e^{x+1} + 1$
2. $y = e^x \cos e^x$
3. $y = \sin e^{x^2}$
4. $y = \cos x e^{\sec x}$
5. $y = e^{\sin x - \tan x}$
6. $y = e^{x^3 - x^2}$
7. $y = e^x - e^7$
8. $y = \dfrac{e^x}{\cos x + \sin x}$

6.5 Logarithmic Functions

In semester one, we discussed logarithmic functions as functions which have the form

$$y = \log_a f(x)$$

where a is a positive constant such that $a \neq 1$. In this section, we will look at derivatives of logarithmic functions. Our main result is in Theorem 6.7.

Theorem 6.7. *If $f(x)$ is a differentiable function, then*

$$\frac{d}{dx}\left[\log_a f(x)\right] = f'(x) \times \frac{1}{f(x)\ln a}$$

where $\ln a = \log_e a$, the natural logarithm.

The following result follows immediately from Theorem 6.7:

$$\frac{d}{dx}[\log_a x] = \frac{1}{x \ln a}.$$

Also since $\ln e = \log_e e = 1$, Theorem 6.7 implies the following:

$$\frac{d}{dx}[\ln x] = \frac{1}{x \ln e} = \frac{1}{x}$$

and

$$\frac{d}{dx}[\ln f(x)] = f'(x) \times \frac{1}{f(x)\ln e} = f'(x) \times \frac{1}{f(x)} = \frac{f'(x)}{f(x)}.$$

Example 6.8.

Find y' for each function below.

1. $y = \log_2(2 + \sin x)$
2. $y = \ln(x^3 + 1)$
3. $y = \ln \cos x$
4. $y = \log_{10}(4x^2 - 5x)$

Solution

1. $y' = \dfrac{1}{(2 + \sin x)\ln 2} \times \dfrac{d}{dx}[2 + \sin x] = \dfrac{\cos x}{(2 + \sin x)\ln 2}.$
2. $y' = \dfrac{1}{x^3 + 1} \times \dfrac{d}{dx}[x^3 + 1] = \dfrac{3x^2}{x^3 + 1}.$
3. $y' = \dfrac{1}{\cos x} \times \dfrac{d}{dx}[\cos x] = \dfrac{-\sin x}{\cos x} = -\tan x.$
4. $y' = \dfrac{1}{(4x^2 - 5x)\ln 10} \times \dfrac{d}{dx}[4x^2 - 5x] = \dfrac{8x - 5}{(4x^2 - 5x)\ln 10}.$

Activity 6.9.

Find the derivative of each function below with respect to x.

1. $y = \log_7(x^2 - 4x)$
2. $y = \ln(x^3 - 4x)$
3. $y = \log_{10}(\sin x + e^x \cos x)$
4. $y = \ln x^3$
5. $y = \sqrt[5]{\ln 5x} + (\ln x)^3$

6. $y = \ln\left[\dfrac{x^2}{3x-4}\right]$

7. $y = \ln(x^4 - 5x + 3)$

Unit Summary

In this unit, you have learnt differentiation of transcendental functions. These are trigonometric functions, natural exponential functions and logarithmic functions.

Activity 6.10 (Unit Test).

Find $\dfrac{dy}{dx}$ for each function below.

1. $y = 5\sin 4x + 3\cos 2x$
2. $y = \tan^2(x^2)$
3. $y = \sec\sqrt{2x^2+3}$
4. $y = \cot\left(\dfrac{x+1}{x^2-3}\right)$
5. $y = \ln(\cos 3x)$
6. $y = \sqrt{\dfrac{\cos x}{x}}$
7. $y = \cos^3 x \sin x$
8. $y = e^{-2x}\cos 4x$
9. $y = x^2 e^{-1/x}$
10. $y = e^x \ln x^2$
11. $y = 4x^2 \ln(\sin x)$
12. $y = x^2 \sin^2(2x^2)$

Unit Answers

Suggested Solutions to Activity 6.3

3. (a) $5\cos x - 3\sin x - \pi\sec^2 \pi x$ (b) $4\sec x\tan x - 7\cos x$ (c) $\dfrac{2\sin x - 2x\cos x - 1}{\sin^2 x}$

(e) $-5x\sin x + 5\cos x - 2x\sin x - x^2\cos x$ (f) $\dfrac{x\cos\sqrt{x^2+1}}{\sqrt{x^2+1}}$

(g) $-\sin x + \sqrt{2}\csc x\cot x$ (i) $4\tan^3 x\sec^2 x - 4x^3\sec^2 x^4$

Suggested Solutions to Activity 6.6

1. $y = e^{x+1}$ 2. $y' = e^x\cos e^x - e^{2x}\sin e^x$ 3. $y' = 2xe^{x^2}\cos e^{x^2}$

5. $(\cos x - \sec^2 x)e^{\sin x - \tan x}$ 6. $(3x^2 - 2x)e^{x^3 - x^2}$ 7. e^x

Suggested Solutions to Activity 6.9

1. $y' = \dfrac{2x-4}{(x^2-4x)\ln 7}$ 2. $y' = \dfrac{3x^2-4}{x^3-4x}$ 3. $y' = \dfrac{\cos x + e^x\cos x - e^x\sin x}{(\sin x + e^x\cos x)\ln 10}$

4. $y' = \dfrac{3}{x}$

Suggested Solutions to Unit Test

1. $y' = 20\cos(4x) - 6\sin(2x)$ 3. $y' = \dfrac{2x\sec\sqrt{2x^2+3}\tan\sqrt{2x^2+3}}{\sqrt{2x^2+3}}$

5. $y' = \dfrac{-3\sin 3x}{\cos 3x}$ 7. $y' = \cos^4 x - 3\cos^2 x\sin^2 x$

8. $-2e^{-2x}\cos(4x) - 4\sin(4x)e^{-2x}$ 10. $e^x\left(\ln x^2 + \dfrac{1}{x}\right)$

12. $4x^3\sin^2 x + 8x^3\sin(2x^2)\cos(2x^2)$

Unit 7

Applications of Derivatives

7.1 Introduction

The derivatives of various functions, discussed in unit 2 and unit 3, can be applied in a number of areas. In this unit, we will look at applications of derivatives in selected areas. We will discuss how to use derivatives to find rates of change and related rates of change in natural science. We will also look at how derivatives are used to discuss the shape of the graph of a function. That is finding extreme values and identifying intervals where the graph of a given function is increasing or decreasing and intervals where the graph is concave upward or concave downward.

7.2 Objectives

By the end of this unit, you should be able to:

- use derivatives to find rates of change and related rates of change.
- use derivatives to find intervals where the graph of a function is increasing or decreasing.
- use derivatives to find local and absolute extrema of a given function.
- use derivatives to find intervals where the graph of a function is concave upward or concave downward.

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- instantaneous rate of change
- average rate of change
- velocity
- acceleration
- related rates of change
- local extrema
- absolute extrema
- increasing and decreasing functions
- increments
- differentials
- concavity
- concavity test
- stationery value
- first derivative test
- inflection point
- second derivative test
- critical numbers

7.3 Rates of Change and Related Rates of Change

Many quantities, in natural science and real life in general, change with time. For example, a chemist may be interested in the rate at which the temperature of a substance rises when heated. A physicist may want to know the rate at which the speed of an object changes with time.

Let $y = f(x)$. If x changes from x_1 to x_2 then the change in x is $\Delta x = x_2 - x_1$ and the corresponding change in y is $\Delta y = f(x_2) - f(x_1)$.

Definition 7.1. *The* **average rate of change** *of $y = f(x)$, with respect to x, in the interval $[x_1, x_2]$, is the quotient* $\dfrac{\Delta y}{\Delta x} = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$.

Definition 7.2. *The* **rate of change** *of $y = f(x)$ with respect to x is the derivative*

$$\frac{dy}{dx} = f'(x) = \lim_{\Delta x \to 0} \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

The rate of change in Definition 7.2 is also called the **instantaneous rate of change of y with respect to x.**

Example 7.3.

The area of a circle is given by $A = \pi r^2$ where r is its radius.

1. Find the rate of change of the area A of a circle with respect to its radius r.
2. Evaluate the rate of change of A at $r = 5$ and at $r = 10$.

Solution

1. The rate of change of A with respect to r is

$$\frac{dA}{dr} = \frac{d}{dr}[\pi r^2] = 2\pi r.$$

2. At $r = 5$, the rate is $2(5)\pi = 10\pi \approx 31.42$ and at $r = 10$, the rate is $2(10)\pi = 20\pi \approx 62.83$.

Example 7.4.

The current I (in amperes) in a certain electrical circuit is given by $I = \dfrac{100}{R}$, where R is resistance (in ohms). Find the rate of change of I with respect to R when the resistance is 15 ohms.

Solution

The rate of change of current with respect to resistance is

$$\frac{dI}{dR} = \frac{R\dfrac{d}{dR}[100] - 100\dfrac{d}{dR}[R]}{R^2} = -\frac{100}{R^2}.$$

If $R = 15$, then $\dfrac{dI}{dR} = -\dfrac{100}{(15)^2} = -\dfrac{100}{225} = -\dfrac{4}{9}$.

Hence the current is decreasing at a rate of $\dfrac{4}{9}$ amperes per ohms when $R = 15$.

7.3.1 Velocity and Acceleration

Consider an object moving in a straight line. A **position function** gives the position for the object as a function of time t. Given the position function $s(x)$ of a particle moving in a straight line, we can find **velocity** and **acceleration** of the particle by calculating the first derivative and the second derivative of $s(x)$ respectively.

Definition 7.5. *Let $s(t)$ be the position of a particle that is moving on a straight line. Then*

i. *The velocity $v(t)$ of the particle is $v(t) = s'(t)$.*

ii. *The speed is the absolute value of velocity at time t, i.e., $|v(t)|$.*

iii. *Acceleration $a(t)$ is given by $a(t) = v'(t) = s''(t)$.*

Example 7.6.

A car moves such that after t seconds its distance (in metres) is given by $s(t) = t^3 - 4t^2 + 12t - 3$.

1. Find the distance travelled after 5 s.

2. What is the velocity after 2 seconds?

3. Find acceleration after 4 seconds.
4. After how many seconds is the velocity of the car equals 47 m/s?
5. Find velocity when acceleration is $10ms^{-2}$.

Solution

1. The distance after 5 seconds is

$$s(5) = 5^3 - 4(5)^2 + 12(5) - 3 = 125 - 100 + 60 - 3 = 82m.$$

2. The velocity function is $v(t) = s'(t) = 3t^2 - 8t + 12$.
   The velocity after 2 seconds is

$$v(2) = 3(2)^2 - 8(2) + 12 = 12 - 16 + 12 = 8m/s.$$

3. Acceleration function is $6t - 8$.
   Acceleration after 4 seconds is

$$6(4) - 8 = 16m/s^2.$$

4. We equate the velocity function to $47m/s^2$ and solve for t.

$$\begin{aligned} 3t^2 - 8t + 12 &= 47 \\ 3t^2 - 8t + 12 - 47 &= 0 \\ 3t^2 - 8t - 35 &= 0 \\ (t-5)(3t+7) &= 0. \end{aligned}$$

   This gives $t = 5$ and $t = -\frac{7}{3}$. We ignore the negative number and take the positive number (why).
   Hence the velocity of the car is 47 m/s after 5 seconds.

5. First we have to find time after which the acceleration is 10 m/s^2.

$$6t - 8 = 10 \Rightarrow 6t = 18 \Rightarrow t = 3.$$

   Hence the acceleration is 10 m/s after 3 seconds.
   Now velocity after 3 seconds is

$$3(3)^2 - 8(3) + 12 = 27 - 24 + 12 = 15m/s.$$

Example 7.7.

A particle moves along a line so that its position at any time t is given by the function $s(t) = t^2 - 4t + 3$, where s is measured in meters and t is measured in seconds.

1. Find the displacement of the particle during the first 3 seconds.
2. Find the average velocity of the particle during the first 5 seconds.
3. Find the instantaneous velocity of the particle when $t = 4$.
4. Find the acceleration of the particle when $t = 5$.

Solution

1. The displacement is

$$s(3) - s(0) = [3^2 - 4(3) + 3] - [0^2 - 4(0) + 3] = [9 - 12 + 3] - 3 = -3.$$

   Hence the particle is 3 units left of where it started.

2. Average velocity is

$$\frac{s(5) - s(0)}{5 - 0} = \frac{(5^2 - 4(5) + 3) - (0^2 - 4(0) + 3)}{5} = \frac{8 - 3}{5} = 1m/s.$$

3. Velocity function is $v(t) = s'(t) = 2t - 4$.

   Hence instantaneous velocity when $t = 4$ is $2(4) - 4 = 4m/s$.

4. Acceleration function is $a(t) = v'(t) = s''(t) = 2$.

   Hence acceleration when $t = 5$ is $2m/s^2$.

7.3.2 Related Rates of Change

If two variables, x and y, are related by an equation and both are functions of a third variable, say t, i.e., $x = f(t)$ and $y = g(t)$, we can find a relation between their rates of change. We say the rates are related and we can compute one rate if we know the other.

We follow the following steps;

1. Find an equation that relates the dependent variables, x and y.

2. Differentiate both sides of the equation with respect to t (using Chain Rule if necessary).

3. Substitute the given information into the related rates equation and solve for the unknown rate.

Example 7.8.

A ladder 25 cm long is leaning against a vertical wall. The floor is slightly slippery and the foot of the ladder slips away from the wall at the rate of 0.2 cm/sec. How fast is the top of the ladder sliding down the wall when the top is 20 cm above the floor?

Solution

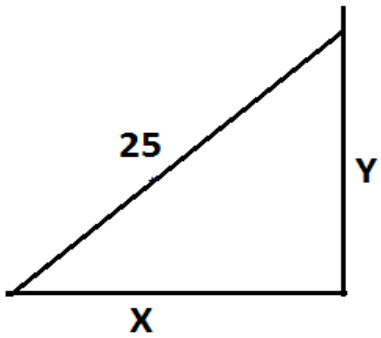

Figure 7.1:

1. In Figure 7.1, by Pythagorus Theorem we have,

$$x^2 + y^2 = 25^2$$

giving

$$x^2 + y^2 = 625.$$

2. Differentiating both sides with respect to t we have,

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0. \tag{7.1}$$

3. We need to find $\frac{dy}{dt}$ given that $\frac{dx}{dt} = 0.2\ cm/sec$ and $y = 20\ cm$.
   Now when $y = 20\ cm$,

$$x^2 + 20^2 = 25^2 \Rightarrow x^2 + 400 = 625 \Rightarrow x^2 = 625 - 400 = 225$$

   giving $x = \sqrt{225} = 15\ cm$.

4. Substituting in Equation 7.1 we have

$$2(15cm)(0.2cm/sec) + 2(20cm)\frac{dy}{dt} = 0,$$

$$6cm^2/sec + 40\frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = \frac{-6cm^2/sec}{40cm} = -0.15cm/sec.$$

   (The negative sign indicates that y is decreasing).

Example 7.9.

The radius of a right circular cylinder is increasing at a rate of 2 cm /min and the height is decreasing at a rate of 3 cm/min. At what rate is the volume changing when the radius is 8 cm and the height is 12 cm? Is the volume increasing or decreasing?

Solution

The volume of the cylinder is $V = \pi r^2 h$ where r and h are radius and height respectively. We need to find $\frac{dV}{dt}$.

Now $\frac{dr}{dt} = 2$ and $\frac{dh}{dt} = -3$.
Therefore

$$\begin{aligned}\frac{dV}{dt} &= 2\pi rh\frac{dr}{dt} + \pi r^2\frac{dh}{dt}\\ &= 2\pi(8)(12)(2) + \pi(64)(-3)\\ &= 384\pi - 192\pi\\ &= 192\pi cm^3/min.\end{aligned}$$

Since the rate is positive, the volume is increasing.

7.3.3 Increments and Differentials

Consider the function $y = f(x)$. If x changes from x_1 to x_2, then the amount of change

$$\Delta x = x_2 - x_1$$

is called an **increment** of x. Note that $x_2 = \Delta x + x_1$. The **increment** of y is given by

$$\Delta y = f(x_2) - f(x_1) = f(x_1 + \Delta x) - f(x_1) \text{ (See Figure 7.2).}$$

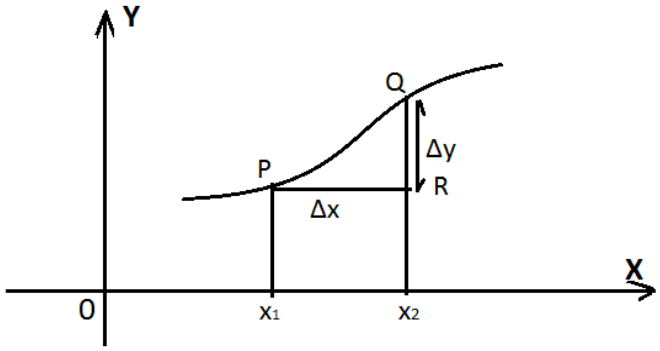

Figure 7.2: Increments

Example 7.10.

Suppose $f(x) = 4x^2 - 6$.

1. If x is given an increment Δx, find Δy.
2. Use Δy to calculate the numerical change in y if x changes from 3 to 3.1.

Solution

1.

$$\begin{aligned}
\Delta y &= f(x + \Delta x) - f(x) \\
&= [4(x + (\Delta))^2 - 6] - [4x^2 - 6] \\
&= 4(x^2 + 2x(\Delta x) + (\Delta x)^2) - 6 - 4x^2 + 6 \\
&= 4x^2 + 8x(\Delta x) + 4(\Delta x)^2 - 6 - 4x^2 + 6 \\
&= 8x(\Delta x) + 4(\Delta x)^2.
\end{aligned}$$

2. $\Delta x = 3.1 - 3 = 0.1$.

   Hence
$$\Delta y = 8(3)(0.1) + 4(0.1)^2 = 2.4 + 0.04 = 2.44.$$

Definition 7.11. *Let $y = f(x)$ where f is differentiable, and let Δx be an increment of x.*

i. *The* **differential** *dx of the independent variable x is $dx = \Delta x$.*

ii. *The* **differential** *dy of the dependent variable y is $dy = f'(x)\Delta x = f'(x)dx$.*

The differential dy in Definition 7.11 is used to approximate Δy, the increment of y.

Example 7.12.

1. If $y = 4x^2 - 6$, use dy to approximate Δy if x changes from 3 to 3.1.
2. Compare the values of Δy and dy if $y = f(x) = x^3 + x^2 - 2x + 1$ and x changes from 2 to 2.05.

Solution

1. By Example 7.10, $\Delta y = 2.44$.

   Now $dy = f'(x)dx = 8xdx$.

   $x = 3$ and $\Delta x = dx = 0.1$, so $dy = 8(3)(0.1) = 2.4$.

2. $dy = f'(x)dx$, $dx = 2.05 - 2 = 0.05$ and $x = 2$.
$$\begin{aligned} dy &= (3x^2 + 2x - 2)dx \\ &= (3(2)^2 + 2(2) - 2)(0.05) \\ &= (12 + 4 - 2)(0.05) \\ &= 14 \times 0.05 \\ &= 0.7. \end{aligned}$$

$$\begin{aligned} \Delta y &= f(x + \Delta x) - f(x) \\ &= f(2.05) - f(2) \\ &= [(2.05)^3 + (0.05)^2 - 2(2.05) + 1] - [2^3 + 2^2 - 2(2) + 1] \\ &= [8.62 + 4.20 - 4.10 + 1] - [8 + 4 - 4 + 1] \\ &= 9.72 - 9 \\ &= 0.72. \end{aligned}$$

Hence the error involved in using dy to approximate Δy is roughly

$$0.72 - 0.7 = 0.02.$$

Activity 7.13.

1. Gas is escaping from a spherical balloon at the rate of 2 cm^3/min. How fast is the surface area changing when the radius is 12 cm? [$V = \frac{4}{3}\pi r^3$ and $S = 4\pi r^2$].

2. Oil spilled from a ruptured tanker spreads out in a circle whose area increases at a constant rate of 6 cm^2/hr. How fast is the radius of the spill increasing when the area is 9 cm^2?

3. A particle moving in a straight line starts from rest and its displacement s m from a fixed point of the line is given by $s(t) = t^2 - kt$ where k is a constant and t is the time (in seconds) after the start. If it comes to instantaneous rest after 2 seconds, find

   (a) the value of k.

   (b) the initial velocity of the particle.

4. The distance (in metres) of a particle moving in a straight line measured from a fixed point O on the line is given by $s = t^2 - 3t + 2$ where t is the time in seconds from start. Find

   (a) its initial distance from O.

   (b) the distance after 4 seconds.

   (c) its initial velocity and in which direction.

   (d) velocity after 5 seconds.

   (e) time when velocity is 10 m/s

   (f) its initial acceleration.

   (g) acceleration after 8 seconds.

5. Use Δy to calculate the change in y corresponding to the stated values of x and Δx.

   (a) $y = 2x^2 - 4x + 5$, $x = 2$ and $\Delta x = -0.2$

   (b) $y = x^3 - 4$, $x = -1$ and $\Delta x = 0.1$

(c) $y = \dfrac{1}{x^2}$, $x = 3$ and $\Delta x = 0.03$

6. Use differentials to approximate the change in $f(x)$ if x changes from a to b given.

   (a) $f(x) = 4x^5 - 6x^4 + 3x^2 - 5$, $a = 1$ and $b = 1.03$

   (b) $f(x) = -3x^3 + 8x - 7$, $a = 4$ and $b = 3.96$

   (c) $f(x) = 8x^3 - 4x^2 + 6x + 9$, $a = 6$ and $b = 6.004$

7.4 Derivatives and the Shape of a Graph

7.4.1 Increasing and Decreasing Functions

Definition 7.14. *Let a function f be defined on an open interval I and let x_1 and x_2 be numbers in the interval. Then*

i. *f is increasing on I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.*

ii. *f is decreasing on I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$.*

iii. *f is constant on I if $f(x_1) = f(x_2)$ for every x_1 and x_2.*

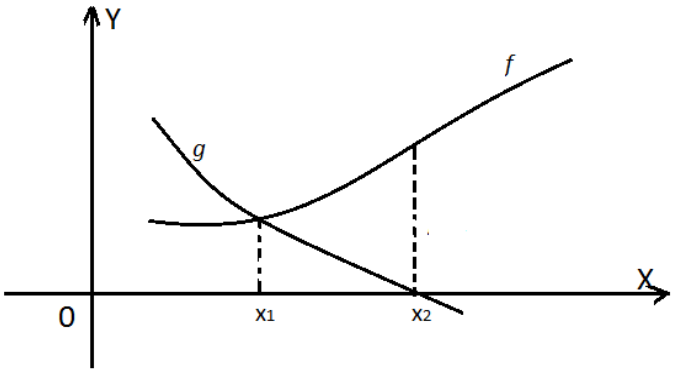

Figure 7.3: Increasing and decreasing functions

In Figure 7.3, f is an increasing function and g is a decreasing function in the interval $[x_1, x_2]$.

The curve of an increasing function $y = f(x)$ slopes upwards so that $\frac{dy}{dx} > 0$. The curve of a decreasing function slopes downwards such that $\frac{dy}{dx} < 0$. At any point where $\frac{dy}{dx} = 0$, $f(x)$ has a **stationary** value and $f(x)$ is neither increasing nor decreasing. This is typical of the interval where the function is constant.

Theorem 7.15 (Increasing/decreasing)**.** *Let a function f be defined on an open interval I.*

i. *If $\frac{df}{dx} > 0$ on an interval I, then f is increasing on I.*

ii. *If $\frac{df}{dx} < 0$ on I, then f is decreasing on I.*

Example 7.16.

For what range of values of x is the function $y = x^3 - 3x^2 - 9x + 23$

1. increasing?
2. decreasing?

Solution

$$\frac{dy}{dx} = 3x^2 - 6x - 9 = 3(x-3)(x+1).$$

We need to solve inequalities

$$3(x-3)(x+1) > 0 \Rightarrow (x-3)(x+1) > 0$$

and

$$(x-3)(x+1) < 0.$$

Key numbers are -1 and 3 and intervals of interest are

$$x < -1, -1 < x < 3 \text{ and } x > 3.$$

The test table is Table 7.1.

1. $\frac{dy}{dx} > 0$ in the intervals $x < -1$ and $x > 3$.
   Hence $y = x^3 - 3x^2 - 9x + 23$ is increasing in these intervals.

Interval	**Test Number**	$(x-3)$	$(x+1)$	$(x-3)(x+1)$
$x < -1$	-2	$-$	$-$	$+$
$-1 < x < 3$	0	$-$	$+$	$-$
$x > 3$	4	$+$	$+$	$+$

Table 7.1: Test table for the inequality $(x-3)(x+1) > 0$

2. $y = x^3 - 3x^2 - 9x + 23$ is decreasing in the interval $-1 < x < 3$.

Example 7.17.

For what range of values of x is the function $y = 4x^3 - 3x$

1. increasing?
2. decreasing?

Solution

$y' = 12x^2 - 3$. To find the interval where the function is increasing, we solve the inequality $y' > 0$. To find the interval where the function is decreasing, we solve th inequality $y' < 0$.

1. $$12x^2 - 3 > 0 \Rightarrow x^2 - \frac{1}{4} > 0 \Rightarrow \left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right) > 0$$

   giving $x < -\frac{1}{2}$ and $x > \frac{1}{2}$.

   Hence the function $y = 4x^3 - 3x$ is increasing in the intervals $x \in \left(-\infty, \frac{1}{2}\right)$ and $x \in \left(\frac{1}{2}, \infty\right)$.

2. $$12x^2 - 3 < 0 \Rightarrow x^2 - \frac{1}{4} < 0 \Rightarrow \left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right) < 0$$

   giving $-\frac{1}{2} < x < \frac{1}{2}$.

   Hence the function $y = 4x^3 - 3x$ is decreasing in the interval $x \in \left(-\frac{1}{2}, \frac{1}{2}\right)$.

7.4.2 Absolute Extrema of a Function on a Closed Interval

Let $y = f(x)$ be a given function. Suppose x_1 is in the domain of f. Then $f(x_1)$ is the maximum value of f if it is greater than all the other function values. Similarly, $f(x_1)$ is the minimum value of f if it is smaller than all the function values.

If the domain of a function f has a countable number of elements, then we can calculate a function value $f(x)$ for each x in the domain of f and identify as the maximum the largest function value. The smallest function value will be the minimum value for the function. However, this is not possible for a function whose domain is an infinite set.

We will use derivatives to find the maximum or the minimum value of a given function.

Definition 7.18. *Let c be a number in the domain D of a function f. Then $f(c)$ is the*

i. **absolute maximum** *value of f on D if $f(c) \geq f(x)$ for all $x \in D$.*

ii. **absolute minimum** *value of f on D if $f(c) \leq f(x)$ for all $x \in D$.*

Not every function has a maximum value and/or a minimum value. That is some functions have maximum value only while others have minimum value but no maximum. There are also other functions which have no maximum and no minimum values.

The maximum and minimum values of f are collectively called **extreme values** or **extrema** of f. Sometimes we are interested in local extrema of f defined in Definition 7.19.

Definition 7.19. *The number $f(c)$ is a*

i. **local maximum** *value of f if there exists an open interval (a, b) containing c such that $f(c) \geq f(x)$ for all x in (a, b).*

ii. **local minimum** *value of f if there exists an open interval (a, b) containing c such that $f(c) \leq f(x)$ for all x in (a, b).*

The word 'local' is used to show that our interest is on a small interval containing c.

Definition 7.20. *A number c in the domain of f is a* **critical number** *of f if either $f'(c) = 0$ or $f'(c)$ does not exist.*

We state in Theorem 7.21 that a maximum or a minimum value of a function occurs at a critical number.

Theorem 7.21. *If a function has a local extremum at a number c in an open interval, then either $f'(c) = 0$ or $f'(c)$ does not exist.*

To find absolute extrema of a function on a closed interval $[a, b]$ we follow the following steps:

i. Find all the critical numbers of f.

ii. Calculate $f(c)$ for each critical number c.

iii. Calculate $f(a)$ and $f(b)$.

iv. The absolute maximum of f on $[a, b]$ is the largest of the values obtained in (ii) and (iii). The absolute minimum of f on $[a, b]$ is the smallest of the values obtained in (ii) and (iii).

Example 7.22.

Find the absolute maximum and minimum values of the function
$f(x) = x^3 + x^2 - 8x - 12$ in the interval $[-4, 2]$.

Solution

$f'(x) = 3x^2 + 2x - 8 = (3x - 4)(x + 2)$.

i. Solving $f'(x) = 0$ gives the two critical numbers $\frac{4}{3}$ and -2.

ii.
$$f(-2) = (-2)^3 + (-2)^2 - 8(-2) - 12 = -8 + 4 + 16 - 12 = 0$$
and
$$\begin{aligned} f\left(\frac{4}{3}\right) &= \left(\frac{4}{3}\right)^3 + \left(\frac{4}{3}\right)^2 - 8\left(\frac{4}{3}\right) - 12 \\ &= \frac{64}{27} + \frac{16}{9} - \frac{32}{3} - 12 \\ &= \frac{64}{27} + \frac{48}{27} - \frac{288}{27} - \frac{324}{27} \\ &= -\frac{500}{27} \approx -18.5. \end{aligned}$$

iii.
$$f(2) = 2^3 + 2^2 - 8(2) - 12 = 8 + 4 - 16 - 12 = -16$$
and
$$f(-4) = (-4)^3(-4)^2 - 8(-4) - 12 = -64 + 16 + 32 - 12 = -28.$$

iv. Therefore the absolute maximum value of $f(x) = x^3 + x^2 - 8x - 12$ on $[-4, 2]$ is $f(-2) = 0$ and the absolute minimum value is $f(-4) = -28$.

Example 7.23.

Find the absolute maximum and minimum values of the function $f(x) = x^3 - 3x^2 - 144x$ in the interval $[-7, 10]$.

Solution

$f'(x) = 3x^2 - 6x - 144 = (3x + 18)(x - 8)$.

i. Solving $f'(x) = 0$ gives the two critical numbers -6 and 8.

ii.
$$f(-6) = (-6)^3 - 3(-6)^2 - 144(-6) = -216 - 108 + 864 = 540$$
and
$$f(8) = 8^3 - 3(8)^2 - 144(8) = 512 - 192 - 1152 = -832.$$

iii.
$$f(-7) = (-7)^3 - 3(-7)^2 - 144(-7) = -343 - 147 + 1008 = 518$$
and
$$f(10) = 10^3 - 3(10)^2 - 144(10) = 1000 - 300 - 1440 = -740.$$

iv. Therefore the absolute maximum value of $f(x) = x^3 + 3x^2 - 144x$ on $[-7, 10]$ is $f(-6) = 540$ and the absolute minimum value is $f(8) = -832$.

7.4.3 Local Extrema: The First Derivative Test

The fact that maximum and minimum values of a function occur at a critical number, discussed in previous subsection, does not imply that every critical number gives a maximum or a minimum value. We need to do some test to determine whether a given critical number gives rise to a maximum or minimum value. An example of such a test is **the first derivative test**.

Theorem 7.24 (The first derivative test)**.** *Suppose c is a critical number of a function f which is continuous on $[a, b]$.*

i. *If $f'(x)$ changes from positive to negative at c, that is $f'(x) > 0$ for $a < x < c$ and $f'(x) < 0$ for $c < x < b$, then $f(c)$ is a local maximum value of f.*

ii. *If $f'(x)$ changes from negative to positive at c, then $f(c)$ is a local minimum value of f.*

iii. *If $f'(x)$ does not change sign at c, that is $f'(x) > 0$ or $f'(x) < 0$ on both sides of c, then $f(c)$ is not a local extremum of f.*

For better understanding of Theorem 7.24, see Figure 7.4.

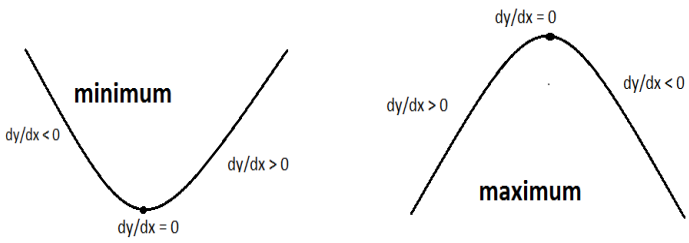

Figure 7.4: Maximum and minimum

Example 7.25.

Assuming that the derivative of a function f is given by $f'(x) = (x - 1)(x + 3)$, identify critical numbers where f has local extrema.

Solution

Solving $f'(x) = 0$ gives two critical numbers 1 and -3. In Table 7.2, we consider the sign of $f'(x)$ on each of the intervals $(-\infty, -3)$, $(-3, 1)$ and $(1, \infty)$.

In Table 7.2, we see that

i. f has a local maximum value at $x = -3$ since $f'(x)$ changes from positive to negative.

ii. f has a local minimum value at $x = 1$ since $f'(x)$ changes from negative to positive.

Interval	**Test Number**	$f'(x)$	Sign of $f'(x)$
$(-\infty, -3)$	-4	5	$+$
$(-3, 1)$	0	-3	$-$
$(1, \infty)$	2	5	$+$

Table 7.2: Test table

Example 7.26.

Find the local maxima and minima of f if $f(x) = x^3 - 5x^2 + 3x + 2$.

Solution

$$f'(x) = 3x^2 - 10x + 3 = (3x - 1)(x - 3).$$

Solving $f'(x) = 0$ gives two critical numbers $\frac{1}{3}$ and 3. In Table 7.3, we consider the sign of $f'(x)$ on each of the intervals $\left(-\infty, \frac{1}{3}\right)$, $\left(\frac{1}{3}, 3\right)$ and $(3, \infty)$.

Interval	**Test Number**	$f'(x)$	Sign of $f'(x)$
$\left(-\infty, \frac{1}{3}\right)$	0	3	$+$
$\left(\frac{1}{3}, 3\right)$	1	-4	$-$
$(3, \infty)$	4	11	$+$

Table 7.3: Test table

From Table 7.3, we observe the following:

i. Since $f'(x)$ changes from positive to negative at $x = \frac{1}{3}$, f has a local maximum value at this critical number. This local maximum value is

$$f\left(\frac{1}{3}\right) = \left(\frac{1}{3}\right)^3 - 5\left(\frac{1}{3}\right)^2 + 3\left(\frac{1}{3}\right) + 2 = \frac{1}{27} - \frac{5}{9} + 1 + 2 = \frac{1}{27} - \frac{15}{27} + \frac{81}{27} = \frac{67}{27} \approx 2.48.$$

ii. f has a local minimum value at $x = 3$ since $f'(x)$ changes from negative to positive . This local minimum value is

$$f(3) = 3^3 - 5(3)^2 + 3(3) + 2 = 27 - 45 + 9 + 2 = -7.$$

7.4.4 Concavity and Its Test

Definition 7.27. *The graph of a function f is*

i. **concave upward** *on an interval I if it lies above all its tangent lines on I.*

ii. **concave downward** *on an interval I if it lies below all its tangent lines on I.*

Figure 7.5 shows the graph of f which is concave upward and the graph of another function g which is concave downward.

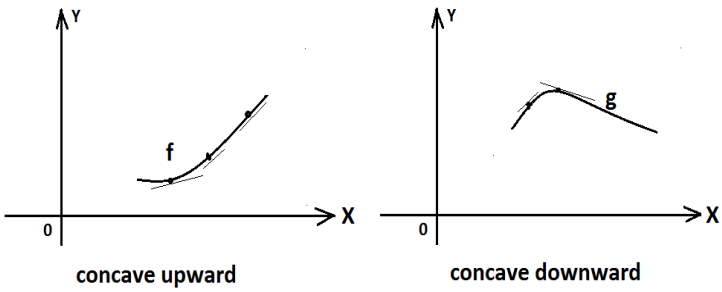

Figure 7.5: concavity

To find the interval where a function is concave upward or concave downward, we do not need to sketch the graph. The second derivative of a given function f is used to determine where the graph of f is concave upward or concave downward.

Theorem 7.28 (Concavity Test)**.** *Let f be a differentiable function.*

i. *If $f''(x) > 0$ for all x in the interval I, then the graph of f is concave upward on I.*

ii. *If $f''(x) < 0$ for all x in I then the graph of f is concave downward.*

A point where a curve changes direction of concavity is called **inflection point** or **point of inflection** (See Definition 7.29).

Definition 7.29. *A point P on a curve is called* **a point of inflection** *if f is continuous at P and the graph of f changes from concave upward to concave downward or from concave downward to concave upward.*

Example 7.30.

If $f(x) = x^3 + x^2 - 5x + 15$,

1. determine intervals on which the graph of f is concave upward and intervals on which the graph is concave downward.

2. Find coordinates of the point(s) of inflection.

Solution

$$\begin{aligned} f(x) &= x^3 + x^2 - 5x + 15 \\ f'(x) &= 3x^2 + 2x - 5 \\ f''(x) &= 6x + 2 = 2(3x + 1). \end{aligned}$$

1. To find intervals where the graph of f is concave upward, we solve the inequality $f''(x) > 0$. Similarly, we solve the inequality $f''(x) < 0$ to find intervals where the graph of f is concave downward.

   Now
   $$2(3x + 1) > 0 \Rightarrow 3x + 1 > 0 \Rightarrow x > -\frac{1}{3}.$$

   Hence the graph of f is concave upwards on the interval $\left(-\frac{1}{3}, \infty\right)$.

   $$f''(x) < 0 \Rightarrow 3x + 1 < 0 \Rightarrow x < -\frac{1}{3}.$$

   Hence the graph of f is concave downward on the interval $\left(-\infty, -\frac{1}{3}\right)$.

2. The point $(-1/3, f(-1/3)) = (-1/3, -88/27)$, at which the concavity changes from downward to upward is a point of inflection.

Example 7.31.

Let $f(x) = 3x^4 + 16x^3 + 24x^2 + 3$.

1. Determine intervals on which the graph of f is concave upward and intervals on which the graph is concave downward.

2. Find the points of inflection.

Solution

$f'(x) = 12x^3 + 48x^2 + 48x$ and $f''(x) = 36x^2 + 96x + 48 = 12(3x^2 + 8x + 4)$.

1. To find interval(s) where f is concave upward, we will solve the inequality

$$f''(x) > 0 \Rightarrow 12(3x^2 + 8x + 4) > 0 \Rightarrow (3x + 2)(x + 2) > 0.$$

Similarly, to find interval(s) where f is concave downward, we will solve the inequality

$$(3x + 2)(x + 2) < 0.$$

Key numbers are -2 and $-\frac{2}{3}$. Intervals of interest are $(-\infty, -2)$, $\left(-2, -\frac{2}{3}\right)$ and $\left(-\frac{2}{3}, \infty\right)$. Test table is Table 7.4.

Interval	**Test Number**	$(3x + 2)$	$(x + 2)$	$(3x + 2)(x + 2)$
$(-\infty, -2)$	-3	$-$	$-$	$+$
$\left(-2, -\frac{2}{3}\right)$	-1	$-$	$+$	$-$
$\left(-\frac{2}{3}, \infty\right)$	0	$+$	$+$	$+$

Table 7.4: Test table for $(3x + 2)(x + 2) < 0$

From Table 7.4, $f''(x) > 0$ in the intervals $(-\infty, -2)$ and $\left(-\frac{2}{3}, \infty\right)$. These are intervals where the graph of the function f is concave upward.

Similarly, the graph of f is concave downward in the interval $\left(-2, -\frac{2}{3}\right)$.

2. The points of inflection are

$$(-2, f(-2)) = (-2, 19)$$

and

$$\left(-\frac{2}{3}, f\left(-\frac{2}{3}\right)\right) = \left(-\frac{2}{3}, \frac{771}{81}\right).$$

7.4.5 Local Extrema: The Second Derivative Test

Apart from the first derivative test, we can also use the second derivative test to find local extrema. We state the second derivative test in Theorem 7.32.

Theorem 7.32 (The Second Derivative Test)**.** *Let a function f be differentiable on an open interval containing c and further suppose that $f'(c) = 0$.*

i. *If $f''(c) > 0$, then $f(c)$ is a local minimum.*

ii. *If $f''(c) < 0$, then $f(c)$ is a local maximum.*

The second derivative test is easy to use than the first derivative test. However, it gives no information when $f''(c) = 0$ and when $f''(c)$ does not exist which makes it less effective. In cases where $f''(c) = 0$ or $f''(c)$ does not exist, the first derivative test is used.

We follow these steps when using the second derivative test to find local extrema of a function f:

i. Find $f'(x)$.

ii. Find every number c such that $f'(c) = 0$.

iii. Find $f''(c)$ for each number c obtained in step ii.

iv. If $f''(c) < 0$, then find $f(c)$ which is the local maximum value of f. If $f''(c) > 0$, $f(c)$ is the local minimum value of f. However, if $f''(c) = 0$ or $f''(c)$ does not exist, then the test can not be used to find local extrema.

Example 7.33.

If $f(x) = 3x^4 + 16x^3 + 24x^2 + 3$, find the local maximum and minimum values of f by the second derivative test.

Solution

i. $f'(x) = 12x^3 + 48x^2 + 48x = 12x(x^2 + 4x + 4) = 12x(x + 2)^2$.

ii. $f'(x) = 0$ implies $x = 0$ or $x = -2$.

iii.

$$\begin{aligned} f''(x) &= 36x^2 + 96x + 48 \\ f''(0) &= 36(0)^2 + 96(0) + 48 = 48 \\ f''(-2) &= 36(-2)^2 + 96(-2) + 48 = 144 - 192 + 48 = 0. \end{aligned}$$

iv. Since $f''(0) = 48$ which is positive, f has a local minimum value at $x = 0$. This minimum value is

$$f(0) = 3(0)^4 + 16(0)^3 + 24(0)^2 + 3 = 3.$$

$f''(-2) = 0$ and the test gives no information on whether there a local maximum value or minimum value at $x = -2$.

Example 7.34.

If $f(x) = x^4 - 2x^2 - 12$, find the local maximum and minimum values of f by the second derivative test.

Solution

i. $f'(x) = 4x^3 - 4x = 4x(x^2 - 1)$.

ii. $f'(x) = 0$ implies $x = 0, 1$ or -1.

iii.

$$\begin{aligned} f''(x) &= 12x^2 - 4, \\ f''(0) &= 12(0)^2 - 4 = -4, \\ f''(1) &= 12(1)^2 - 4 = 8, \\ f''(-1) &= 12(-1)^2 - 4 = 8. \end{aligned}$$

iv. Since $f''(0) = -4$ which is negative, f has a local maximum value at $x = 0$. This maximum value is

$$f(0) = (0)^4 - 2(0)^2 - 12 = -12.$$

Similarly, local minimum value occur at both $x = 1$ and $x = -1$ since both $f''(1)$ and $f''(-1)$ are positive. This minimum value is

$$f(-1) = f(1) = (1)^4 - 2(1)^2 - 12 = -13.$$

Activity 7.35.

1. Find critical numbers for each function below.

   (a) $f(x) = x^{1/4}(x - 1)^3$

(b) $f(x) = x^3(x-2)^{2/3}$

2. Find the intervals where each function is increasing or decreasing;

   (a) $f(x) = 5 - 7x - 4x^2$
   (b) $f(x) = 4x^3 - 3x^4$
   (c) $f(x) = 3x^2 + 6x$

3. Find the absolute maximum and absolute minimum values of f on the given interval;

   (a) $f(x) = 12 + 4x - x^2$, $[0, 5]$
   (b) $f(x) = x^2$, $[-1, 2]$
   (c) $f(x) = x^3$, $[-2, 2]$
   (d) $f(x) = \sin x$, $[-\pi/2, \pi]$
   (e) $f(x) = 2x^3 + 3x^2 - 12x + 4$, $[0, 2]$
   (f) $f(x) = 2x^3 + 3x^2 - 12x + 4$, $[-4, 2]$

4. Find the local extrema [*compare the values obtained by the first and second derivative tests*].

   (a) $f(x) = 2x^3 + 3x^2 - 36x$
   (b) $f(x) = 5x^{2/3} - 2x^{5/3}$
   (c) $f(x) = 4x^3 + 3x^2 - 6x + 1$
   (d) $f(x) = x\sqrt{6-x}$
   (e) $f(x) = x^4 - 2x^2 + 3$

5. The function $f(x) = x^3 + rx^2 - 7x - 1$ has a stationary value where $x = 1$. Find the value of r.

6. Determine interval(s) where the graph of the function f is concave upward and interval(s) where it is concave downward, then find point(s) of inflection.

   (a) $f(x) = 3x^4 - 44x^3 + 144x^2$
   (b) $f(x) = 4x^3 - 18x^2 + 48x - 290$
   (c) $f(x) = x^4 - 8x^2$
   (d) $f(x) = x^3 - 3x^2 - 144x$
   (e) $f(x) = x^4 + 28x^3 + 10x$

Unit Summary

In this unit, you have learnt the use of derivatives to find rates of change and related rates of change. You have also learnt how to use the derivatives of a function to discuss the shape of the graph of a function.

Activity 7.36 (Unit Test).

1. Use the second derivative test to find local maximum and local minimum values of the function $f(x) = 3x^5 - 5x^3 + 3$.

2. The velocity v cms^{-1} of a particle moving in a straight line is given by $v = 6t - ct^2$, where c is a constant and t is time in seconds from start. If its acceleration is 0 cms^{-2}, when $t = 1$, find

   (a) the value of c

   (b) the time when the particle comes to instantaneous rest

   (c) the maximum velocity of the particle

3. Find the intervals where each function is increasing or decreasing;

   (a) $f(x) = 6x^2 - 9x + 5$

   (b) $f(x) = 2x^3 + x^2 - 20x + 1$

4. Find the absolute maximum and absolute minimum values of f on the given interval;

   (a) $f(x) = x^2$, $[-2, 2]$

   (b) $f(x) = 3x(x + 4)^{2/3}$, $[-5, -1]$

   (c) $f(x) = 2x^3 - 3x^2 - 12x + 1$, $[-2, 3]$

   (d) $f(x) = 3x^2 - 10x + 7$, $[-1, 3]$

   (e) $f(x) = 5 + 54x - 2x^3$, $[0, 4]$

5. Where is the function $3x^2 + 4x + 1$ increasing? Where is it decreasing?

6. Determine interval(s) where the graph of the function f is concave upward and interval(s) where it is concave downward, then find point(s) of inflection.

   (a) $f(x) = 3x^4 + 8x^3 - 174x^2 - 360x$

(b) $f(x) = x^2(x+4)$

7. Two parallel sides of a rectangle are being shortened at the rate of 2 cm/sec, while the other two are shortened in such a way that the figure remains a rectangle with constant area of 50 cm^2. What is the rate of change of perimeter when the length of an increasing side is 5 cm? Is the perimeter increasing or decreasing? [$A = lw$ and $P = 2l + 2w$].

8. Sand falling from a chute forms a conical pile whose height is always equal to $\frac{4}{3}$ the radius of the base. How fast is the volume changing when the radius of the base is 3 m and is increasing at the rate of 3 m/min? [$V = \frac{1}{3}\pi r^2 h$ and $h = \frac{4}{3}r$].

Unit Answers

Suggested Solutions to Activity 7.13

1. $\frac{dS}{dt} = -\frac{1}{3}$ cm^2/min 2. $\frac{dr}{dt} = \frac{1}{\sqrt{\pi}}$ cm/hr

3. (a) $k = 4$ (b) initial velocity is -4 m/s.

4. (a) 2 m (b) 6 m (c) -3 m/s (d) 6 m/s
   (e) 7 s (f) 2 m/s^2 (g) 2 m/s^2

5. (a) $\Delta y = -0.72$ (b) $\Delta y = 0.271$

6. (a) $\Delta y \approx dy = 0.06$ (b) $\Delta y \approx dy = 5.44$

Suggested Solutions to Activity 7.35

1. (a) Critical numbers are $\frac{1}{13}$, 0 and 1.
   (b) Critical numbers are $\frac{18}{11}$, 0 and 2.

2. (a) The function is increasing on the interval $\left(-\infty, -\frac{7}{8}\right)$ and it is decreasing on the interval $\left(-\frac{7}{8}, \infty\right)$.

(b) f is increasing on $(-\infty, 1)$ and decreasing on $(1, \infty)$.

(c) f is increasing on $(-\infty, -2)$ and $(0, \infty)$ and is decreasing on $(-2, 0)$.

3. (a) Absolute maximum is 16 and absolute minimum is 7.

   (b) Absolute maximum is 4 and absolute minimum is 0.

   (c) Absolute maximum is 8 and absolute minimum is -8.

   (e) Absolute maximum is 8 and absolute minimum is -3.

   (f) Absolute maximum is 24 and absolute minimum is -28.

4. (a) Local maximum is $f(-3) = 81$ and local minimum is $f(2) = -44$.

   (c) Local maximum is $f(-1) = 6$ and local minimum is $f\left(\frac{1}{2}\right) = -\frac{3}{4}$.

   (d) Local maximum is $f(4) = 4\sqrt{2}$.

   (e) Local maximum is $f(0) = 3$ and local minimum is $f(1) = 2$.

5. $r = 2$.

6. (a) concave upward: $\left(-\infty, \frac{4}{3}\right)$, $(6, \infty)$, concave downward: $\left(\frac{4}{3}, 6\right)$

   (b) concave upward: $\left(\frac{3}{2}, \infty\right)$, concave downward: $\left(-\infty, \frac{3}{2}\right)$ and point of inflection is $\left(\frac{3}{2}, -245\right)$.

   (c) concave upward: $\left(-\infty, -\frac{2}{\sqrt{3}}\right)$, $\left(\frac{2}{\sqrt{3}}, \infty\right)$, concave downward: $\left(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$ and point of inflection is $\left(-\frac{2}{\sqrt{3}}, -\frac{80}{9}\right)$.

   (d) concave upward: $(1, \infty)$, concave downward: $(-\infty, 1)$ and point of inflection is $(1, -146)$.

   (e) concave upward: $(-\infty, -14)$, $(0, \infty)$, concave downward: $(-14, 0)$ and points of inflection $(0, 0)$ and $(-14, -1316)$.

Suggested Solutions to Unit Test

1. Local maximum: -5, local minimum: 1

2. (a) $c = 3$ (b) $t = 2$ s (c) 3 m/s

3. (a) The function is increasing on $\left(\frac{3}{4}, \infty\right)$ and is decreasing on $\left(-\infty, \frac{3}{4}\right)$.

   (b) The function is increasing on $(-\infty, -2)$ and $\left(\frac{5}{3}, \infty\right)$, it is decreasing on $\left(-2, \frac{5}{3}\right)$.

4. (a) Absolute maximum is $f(2) = 4$ and absolute minimum is $f(0) = 0$.

   (b) Absolute maximum is 0 and absolute minimum is -15.

   (c) Absolute maximum is 8 and absolute minimum is -19.

5. The function is increasing on $(-\infty, -1)$ and $\left(-\frac{1}{3}, \infty\right)$, it is decreasing on $(\left(-1, -\frac{1}{3}\right))$.

7. $\frac{dP}{dt} = -4$ cm/s. The perimeter is decreasing.

Unit 8

Integration

8.1 Introduction

In this unit, we will introduce integration as a reverse operation to differentiation. We will discuss both indefinite and definite integrals. Basic techniques of integration, such as constant rule and power rule, will also be discussed. We will also discuss integrals of natural exponential functions and trigonometric functions.

8.2 Objectives

By the end of this unit, you should be able to:

- find indefinite and definite integrals of polynomial functions.
- find indefinite and definite integrals of trigonometric functions.
- find indefinite and definite integrals of natural exponential functions.

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- integrals

- integrand
- indefinite integrals
- definite integrals
- constant rule of integration
- power rule of integration
- properties of indefinite integrals
- properties of definite integrals
- integrals of trigonometric integrals
- integrals of natural exponential functions

8.3 Indefinite Integrals

Given a differentiable function $f(x)$, we can differentiate it to find its derivative $f'(x)$. Suppose we are given $f'(x)$ and asked to find the original function $f(x)$. To find $f(x)$ we apply the reverse operation to **differentiation** called **integration** to $f'(x)$.

In integration, the function that gets integrated is called an **integrand**. The function, which results from integrating another function, is called an **integral** or an **anti-derivative**. The symbol for integration is $\int$. Thus

$$\int f(x)dx = F(x) \text{ means } F'(x) = f(x).$$

Consider the two functions $f(x) = x^3 + 4x^2 + 8x + 7$ and $g(x) = x^3 + 4x^2 + 8x + 18$. Clearly $f(x) \neq g(x)$ but $f'(x) = g'(x) = 3x^2 + 8x + 8$. In general, the derivative of a function $x^3 + 4x^2 + 8x + C$, where C is any constant, is $3x^2 + 8x + 8$. This is so because the derivative of any constant is zero.

The preceding discussion implies that

$$\int (3x^2 + 8x + 8)dx = x^3 + 4x^2 + 8x + C$$

since integration is the reverse of differentiation and

$$\frac{d}{dx}[x^3 + 4x^2 + 8x + C] = 3x^2 + 8x + 8.$$

The integral $3x^2 + 8x + 8 + C$ is called **indefinite integral** since it represents the whole family of infinitely many functions ($C \in \mathbb{R}$ and $\mathbb{R}$ is an infinite set).

8.3.1 The Constant Rule of Integration

Theorem 8.1 (Constant Rule)**.** *If k is a constant, then $\int k dx = kx + C$.*

Example 8.2.

1. $\int 2dx = 2x + C$
2. $\int 7dx = 7x + C$
3. $\int dx = x + C$
4. $\int -4dx = -4x + C$

8.3.2 The Power Rule of Integration

Theorem 8.3 (Power Rule)**.** *If n is any real number, then*

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C.$$

Example 8.4.

1. $\int x^2 dx = \dfrac{x^{2+1}}{2+1} + C = \dfrac{x^3}{3} + C.$
2. $\int \sqrt{x} dx = \int x^{1/2} dx = \dfrac{x^{1/2+1}}{1/2+1} + C = \dfrac{x^{3/2}}{(3/2)} + C = \dfrac{2\sqrt{x^3}}{3} + C.$
3. $\int x^{-4} dx = \dfrac{x^{-3}}{-3} + C = -\dfrac{1}{3x^3} + C.$

8.3.3 Properties of Indefinite Integrals

Theorem 8.5. *Let $f(x)$ and $g(x)$ be two functions and let k be a constant. Then*

i. $\int kf(x)dx = k\int f(x)dx$.

ii. $\int[f(x)+g(x)]dx = \int f(x)dx + \int g(x)dx$.

iii. $\int[f(x)-g(x)]dx = \int f(x)dx - \int g(x)dx$.

In words, Theorem 8.5 states that:

i. the integral of a constant times a function is the constant times the integral of the function.

ii. the integral of the sum of two (or more) functions is the sum of the integrals of the two functions.

iii. the integral of the difference between two functions is the difference between the integrals of the to functions.

Example 8.6.

Evaluate the following integrals;

1. $\int(2x^3+3x-4)dx$
2. $\int(7x^6-3x^2+2x)dx$
3. $\int(19x^7+1)dx$

Solution

1.

$$\begin{aligned}\int(2x^3+3x-4)dx &= 2\int x^3dx + 3\int xdx - 4\int dx \\ &= 2\left(\frac{x^4}{4}\right) + 3\left(\frac{x^2}{2}\right) - 4x + C \\ &= \frac{x^4}{2} + \frac{3x^2}{2} - 4x + C.\end{aligned}$$

2.

$$\begin{aligned}\int (7x^6 - 3x^2 + 2x)dx &= 7\int x^6 dx - 3\int x^2 dx + 2\int x dx \\ &= 7\left(\frac{x^7}{7}\right) - 3\left(\frac{x^3}{3}\right) + 2\left(\frac{x^2}{2}\right) + C \\ &= x^7 - x^3 + x^2 + C.\end{aligned}$$

3.

$$\int (19x^7 + 1)dx = \frac{19}{8}x^8 + x + C.$$

8.3.4 Integrals of Natural Exponential Functions

Since $\frac{d}{dx}[e^x] = e^x$ and $\frac{d}{dx}[e^{ax}] = ae^x$ $(a \in \mathbb{R})$, the following result is immediate.

Theorem 8.7. *Let k be a constant. Then*

i. $\int e^x dx = e^x + C$.

ii. $\int e^{kx} dx = \frac{1}{k}e^{kx} + C$.

Example 8.8.

1. $\int e^{2x} dx = \frac{1}{2}e^{2x} + C$.

2. $\int 3e^{-4x} dx = -\frac{3}{4}e^{-4x} + C$.

8.3.5 Integrals of Trigonometric Functions

Integrals of trigonometric functions are given in Table 8.1. Please compare Table 8.1 with Table 6.1 which gives derivatives of basic trigonometric functions in Unit 3.

$f(x)$	$\sin x$	$\cos x$	$\sec^2 x$	$\csc^2 x$	$\csc x \cot x$	$\sec x \tan x$
$\int f(x)dx$	$-\cos x + C$	$\sin x + C$	$\tan x + C$	$-\cot x + C$	$-\csc x + C$	$\sec x + C$

Table 8.1: Integrals of trigonometric functions

Example 8.9.

Evaluate the following integrals.

1. $\int (x - \csc x \cot x)dx$
2. $\int \sec x(\sec x + \tan x)dx$
3. $\int \dfrac{\sin x + \sin x \tan^2 x}{\sec^2 x}dx$
4. $\int (5e^x + 3\cos x)dx$

Solution

1. $\int (x - \csc x \cot x)dx = \dfrac{x^2}{2} - (-\csc x) + C = \dfrac{x^2}{2} + \csc x + C.$
2. $\int \sec x(\sec x + \tan x)dx = \int (\sec^2 x + \sec x \tan x)dx = \tan x + \sec x + C.$
3. $$\begin{aligned}\int \frac{\sin x + \sin x \tan^2 x}{\sec^2 x}dx &= \int \frac{\sin x(1 + \tan^2 x)}{\sec^2 x}dx \\ &= \int \frac{\sin x \sec^2 x}{\sec^2 x}dx \\ &= \int \sin x dx \\ &= -\cos x + C.\end{aligned}$$

   Notice that we have used the trigonometric identity

   $$\tan^2 x + 1 = \sec^2 x$$

   whch is true for all angles x.
4. $\int (5e^x + 3\cos x)dx = 5e^x + 3\sin x + C.$

Activity 8.10.

Evaluate each of the following integrals.

1. $\int (\tan^2 x + 1)dx$

2. $\int (x^3 - 4\sin x)dx$

3. $\int \sqrt{x}(x^2 - 3)dx$

4. $\int \left(1 - \frac{1}{\sqrt[3]{x^4}}\right) dx$

5. $\int (x^5 - 6x^2 + x - 1)dx$

6. $\int \left(\frac{1 + \cos^2 x}{\cos^2 x}\right) dx$

7. $\int \frac{e^{2x}}{e^x} dx$

8. $\int \left(\sin x + \frac{1}{2}x\right) dx$

8.4 Definite Integrals

In Section 5.3, you learnt the integral of a function, called indefinite integral, which is another function. However, if we are given the interval in which the integral is to be evaluated, we find **definite integral** which is a number. The notation

$$\int_a^b f(x)dx$$

means the integral of $f(x)$ is to be evaluated in the interval $[a, b]$.

The numbers a and b are called **limits of integration**. a is the **lower limit** while b is the **upper limit**. We will use Theorem 8.11 in evaluating definite integrals. Theorem 8.11 is also called **fundamental theorem of calculus**.

Theorem 8.11 (Fundamental Theorem of Calculus)**.** *Suppose that* $\int f(x)dx = F(x)$*, i.e.,* $F(x)$ *is an anti-derivative of* $f(x)$*. If* $f(x)$ *is continuous on* $[a, b]$ *then*

$$\int_a^b f(x)dx = F(b) - F(a).$$

By Theorem 8.11, the definite integral is found by following these steps:

i. Find the integral $F(x)$ of $f(x)$ (of course without adding a constant C).

ii. Substitute the upper limit b for x in $F(x)$, i.e., find $F(b)$.

iii. Substitute the lower limit a for x in $F(x)$, i.e., find $F(a)$.

iv. The definite integral is $F(b) - F(a)$.

We state properties of definite integrals in Theorem 8.12.

Theorem 8.12. *Let f and g be functions and let a, b and c be constants. Then*

i. $\int_a^b k dx = k(b-a)$.

ii. $\int_a^b kf(x)dx = k\int_a^b f(x)dx$.

iii. $\int_a^a f(x)dx = 0$.

iv. $\int_a^b f(x)dx = -\int_b^a f(x)dx$.

v. $\int_a^b [f(x) \pm g(x)]dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$.

vi. $\int_a^c f(x)dx + \int_c^b f(x)dx = \int_a^b f(x)dx$, *provided that* $a < c < b$.

Example 8.13.

Evaluate the following integrals

1. $\int_1^4 (3x^2 - 2x + 6)dx$
2. $\int_0^\pi \sec x \tan x dx$

Solution

1. i. $F(x) = \int (3x^2 - 2x + 6)dx = x^3 - x^2 + 6x$.
   ii. $F(4) = 4^3 - 4^2 + 6(4) = 64 - 16 + 24 = 72$.
   iii. $F(1) = 1^3 - 1^2 + 6(1) = 1 - 1 + 6 = 6$.
   iv. $\int_1^4 (3x^2 - 2x + 6)dx = F(4) - F(1) = 72 - 6 = 66$.

2. i. $F(x) = \int \sec x \tan x dx = \sec x = \dfrac{1}{\cos x}$.
   ii. $F(\pi) = \dfrac{1}{\cos \pi} = \dfrac{1}{(-1)} = -1$.
   iii. $F(0) = \dfrac{1}{\cos 0} = \dfrac{1}{1} = 1$.

iv. $\int_0^{\pi} \sec x \tan x dx = F(\pi) - F(0) = -1 - 1 = -2.$

The steps followed in finding definite integrals can be combined, in order to save time and space, as in Example 8.14 below.

Example 8.14.

Evaluate each integral below.

1. $\int_1^2 (4x^5 + 6x^2 - 4x)dx$
2. $\int_0^1 (2x^4 + 3x^3 - 21)dx$

Solution

1.
$$\begin{aligned}
\int_1^2 (4x^5 + 6x^2 - 4x)dx &= \left[\frac{2}{3}x^6 + 2x^3 - 2x^2\right]_1^2 \\
&= \left(\frac{2}{3}(2)^6 + 2(2)^3 - 2(2)^2\right) - \left(\frac{2}{3}(1)^6 + 2(1)^3 - 2(1)^2\right) \\
&= \left(\frac{128}{3} + 16 - 8\right) - \left(\frac{2}{3} + 2 - 2\right) \\
&= \frac{128}{3} + 16 - 8 - \frac{2}{3} \\
&= \frac{126}{3} + 8 \\
&= 42 + 8 \\
&= 50.
\end{aligned}$$

2.
$$\begin{aligned}
\int_0^1 (2x^4 + 3x^3 - 21)dx &= \left[\frac{2}{5}x^5 + \frac{3}{4}x^2 - 21x\right]_0^1 \\
&= \left(\frac{2}{3}(1)^5 + \frac{3}{4}(1)^2 - 21(1)\right) - \left(\frac{2}{3}(0)^5 + \frac{3}{4}(0)^2 - 21(0)\right) \\
&= \frac{2}{5} + \frac{3}{4} - 21 \\
&= \frac{8}{20} + \frac{15}{20} - \frac{420}{20} \\
&= -\frac{397}{20}.
\end{aligned}$$

Activity 8.15.

Evaluate each definite integral below.

1. $\int_{-1}^{2}(x^2 - 2x + 3)dx$
2. $\int_{0}^{1} x(x^2 + 1)dx$
3. $\int_{1}^{4}\left(x^2 + \frac{1}{2\sqrt{x}}\right)dx$
4. $\int_{0}^{2} x(2 - 3x)^2 dx$
5. $\int_{1}^{2} \frac{x^2 + 1}{\sqrt{x}}dx$
6. $\int_{-2}^{2}(x^4 - 3x^2 + 5)dx$
7. $\int_{4}^{4}(x - 5x^4)dx$
8. $\int_{4}^{1}\left(x^2 + \frac{1}{2\sqrt{x}}\right)dx$
9. $\int_{0}^{\pi} \sin x dx$

Unit Summary

In this unit, you have learnt calculation of integrals of basic functions. You have learnt definite and indefinite integrals of polynomials, exponential functions and trigonometric functions.

Activity 8.16 (Unit Test).

Evaluate each integral below.

1. $\int(3x^2 - \sqrt{5x} + 2)dx$
2. $\int\left(\frac{3}{\sqrt{x}} - \frac{2}{x^2}\right)dx$

3. $\int \frac{x^2+3x-2}{\sqrt{x}}dx$

4. $\int (x^3-2x^2)\left(\frac{1}{x}-5\right)dx$

5. $\int_3^7 5dx$

6. $\int \left(x^{1/2}-3x^{2/3}+6\right)dx$

7. $\int_0^3 (6x^3-15x^2-3x)dx$

8. $\int_{-1}^2 [(x+2)-x^2]dx$

9. $\int_0^2 (x^3-5x^2+6x)dx$

10. $\int (2\cos x+3e^x)dx$

11. $\int_0^{\pi/2} 3\sin x dx$

Unit Answers

Suggested Solutions to Activity 8.10

1. $\tan x + C$ 2. $\frac{x^4}{4}+4\cos x+C$ 3. $\frac{2}{7}x^{7/2}-2\sqrt{x^3}+C$ 4. $x+\frac{3}{\sqrt[3]{x}}+C$

5. $\frac{x^6}{6}-2x^3+\frac{x^2}{2}-x+C$ 6. $\tan x+x+C$ 7. e^x+C 8. $-\cos x+\frac{1}{4}x^2+C$

Suggested Solutions to Activity 8.15

1. $\int_{-1}^2 (x^2-2x+3)dx = 9$ 2. $\int_0^1 x(x^2+1)dx = \frac{7}{12}$

3. $\int_1^4 \left(x^2+\frac{1}{2\sqrt{x}}\right)dx = 22$ 4. $\int_0^2 x(2-3x)^2dx = 12$

7. $\int_4^4 (x-5x^4)dx = 0$ 8. $\int_4^1 \left(x^2+\frac{1}{2\sqrt{x}}\right)dx = -22$ 9. $\int_0^\pi \sin x dx = 2$

Suggested Solutions to Unit Test

1. $\int(3x^2 - \sqrt{5x} + 2)dx = x^3 - \frac{2}{3}x\sqrt{5x} + 2x + C$

2. $\int \left(\frac{3}{\sqrt{x}} - \frac{2}{x^2}\right) dx = 6\sqrt{x} + \frac{2}{x} + C$

3. $\int \frac{x^2 + 3x - 2}{\sqrt{x}} dx = \frac{2}{5}x^2\sqrt{x} + 2x\sqrt{x} - 4\sqrt{x} + C$

4. $\int (x^3 - 2x^2)\left(\frac{1}{x} - 5\right) dx = -\frac{5}{4}x^4 + \frac{11}{3}x^3 - x^2 + C$

5. $\int_3^7 5dx = 20$

6. $\int \left(x^{1/2} - 3x^{2/3} + 6\right) dx = \frac{2}{3}x^{3/2} - \frac{9}{5}x^{5/3} + 6x + C$

7. $\int_0^3 (6x^3 - 15x^2 - 3x)dx = -18$

8. $\int_{-1}^2 [(x + 2) - x^2]dx = \frac{9}{2}$

9. $\int_0^2 (x^3 - 5x^2 + 6x)dx = \frac{8}{3}$

10. $\int(2\cos x + 3e^x)dx = 2\sin x + 3e^x + C$

11. $\int_0^{\pi/2} 3\sin x dx = 3$

Unit 9

Integration Techniques

9.1 Introduction

Having introduced integration in Unit 5, we will look at further techniques of integration in this unit. These techniques are the substitution rule, integration by parts and integration by partial fractions. Under substitution rule, we will also consider integrals of the form $\int \frac{g'(x)}{g(x)} dx$.

9.2 Objectives

By the end of this unit, you should be able to:

- use the substitution rule to find integrals of appropriate functions.
- use integration by parts to find integrals of appropriate functions.
- use integration by partial fractions to find integrals of appropriate functions.

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- partial fractions
- the substitution rule
- integration by parts
- integration by partial fractions
- substitution rule for definite integrals
- the integral $\dfrac{g'(x)}{g(x)}dx$
- improper fractions
- linear factor
- repeated factor
- irreducible quadratic factor

9.3 The Substitution Rule

Integrals for some functions are difficult to evaluate in the function's original representation. In some cases we have to make a **substitution**, i.e., we have to change the variable. We calculate the integral and then change back to the original variable. The change of the variable is made in such a way that the integral looks different from the original one. The most important thing is that the integral becomes easy to evaluate than it was before changing the variable.

The **Substitution Rule** is the method used to find integrals of the form $\int f(g(x))g'(x)dx$. For example, in $\int 2x\sqrt{x^2+4}dx$ notice that $\dfrac{d}{dx}[x^2+4] = 2x$. We state the rule in Theorem 9.1.

Theorem 9.1 (The Substitution rule)**.** *If $u = g(x)$ is a differentiable function whose range is an interval I and if f is continuous on I, then*

$$\int f(g(x))g'(x)dx = \int f(u)du.$$

By Theorem 9.1, to find the integral

$$\int f(g(x))g'(x)dx,$$

we substitute $u = g(x)$ and $du = g'(x)dx$ to get

$$\int f(u)du,$$

i.e., we change the variable from x to u. We calculate the integral with respect to u after which we revert back to the original variable x.

For example, to find the integral

$$\int 2x\sqrt{x^2+4}dx,$$

we first put $u = x^2 + 4$. Differentiating we have $du = 2xdx$.

Therefore

$$\begin{aligned}\int 2x\sqrt{x^2+4}dx &= \int \left(\sqrt{x^2+4}\right)(xdx)\\ &= \int \sqrt{u}du\\ &= \int u^{1/2}du\\ &= \frac{2}{3}u^{3/2} + C.\end{aligned}$$

Writing it in terms of the original variable x we have

$$\frac{2}{3}u^{3/2} + C = \frac{2}{3}(x^2+4)^{3/2} + C.$$

Example 9.2.

Evaluate each integral below.

1. $\int \frac{4x}{\sqrt{2x^2+1}}dx$
2. $\int \sin(7x-3)dx$
3. $\int 2xe^{x^2-5}dx$

4. $\int \frac{\sin\sqrt{x}}{\sqrt{x}}dx$

5. $\int e^{\cos x}\sin x dx$

Solution

1. Let $u = 2x^2 + 1$. Then $du = 4xdx$.

   Therefore

$$\begin{aligned}\int \frac{4x}{\sqrt{2x^2+1}}dx &= \int \left(\frac{1}{\sqrt{2x^2+1}}\right) 4xdx \\ &= \int \frac{1}{\sqrt{u}}du \\ &= \int u^{-1/2}du \\ &= \frac{u^{1/2}}{1/2} + C \\ &= 2\sqrt{u} + C \\ &= 2\sqrt{2x^2+1} + C.\end{aligned}$$

2. Let $u = 7x - 3$. Then $du = 7dx$ giving $\frac{du}{7} = dx$.

   Therefore

$$\begin{aligned}\int \sin(7x-3)dx &= \int (\sin u)\frac{du}{7} \\ &= \frac{1}{7}\int \sin u du \\ &= -\frac{1}{7}\cos u + C \\ &= -\frac{1}{7}\cos(7x-3) + C.\end{aligned}$$

3. Let $u = x^2 - 5$. Then $du = 2xdx$.

Therefore

$$\begin{aligned}\int 2xe^{x^2-5}dx &= \int \left(e^{x^2-5}\right) 2xdx \\ &= \int e^u du \\ &= e^u + C \\ &= e^{x^2-5} + C.\end{aligned}$$

4. Let $u = \sqrt{x} = x^{1/2}$. Then $du = \frac{1}{2}x^{-1/2}dx = \frac{1}{2\sqrt{x}}dx$ giving $2du = \frac{dx}{\sqrt{x}}$.

   Therefore

$$\begin{aligned}\int \frac{\sin\sqrt{x}}{\sqrt{x}}dx &= \int (\sin\sqrt{x})\frac{dx}{\sqrt{x}} \\ &= \int \sin u(2du) \\ &= 2\int \sin u du \\ &= -2\cos u + C \\ &= -2\cos\sqrt{x} + C.\end{aligned}$$

5. Let $u = \cos x$. Then $du = -\sin x dx$ giving $-du = \sin x dx$.

   Therefore

$$\begin{aligned}\int e^{\cos x}\sin x dx &= \int e^{\cos x}\sin x dx \\ &= \int e^u(-du) \\ &= -\int e^u du \\ &= -e^u + C \\ &= -e^{\cos x} + C.\end{aligned}$$

9.3.1 The Substitution Rule for Definite Integrals

Recall that the definite integral

$$\int_a^b f(x)dx$$

is the number $F(b)-F(a)$ where $F(x)$ is the integral $\int f(x)dx$. To evaluate definite integrals using the substitution rule, we will change the variable from x to u as before but we do not need to revert back to x at the end. Instead we will change the limits of integration from a and b to $g(a)$ and $g(b)$ respectively (see Theorem 9.3). Then the new limits of integration, $g(a)$ and $g(b)$, will be substituted into the integral $\int f(u)du$.

Theorem 9.3. *If g' is continuous on $[a, b]$ and if f is continuous on the range of $u = g(x)$, then*

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Example 9.4.

Evaluate each integral below.

1. $\int_1^{e^\pi} \dfrac{\sin(\ln x)}{x} dx$
2. $\int_0^1 (x+1)e^{x^2+2x} dx$
3. $\int_0^5 x^3\sqrt{x^4+1}dx$

Solution

1. Let $u = \ln x$. Then $du = \dfrac{1}{x}dx = \dfrac{dx}{x}$.

   Now when $x = 1$, $u = \ln 1 = 0$ and when $x = e^\pi$, $u = \ln e^\pi = \pi \ln e = \pi$ (since $\ln e = 1$).

   Therefore

$$\begin{aligned}
\int_1^{e^\pi} \frac{\sin(\ln x)}{x} dx &= \int_1^{e^\pi} \sin(\ln x)\frac{dx}{x} \\
&= \int_0^\pi \sin u du \\
&= [-\cos u]_0^\pi \\
&= -\cos\pi - (-\cos 0) \\
&= -(-1) - (1) \\
&= 2.
\end{aligned}$$

2. Let $u = x^2 + 2x$. Then $du = (2x+2)dx = 2(x+1)dx$ giving $\dfrac{du}{2} = (x+1)dx$.

   Now, when $x = 0$, $u = 0^2 + 2(0) = 0$ and when $x = 1$, $u = 1^2 + 2(1) = 3$.

   Therefore

$$\begin{aligned}
\int_0^1 (x+1)e^{x^2+2x}dx &= \int_0^1 \left(e^{x^2+2x}\right)(x+1)dx \\
&= \int_0^3 e^u \frac{du}{2} \\
&= \frac{1}{2}\int_0^3 e^u du \\
&= \frac{1}{2}\left[e^u\right]_0^3 \\
&= \frac{1}{2}[e^3 - e^0] \\
&= \frac{1}{2}\left(e^3 - 1\right).
\end{aligned}$$

3. Let $u = x^4 + 1$. Then $du = 4x^3dx$ giving $\dfrac{du}{4} = x^3dx$.

   Now when $x = 0$, $u = 0^4 + 1 = 1$ and when $x = 5$, $u = 5^4 + 1 = 626$.

   Therefore

$$\begin{aligned}
\int_0^5 x^3\sqrt{x^4+1}dx &= \int_0^5 \left(\sqrt{x^4+1}\right)x^3dx \\
&= \int_1^{626} \sqrt{u}\frac{du}{u} \\
&= \frac{1}{4}\int_1^{626} u^{1/2}du \\
&= \frac{1}{4}\left[\frac{2}{3}u^{3/2}\right]_1^{626} \\
&= \frac{2}{12}\left[626^{3/2} - 1^{3/2}\right] \\
&= \frac{1}{6}\left[\sqrt{626^3} - 1\right] \\
&\approx 2610.25.
\end{aligned}$$

9.3.2 The Integral $\int \frac{g'(x)}{g(x)}dx$

We consider the integral of the fractional function in which the numerator is the derivative of the denominator. The simplest form of this function is

$$\frac{1}{x}.$$

Recall, from Unit 3, that

$$\frac{d}{dx}[\ln x] = \frac{1}{x}.$$

Working in reverse we have

$$\int \frac{1}{x}dx = \ln x + C.$$

Also recall that if g is a differentiable function, then

$$\frac{d}{dx}[\ln g(x)] = \frac{g'(x)}{g(x)}.$$

Now, working in reverse, we must have

$$\int \frac{g'(x)}{g(x)}dx = \ln g(x) + C.$$

It works as follows. Put $u = g(x)$ and then $du = g'(x)dx$ to obtain

$$\int \frac{g'(x)}{g(x)}dx = \int \frac{du}{u} = \ln u + C = \ln g(x) + C.$$

Recall that the domain of a logarithmic function is the set of positive numbers only. Therefore $\ln g(x)$ is true only for those values of x such that $g(x)$ is positive. In general, we will write

$$\int \frac{g'(x)}{g(x)}dx = \ln |g(x)| + C$$

where $\ln |g(x)|$ makes sure that we do not find logarithms of non-positive numbers ($|,|$ is the absolute value function discussed last semester).

Similarly we have

$$\int \frac{1}{x}dx = \ln |x| + C.$$

Example 9.5.

Evaluate each integral below.

1. $\int \cot x dx$
2. $\int \frac{2x^4}{x^5+1} dx$
3. $\int \frac{3x^4+12x^3+6}{x^5+5x^4+10x+12} dx$

Solution

1. We note that $\int \cot x dx = \int \frac{\cos x}{\sin x} dx$.

   Let $u = \sin x$. Then $du = \cos x dx$.

   Therefore

$$\begin{aligned}\int \cot x dx &= \int \frac{\cos x}{\sin x} dx \\ &= \int \frac{du}{u} \\ &= \ln|u| + C \\ &= \ln|\sin x| + C.\end{aligned}$$

2. Let $u = x^5 + 1$. Then $du = 5x^4 dx$ giving $\frac{du}{u} = x^4 dx$.

   Therefore

$$\begin{aligned}\int \frac{2x^4}{x^5+1} dx &= \int 2\left(\frac{1}{x^5+1}\right) x^4 dx \\ &= 2\int \left(\frac{1}{u}\right) \frac{du}{5} \\ &= \frac{2}{5}\int \frac{du}{u} \\ &= \frac{2}{5}\ln|u| + C \\ &= \frac{2}{5}\ln|x^5+1| + C.\end{aligned}$$

3. Let $u = x^5+5x^4+10x+12$. Then $du = (5x^4+20x^3+10)dx = 5(x^4+4x^3+2)dx$ giving $\dfrac{du}{5} = (x^4+4x^3+2)$.

   Therefore

$$\begin{aligned}
\int \frac{3x^4+12x^3+6}{x^5+5x^4+10x+12}dx &= \int 3\left(\frac{1}{x^5+5x^4+10x+12}\right)(x^4+4x^3+2)dx \\
&= 3\int \left(\frac{1}{u}\right)\left(\frac{du}{5}\right) \\
&= \frac{3}{5}\int \frac{du}{u} \\
&= \frac{3}{5}\ln|u| + C \\
&= \frac{3}{5}\ln|x^5+5x^4+10x+12| + C.
\end{aligned}$$

Activity 9.6.

Evaluate each integral below.

1. $\int_0^1 \dfrac{dx}{3x+5}$
2. $\int_3^4 \dfrac{6x-3}{4x^2-4x+1}dx$
3. $\int \dfrac{10x^3-5x}{\sqrt{x^4-x^2+6}}dx$
4. $\int \dfrac{dx}{x\ln x}$
5. $\int \dfrac{\ln x^2}{x}dx$
6. $\int (x+1)(x-2)^9 dx$ [Let $u = x-2$. Then $u+3 = x+1$.]
7. $\int \dfrac{2x\ln(x^2+1)}{x^2+1}dx$
8. $\int \tan x dx$
9. $\int \dfrac{3x}{x^2+7}dx$

9.4 Integration by Parts

Recall (the product rule of differentiation) that if f and g are differentiable functions, then

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + g'(x)f(x). \tag{9.1}$$

Integrating both sides of Equation 9.1, gives

$$\begin{aligned} f(x)g(x) &= \int [f'(x)g(x) + g'(x)f(x)]dx \\ &= \int f'(x)g(x)dx + \int g'(x)f(x)dx. \end{aligned}$$

Rearranging, we have

$$\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx. \tag{9.2}$$

In Equation 9.2 we have what is called the **formula for integration by parts**. If we put $u = f(x)$ and $v = g(x)$ in 9.2, we have the following form which is easy to remember;

$$\int uv' = uv - \int vu'.$$

The key in integration by parts is in making right substitutions for both u and v'. Generally, we choose to be u the function whose derivative u' is simpler than u itself. v' is the function whose integral is found easily.

Example 9.7.

Evaluate each integral below.

1. $\int x^3 \ln x dx$
2. $\int \sin x \ln(\cos x) dx$
3. $\int e^x \sin x dx$
4. $\int_0^1 x^2 e^{-x} dx$

Solution

1. Let $u = \ln x$ and $v' = x^3$. Then $u' = \frac{d}{dx}[\ln x] = \frac{1}{x}dx$ and $v = \int x^3 dx = \frac{x^4}{4}$.

   Therefore

$$\begin{aligned}
\int uv' &= uv - \int vu' \\
\int x^3 \ln x dx &= \frac{x^4}{4}(\ln x) - \int \left(\frac{x^4}{4}\right)\left(\frac{1}{x}\right) dx \\
&= \frac{x^4}{4}(\ln x) - \frac{1}{4}\int x^3 dx \\
&= \frac{x^4}{4}(\ln x) - \frac{1}{16}x^4 + C.
\end{aligned}$$

2. Let $u = \ln(\cos x)$ and $v' = \sin x$.

   Then $u' = \frac{d}{dx}[\ln(\cos x)] = \frac{-\sin x}{\cos x}dx = -\tan x dx$ and
   $v = \int \sin x dx = -\cos x$.

   Therefore

$$\begin{aligned}
\int uv' &= uv - \int vu' \\
\int \sin x \ln(\cos x) dx &= -\cos x \ln(\cos x) - \int (-\cos x)(-\tan x) dx \\
&= -\cos x \ln(\cos x) - \int \cos x \left(\frac{\sin x}{\cos x}\right) dx \\
&= -\cos x \ln(\cos x) - \int \sin x dx \\
&= -\cos x \ln(\cos x) - (-\cos x) + C \\
&= -\cos x \ln(\cos x) + \cos x + C.
\end{aligned}$$

3. Let $u = e^x$ and $v' = \sin x$. Then $u' = \frac{d}{dx}[e^x] = e^x dx$ and
   $v = \int \sin x dx = -\cos x$.

   Therefore

$$\begin{aligned}
\int uv' &= uv - \int vu' \\
\int e^x \sin x dx &= -e^x \cos x - \int e^x(-\cos x) dx \\
&= -e^x \cos x + \int e^x \cos x dx.
\end{aligned}$$

We apply integration by parts again to get the integral $\int e^x \cos x dx$.

Let $u = e^x$ and $v' = \cos x$. Then $u' = \dfrac{d}{dx}[e^x] = e^x dx$ and
$v = \int \cos x dx = \sin x$.
So

$$\int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx.$$

Therefore

$$\int e^x \sin x dx = -e^x \cos x + e^x \sin x - \int e^x \sin x dx.$$

Hence

$$2\int e^x \sin x dx = -e^x \cos x + e^x \sin x = e^x(\sin x - \cos x)$$

giving

$$\int e^x \sin x dx = \frac{1}{2}e^x(\sin x - \cos x).$$

4. Let $u = x^2$ and $v' = e^{-x}$. Then $u' = 2x dx$ and $v = -e^{-x}$.

$$\begin{aligned}\int x^2 e^{-x} dx &= -x^2 e^{-x} - \int 2x(-e^{-x}) dx \\ &= -x^2 e^{-x} + 2\int x e^{-x} dx.\end{aligned}$$

For $\int x e^{-x} dx$, let $u = x$ and $v' = e^{-x}$. Then $u' = 1$ and $v = -e^{-x}$.
So

$$\begin{aligned}\int x e^{-x} dx &= -x e^{-x} - \int (1)(-e^{-x}) dx \\ &= -x e^{-x} + \int e^{-x} dx \\ &= -x e^{-x} - e^{-x}.\end{aligned}$$

Hence

$$\begin{aligned}\int_0^1 x^2 e^{-x} dx &= \left[-x^2 e^{-x} - 2x e^{-x} - 2e^{-x}\right]_0^1 \\ &= -e^{-1} - 2e^{-1} - 2e^{-1} + 2 \\ &= 2 - \frac{5}{e}.\end{aligned}$$

Activity 9.8.

Evaluate integrals below.

1. $\int x^2 e^{2x} dx$
2. $\int e^{2x} \cos x dx$
3. $\int x e^x dx$
4. $\int x \sec^2 x dx$
5. $\int x \sec x \tan x dx \quad$ [Let $u = x$]
6. $\int x^5 \ln x dx$
7. $\int e^x \cos x dx$
8. $\int x^2 \sin x dx$
9. $\int x \ln x^2 dx \quad$ [Let $u = \ln x^2$]

9.5 Integration by Partial Fractions

In Subsection 9.3.2, we discussed how to find integrals of the form

$$\int \frac{g'(x)}{g(x)} dx,$$

i.e., integrals of fractions in which the numerator is the derivative of the denominator. In this section we will look at integrals of the form

$$\int \frac{p(x)}{q(x)} dx \tag{9.3}$$

where both $p(x)$ and $q(x)$ are **polynomials** which are not related as above. The function $\dfrac{p(x)}{q(x)}$ is called a **rational function**.

The integral of the form 9.3 will be found by first decomposing

$$\frac{p(x)}{q(x)}$$

into smaller fractions called **partial fractions**. Then we will integrate each partial fraction and put the results together. Generally the partial fractions will be easier to integrate than the whole function in its original form.

9.5.1 Strategy for Decomposing Rational Functions into Partial Fractions

Consider a rational function $\dfrac{p(x)}{q(x)}$.

1. **Divide if improper**: Before expressing $\dfrac{p(x)}{q(x)}$ into partial fractions, we have to make sure that the **degree** (highest power of x) of $q(x)$ is greater than that of $p(x)$. If the degree of $p(x)$ is greater than or equal to that of $q(x)$, then we have an **improper fraction**. In this case, we divide $q(x)$ into $p(x)$ to obtain
$$\frac{p(x)}{q(x)} = f(x) + \frac{g(x)}{q(x)}$$
where $f(x)$ is the quotient and $g(x)$ is the remainder.

2. **Factorise the denominator**: Completely factorise the denominator into factors of the form
$$(ax+b)^m \text{ or } (ax^2+bx+c)^n$$
where ax^2+bx+c is an **irreducible** quadratic factor, i.e., a quadratic factor which can not be factorised into linear factors.

3. **Linear factors**:

   i. A linear factor $ax+b$ in the denominator gives rise to a partial fraction of the form
   $$\frac{A}{ax+b}$$
   where $A \in \mathbb{R}$.

   ii. A linear factor $ax+b$, repeated r times in the denominator, gives rise to partial fractions of the form
   $$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + ... + \frac{A_r}{(ax+b)^r}$$
   where $A_1, A_2, ..., A_r \in \mathbb{R}$.

4. **Quadratic factors**:

   i. A quadratic factor ax^2+bx+c in the denominator gives rise to a partial fraction of the form
   $$\frac{Ax+B}{ax^2+bx+c}$$
   where $A, B \in \mathbb{R}$.

ii. A quadratic factor $ax^2 + bx + c$, repeated r times in the denominator, gives rise to partial fractions of the form

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + ... + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$$

where $A_1, A_2, ..., A_r, B_1, B_2, ..., B_r \in \mathbb{R}$.

Table 9.1 is a summary of the preceding discussion.

Factor in Denominator	**Factor Name**	**Corresponding Partial Fraction(s**
$ax + b$	linear Factor	$\frac{A}{ax + b}$
$(ax + b)^r$	repeated linear factor	$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + ... + \frac{A_r}{(ax + b)^r}$
$ax^2 + bx + c$	quadratic factor	$\frac{Ax + B}{ax^2 + bx + c}$
$ax^2 + bx + c$	repeated quadratic factor	$\frac{A_1x + B_1}{ax^2 + bx + c} + ... + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$

Table 9.1: Partial Fraction Decomposition

Example 9.9.

Evaluate each integral below.

1. $\int \frac{3x + 11}{x^2 - x - 6} dx$
2. $\int \frac{dx}{x^2 - 1}$
3. $\int \frac{dx}{(x - 1)^2(x + 1)}$
4. $\int \frac{x}{x^4 - 16} dx$
5. $\int \frac{2x^3 + x^2 - 7x + 7}{x^2 + x - 2} dx$

Solution

1. Factorising the denominator we have

$$\frac{3x+11}{x^2-x-6} = \frac{3x+11}{(x-3)(x+2)}.$$

Now lets decompose this into partial fractions.

Suppose that

$$\frac{3x+11}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2} = \frac{A(x+2)+B(x-3)}{(x-3)(x+2)},$$

i.e.,

$$3x+11 = A(x+2)+B(x-3).$$

Let $x=-2$. Then

$$3(-2)+11 = A(-2+2)+B(-2-3) \Rightarrow -6+11 = -5B \Rightarrow 5 = -5B$$

giving $B=-1$.

Let $x=3$. Then

$$3(3)+11 = A(3+2)+B(3-3) \Rightarrow 20 = 5A \Rightarrow A = 4.$$

Therefore

$$\frac{3x+11}{(x-3)(x+2)} = \frac{4}{x-3} - \frac{1}{x+2}.$$

Integrating we have

$$\begin{aligned}\int \frac{3x+11}{(x-3)(x+2)}dx &= \int \left[\frac{4}{x-3} - \frac{1}{x+2}\right] dx \\ &= \int \frac{4dx}{x-3} - \int \frac{dx}{x+2} \\ &= 4\ln|x-3| - \ln|x+2| + C \\ &= \ln\left|\frac{(x-3)^4}{x+2}\right| + C.\end{aligned}$$

Notice that we have used laws of logarithms in coming up with the last step.

2. Factorising the denominator we have

$$\frac{1}{x^2-1} = \frac{1}{(x+1)(x-1)}.$$

Suppose

$$\frac{1}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1} = \frac{A(x-1)+B(x+1)}{(x+1)(x-1)},$$

i.e.,

$$1 = A(x-1) + B(x+1).$$

Let $x = 1$. Then $1 = A(1-1) + B(1+1) \Rightarrow 1 = 2B \Rightarrow B = \frac{1}{2}$.

let $x = -1$. Then $1 = A(-1-1) + B(-1+1) \Rightarrow 1 = -2A \Rightarrow A = -\frac{1}{2}$.

Therefore

$$\frac{1}{(x+1)(x-1)} = \frac{-1/2}{x+1} + \frac{1/2}{x-1} = \frac{1}{2}\left[\frac{1}{x-1} - \frac{1}{x+1}\right].$$

Integrating we have

$$\begin{aligned}
\int \frac{dx}{x^2-1} &= \int \frac{1}{(x+1)(x-1)} \\
&= \frac{1}{2}\int \left[\frac{1}{x-1} - \frac{1}{x+1}\right] dx \\
&= \frac{1}{2}\left[\int \frac{dx}{x-1} - \int \frac{dx}{x+1}\right] \\
&= \frac{1}{2}\left[\ln|x-1| - \ln|x+1|\right] + C.
\end{aligned}$$

3. Suppose

$$\frac{dx}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1} = \frac{A(x-1)(x+1)+B(x+1)+C(x-1)^2}{(x-1)^2(x+1)},$$

i.e.,

$$1 = A(x-1)(x+1) + B(x+1) + C(x-1)^2.$$

If we put $x = 1$ we get $1 = 2B$ which gives $B = \frac{1}{2}$.

If we put $x = -1$ we have $1 = C(-1-1)^2 \Rightarrow 1 = 4C$ giving $C = \frac{1}{4}$.

Equating constant terms we have

$$1 = -A + B + C \Rightarrow 1 = -A + \frac{1}{2} + \frac{1}{4} \Rightarrow A = -\frac{1}{4}.$$

Therefore

$$\frac{dx}{(x-1)^2(x+1)} = \frac{-1/4}{x-1} + \frac{1/2}{(x-1)^2} + \frac{1/4}{x+1} = \frac{1}{4}\left[-\frac{1}{x-1} + \frac{2}{(x-1)^2} + \frac{1}{x+1}\right].$$

Integrating we have

$$\begin{aligned}\int \frac{dx}{(x-1)^2(x+1)} &= \frac{1}{4}\int\left(-\frac{1}{x-1} + \frac{2}{(x-1)^2} + \frac{1}{x+1}\right)dx \\ &= \frac{1}{4}\left(-\ln|x-1| - \frac{2}{x-1} + \ln|x+1|\right) + C \\ &= \frac{1}{4}\left(\ln\left|\frac{x+1}{x-1}\right| - \frac{2}{x-1}\right) + C.\end{aligned}$$

4. Factorising the denominator we have

$$\frac{x}{x^4-16} = \frac{x}{(x+2)(x-2)(x^2+4)}.$$

Suppose

$$\begin{aligned}\frac{x}{x^4-16} &= \frac{A}{x+2} + \frac{B}{x-2} + \frac{Cx+D}{x^2+4} \\ &= \frac{A(x-2)(x^2+4) + B(x+2)(x^2+4) + (Cx+D)(x+2)(x-2)}{(x+2)(x-2)(x^2+4)},\end{aligned}$$

i.e.,

$$x = A(x-2)(x^2+4) + B(x+2)(x^2+4) + (Cx+D)(x+2)(x-2).$$

Let $x = 2$. Then $2 = (2+2)(4+4)B \Rightarrow 2 = 32B$ giving $B = \frac{1}{16}$.

Let $x = -2$. Then $-2 = (-2-2)(4+4)A \Rightarrow -2 = -32A$ giving $A = \frac{1}{16}$.

Equating coefficients of x^3 we have $0 = A + B + C \Rightarrow 0 = 1/16 + 1/16 + C$ giving $C = -\frac{2}{16} = -\frac{1}{8}$.

Equating constant terms we have

$0 = -8A + 8B - 4D \Rightarrow 0 = -8\left(\frac{1}{16}\right) + 8\left(\frac{1}{16}\right) - 4D$ giving $D = 0$.

Therefore

$$\frac{x}{x^4-16} = \frac{1/16}{x+2} + \frac{1/16}{x-2} + \frac{-(1/8)x}{x^2+4} = \frac{1}{16}\left[\frac{1}{x+2} + \frac{1}{x-2} - \frac{2x}{x^2+4}\right].$$

Integrating we have

$$\begin{aligned}\frac{x}{x^4-16} &= \frac{1}{16}\int\left[\frac{1}{x+2}+\frac{1}{x-2}-\frac{2x}{x^2+4}\right]dx\\ &= \frac{1}{16}\left[\ln|x+2|+\ln|x-2|-\ln|x^2+4|\right]+C\\ &= \frac{1}{16}\ln\left|\frac{(x+2)(x-2)}{x^2+4}\right|+C.\end{aligned}$$

5. Since the degree of the numerator is greater than that of the denominator, we first need to use long division to simplify the integrand. Please verify that after long division we get

$$\frac{2x^3+x^2-7x+7}{x^2+x-2} = (2x-1)+\frac{-2x+5}{x^2+x-2}.$$

Now we need to express

$$\frac{-2x+5}{x^2+x-2}$$

into partial fractions.

Factorising the denominator we have

$$\frac{-2x+5}{x^2+x-2} = \frac{-2x+5}{(x+2)(x-1)}.$$

Let

$$\frac{-2x+5}{(x+2)(x-1)} = \frac{A}{x+2}+\frac{B}{x-1} = \frac{A(x-1)+B(x+2)}{(x+2)(x-1)},$$

i.e.,

$$-2x+5 = A(x-1)+B(x+2).$$

Equating constant terms we have

$$5 = -A+2B.....(1)$$

Equating coefficients of x we have

$$-2 = A+B.....(ii)$$

Adding (i) and (ii) we have

$$\begin{array}{rrcrcr} & 5 & = & -A & + & 2B\\ + & -2 & = & A & + & B\\ \hline & 3 & = & & & 3B\end{array}$$

giving $B = 1$.

Substituting in (i) we have

$$5 = -A + 2(1) \text{ giving } A = -3.$$

Therefore

$$\frac{-2x+5}{(x+2)(x-1)} = \frac{-3}{x+2} + \frac{1}{x-1}.$$

Combining we have

$$\frac{2x^3+x^2-7x+7}{x^2+x-2} = (2x-1) - \frac{3}{x+2} + \frac{1}{x-1}.$$

Integrating we have

$$\begin{aligned}\int \frac{2x^3+x^2-7x+7}{x^2+x-2}dx &= \int \left[(2x-1) - \frac{3}{x+2} + \frac{1}{x-1}\right] dx \\ &= x^2 - x - \ln|x+2| + \ln|x-1| + C.\end{aligned}$$

Activity 9.10.

Evaluate each integral below.

1. $\int_1^2 \frac{x^4+1}{x^3+2x}dx$
2. $\int \frac{5x-3}{(x-2)(x-3)^2}dx$
3. $\int \frac{dx}{x^2+5x+6}$
4. $\int \frac{5x^2+20x+6}{x^3+2x^2+x}dx$
5. $\int \frac{2x-2}{(x^2+2x-15)(x+2)}dx$
6. $\int \frac{x^2-2x-5}{x^3-5x^2}dx$

7. $\int \frac{x^2+x-3}{(x+1)(x^2-2x+3)}$

Unit Summary

In this unit, you have learnt three techniques of integration: substitution rule, integration by parts and integration by partial fractions. You have learnt that the key in both substitution rule and integration by parts lies in making right substitutions. Under integration by partial fractions, you learnt how to express a rational function into partial fractions before performing integration.

Activity 9.11 (Unit Test).

1. Evaluate each integral below.

   (a) $\int_0^{\pi/4} \frac{\tan^3 x}{\cos^2 x} dx$

   (b) $\int \frac{4}{x(x^2+4)}$

   (c) $\int \frac{10(2x-1)}{(x+1)(x-2)(x+3)}$

   (d) $\int 3xe^{-x} dx$

   (e) $\int_0^2 xe^{x^2} dx$

   (f) $\int x^2 \ln x dx$

   (g) $\int x \cos 2x dx$

   (h) $\int \frac{(x^2-1)e^{x+1/x}}{x^2} dx$

   (i) $\int_2^4 \frac{36}{(x^2-2x+1)(x+5)} dx$

   (j) $\int \frac{3x+1}{2x^2-x-1} dx$

   (k) $\int \frac{dx}{\cos^2 x\sqrt{\tan x}}$

   (l) $\int e^{2x} \sin x dx$

   (m) $\int \frac{5}{x^2+x-6} dx$

(n) $\int e^x \sec^2(e^x)dx$

(o) $\int x^3 e^x dx$

2. Use the Substitution rule to verify the following integrals. [*Use trigonometric identities where necessary*].

   (a) $\int e^{\tan x} \sec^2 x dx = e^{\tan x} + C$

   (b) $\int \sin^3 x \cos x = \frac{1}{4} \sin x + C$

   (c) $\int \sqrt{\sin x} \cos^3 x dx = \frac{2}{3} \sin^{3/2} x - \frac{2}{7} \sin^{7/2} x + C$

   (d) $\int \cos^7 x dx = \sin x - \sin^3 x - \frac{1}{5} \sin^5 x - \frac{1}{6} \sin^6 x + c$

   (e) $\int \tan^3 x \sec^3 x dx = \frac{1}{5} \sec^5 x - \frac{1}{3} \sec^3 x + C$

3. Evaluate the following integrals by the substitution rule and trigonometric identities.

   (a) $\int \cot^3 x \csc^3 x dx$

   (b) $\int \frac{\sec x}{\cot^3 x} dx$

   (c) $\int \csc^4 x \cot^4 x dx$

Unit Answers

Suggested Solutions to Activity 9.6

1. $\frac{1}{3}(\ln 8 - \ln 5)$ 2. $\frac{3}{2}(\ln 7 - \ln 5)$ 3. $5\sqrt{x^4 - x^2 + 6} + C$

4. $\ln|\ln x| + C$ 5. $(\ln x)^2 + C$ 6. $\frac{1}{11}(x-2)^{11} + \frac{3}{10}(x-2)^{10} + C$

7. $\frac{1}{2}(\ln|x^2+1|)^2 + C$ 8. $\ln|\sec x| + C$ 9. $\frac{3}{2} \ln|x^2+7| + C$

Suggested Solutions to Activity 9.8

1. $\frac{1}{2}x^2 e^{2x} - \frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + C$ 2. $\frac{1}{5}e^{2x} \sin x + \frac{2}{5}e^{2x} \cos x + C$

3. $(x-1)e^x + \mathrm{C}$ 4. $x \tan x - \ln|\sec x| + C$ 5. $x \sec x - \ln|\sec x + \tan x| + C$

6. $\frac{1}{6}x^6 \ln x - \frac{1}{36}x^6 + C$ 7. $\frac{1}{2}e^x(\cos x + \sin x) + C$

8. $-x^2 \cos x + 2x \sin x + 2\cos x + C$ 9. $\frac{1}{2}x^2(\ln x^2 - 1) + C$

Suggested Solutions to Activity 9.10

1. $\frac{3(2-\ln 2)}{4}$ 2. $7\ln\left|\frac{x-2}{x+3}\right| - \frac{12}{x-3} + C$ 3. $\ln|x+2| - \ln|x+3| + C$

4. $6\ln|x| - \ln|x+1| - \frac{9}{x+1} + C$ 5. $-\frac{1}{2}\ln|x+5| + \frac{2}{5}\ln|x+2| + \frac{1}{10}\ln|x-3| + C$

6. $\frac{3}{5}\ln|x| + \frac{2}{5}\ln|x-5| - \frac{1}{x} + C$ 7. $-\frac{1}{2}\ln|x+1| + \frac{3}{4}\ln|x^2-2x+3| + C$

Suggested Solutions to Unit Test

1. (a) $\frac{1}{4}$ (b) $\ln|x| - \frac{1}{2}\ln|x^2+4| + C$ (c) $\ln\left|\frac{(x+1)^5(x-2)^2}{(x+3)^7}\right| + C$

   (d)] $-3xe^{-x} - 3e^{-x} + C$ (e) $\frac{1}{2}(e^4 - 1)$ (f) $\frac{1}{3}x^3\ln|x| - \frac{1}{9}x^3 + C$

   (g) $\frac{1}{2}x\sin 2x + \frac{1}{4}\cos 2x + C$ (h) $e^{x+1/x} + C$ (i) $4 + \ln\left(\frac{3}{7}\right)$

   (j) $\frac{4}{3}\ln|x-1| + \frac{1}{6}\ln|2x+1| + C$ (k) $2\sqrt{\tan x} + C$ (l) $\frac{1}{5}e^{2x}(2\sin x - \cos x) + C$

   (m) $\ln\left|\frac{x-2}{x+3}\right| + C$ (n) $\tan(e^x) + C$ (o) $(x^3 - 3x^2 + 6x - 6)e^x + C$

Unit 10

Application of Integration

10.1 Introduction

There are a number of applications of integration. In this unit, we will discuss one such application. This is the calculation of definite integrals to find area between a curve and the x-axis and the area between two curves. In both cases, the areas are bounded by two vertical lines $x = a$ and $x = b$.

10.2 Objectives

By the end of this unit, you should be able to:

- find the area enclosed by a curve, the x-axis and the vertical lines $x = a$ and $x = b$.
- find the area enclosed by two curves and the vertical lines $x = a$ and $x = b$.

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:

- areas

- curves
- x-axis

10.3 Areas

10.3.1 The Area between a Curve and the x-axis

The area between a curve and the x-axis, for x between a and b, is found by the following;

1. If the curve of $f(x)$ is entirely above the x-axis, then the area is

$$\int_a^b f(x)dx \quad \text{9square units).}$$

2. If the curve of $f(x)$ is entirely below the x-axis, then the area is

$$-\int_a^b f(x)dx \quad \text{(square units).}$$

3. If parts of the curve are below the x-axis and the other parts are above the x-axis, then the area is found by subtracting the integral for those parts of the curve that are below and adding the integrals for those lying above.

Example 10.1.

1. Find the area bounded by $y = x^2 - 4$, the x-axis and the lines $x = -1$ and $x = 2$.
2. Find the area bounded by $y = x^2 + x + 4$, the x-axis and the ordinates $x = 1$ and $x = 3$.
3. Find the area between the graph of $y = x^2 - 1$ and the x-axis for x between 0 and 2.

Solution

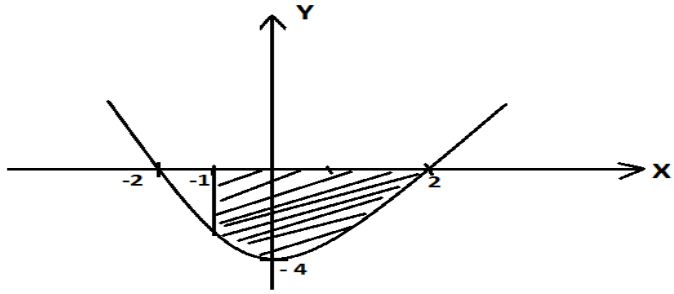

Figure 10.1: Sketch for $y = x^2 - 4$ in Example 10.1

1. See Figure 10.1.

   Since the graph is entirely below the x-axis, the area is;

$$
\begin{aligned}
A &= -\int_{-1}^{2} (x^2 - 4)dx = \int_{-1}^{2} (-x^2 + 4)dx \\
&= \left[-\frac{x^3}{3} + 4x \right]_{-1}^{2} \\
&= \left(-\frac{8}{3} + 8 \right) - \left(\frac{1}{3} - 4 \right) \\
&= -\frac{8}{3} + \frac{24}{3} - \frac{1}{3} + \frac{12}{3} \\
&= \frac{27}{3} \\
&= 9 \ \text{(square units)}.
\end{aligned}
$$

2. See Figure 10.2.

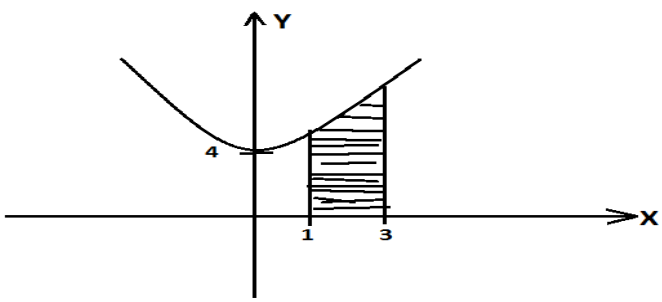

Figure 10.2: Sketch for $y = x^2 + x + 4$ in Example 10.1

The required area is entirely above the x-axis, so

$$\begin{aligned}
A &= \int_1^3 (x^2 + x + 4)dx = \left[\frac{x^3}{3} + \frac{x^2}{2} + 4x\right]_1^3 \\
&= \left(\frac{27}{3} + \frac{9}{2} + 12\right) - \left(\frac{1}{3} + \frac{1}{2} + 4\right) \\
&= \left(\frac{54}{6} + \frac{27}{6} + \frac{72}{6}\right) - \left(\frac{2}{6} + \frac{3}{6} + \frac{24}{6}\right) \\
&= \frac{153}{6} - \frac{29}{6} \\
&= \frac{62}{3} \text{ (square units).}
\end{aligned}$$

3. See Figure 10.3.

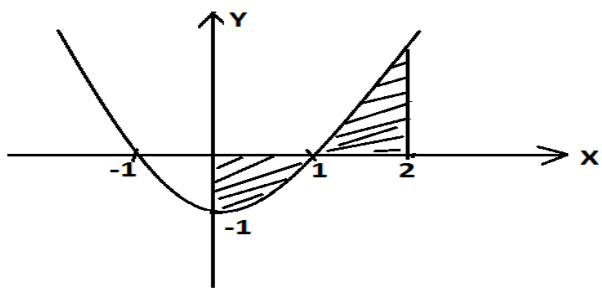

Figure 10.3: Sketch for $y = x^2 - 1$ in Example 10.1

The graph is below the x-axis when $0 \leq x \leq 1$. It is above the x-axis when

$1 \leq x \leq 2$. Hence the area is

$$\begin{aligned}
A &= -\int_0^1 (x^2-1)dx + \int_1^2 (x^2-1)dx \\
&= \int_0^1 (-x^2+1)dx + \int_1^2 (x^2-1)dx \\
&= \left[-\frac{x^3}{3}+x\right]_0^1 + \left[\frac{x^3}{3}-x\right]_1^2 \\
&= \left(-\frac{1}{3}+1\right) + \left(\frac{8}{3}-2\right) - \left(\frac{1}{3}-1\right) \\
&= -\frac{1}{3}+\frac{3}{3}+\frac{8}{3}-\frac{6}{3}-\frac{1}{3}+\frac{3}{3} \\
&= \frac{6}{3} \\
&= 2 \text{ (square units).}
\end{aligned}$$

10.3.2 The Area between Curves

The area between two graphs is found by subtracting the area between the lower graph and the x-axis from the area between the upper graph and the x-axis.

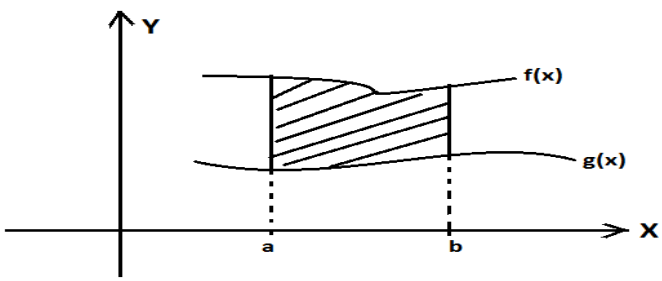

Figure 10.4: Area between curves

Therefore the area of the shaded part in Figure 10.4 is

$$\begin{aligned}
A &= \int_a^b f(x)dx - \int_a^b g(x)dx \\
&= \int_a^b [f(x)-g(x)]dx \text{ (square units).}
\end{aligned}$$

Example 10.2.

1. Calculate the area of the segment cut from the curve $y = x(3 - x)$ and the line $y = x$.

2. Find the area of the region bounded above by $y = x$, bounded below by $y = x^2$ and bounded on the sides by $x = 0$ and $x = 1$.

Solution

1. Solving for x in $x(3 - x) = x$ gives values of x at the points of intersection of the two graphs. So

$$x(3 - x) = x \Rightarrow 3x - x^2 = x \Rightarrow x^2 - 2x = 0$$

giving $x = 0$ and $x = 2$. See Figure 10.5.

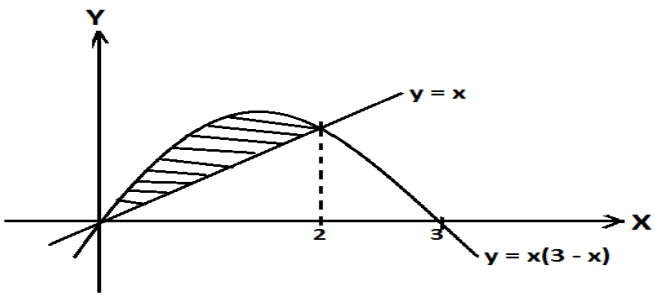

Figure 10.5: Area between $y = x(3 - x)$ and $y = x$

Hence

$$\begin{aligned} A &= \int_0^2 [x(3 - x) - x]dx \\ &= \int_0^2 (-x^2 + 2x)dx \\ &= \left[-\frac{x^3}{3} + x^2\right]_0^2 \\ &= -\frac{8}{3} + 4 \\ &= \frac{4}{3} \text{ (square units).} \end{aligned}$$

2. We do not need to sketch the two graphs to find the stated area, since we know that the graph of $y = x$ is above that of $y = x^2$. In fact the area is

$$\begin{aligned} A &= \int_0^1 \left(x - x^2\right) dx \\ &= \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 \\ &= \frac{1}{2} - \frac{1}{3} \\ &= \frac{3}{6} - \frac{2}{6} \\ &= \frac{1}{6} \text{ (square units).} \end{aligned}$$

Activity 10.3.

1. Find the area of the region included between the parabola $y = \frac{3}{4}x^2$ and the line $3x - 2y + 12 = 0$.

2. Find the area of the region bounded by the curve $y = x^2 + x$, x-axis and the lines $x = 2$ and $x = 5$.

3. Find the area under the curve of $y = x^2$ between $x = 0$ and $x = 4$.

4. Find the area of the region bounded above by the graph of $y = 15xe^{-x^2}$, bounded below by the x-axis and bounded on the sides by lines $x = 0$ and $x = 2$.

5. Find the area of the region bounded above by $y = x^2$, bounded below by $y = x^4$ and bounded on the sides by the lines $x = -1$ and $x = 1$.

6. Find the area of the region enclosed by $y = x^2$ and $y = 2x - x^2$.

Unit Summary

In this unit, you have learnt how to find areas by calculating definite integrals. These are areas between curves and areas between given curves and the x-axis.

Activity 10.4 (Unit Test).

1. Find the area of the region that lies between the x-axis and the graph of $y = x \sin x^2$ for $0 \leq x \leq \sqrt{\pi}$.
2. Find the area of the region that lies between the graphs of $y = x$ and $y = xe^{x^2}$ for $0 \leq x \leq 1$.
3. Find the area of the region bounded above by $y = \sec^2 x$, bounded below by $y = \sin x$ and bounded on the sides by $x = 0$ and $x = \dfrac{\pi}{4}$.
4. Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.
5. Find the area enclosed by the graphs of $f(x) = x^2 + 1$ and $g(x) = 2x + 4$.
6. Find the area of the curve $y = \sin x$ between $x = 0$ and $x = \pi$.

Unit Answers

Suggested Solutions to Activity 10.3

1. 27 sq units 2. $\dfrac{297}{6}$ sq units 3. $\dfrac{64}{3}$ sq units 4. 7.36 sq units

5. $\dfrac{4}{15}$ sq units 6. $\dfrac{1}{3}$ sq units.

Suggested Solutions to Unit Test

1. 1 sq units 2. $\dfrac{1}{2}(e-2)$ sq units 3. $\dfrac{\sqrt{2}}{2}$ sq units

4. $\dfrac{9}{2}$ sq units 5. $\dfrac{32}{3}$ sq units 6. 2 sq units

Bibliography

[1] J. Stewart, *Calculus: Early Transcendentals* (7th ed.) USA: Brooks/Cole, 2008.

[2] E. Swokowski, *Calculus with Analytic Geometry* (9th Ed.), PWS publishers, 1983.

[3] J. F. Talbert and H. H. Heng, *Additional Mathematics: Pure and Applied*, South Asia: Pearson Education, 1990.